

Behera Higher Institute of Engineering & Technology

Department of Computer Engineering

Graduation project

Smart Glasses for The Blind people

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J u n e – 2 0 2 3

Acknowledged:

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Abstract:

This graduation project book presents the development and implementation of a cutting-edge system called "Smart Glasses for the Blind" that utilizes machine learning techniques, including OCR (Optical Character Recognition), face recognition, and object detection. The project leverages the power of Raspberry Pi as a hardware platform and integrates it with a mobile application designed specifically for the visually impaired. Additionally, a web-based interface is created to cater to the needs of facilities and institutions supporting the blind community.

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Chapter 1: UI-UX

1.0. Introduction:

In this chapter the reader will get advanced knowledge about the Design of the project with lots of details System design is the process of defining the architecture, Modules, interfaces, and data for a system to satisfy specified requirements.

1.1.0 Fundamental Defines:

1.1.1 UI (User Interface):

The user interface (UI) It is the point of human and computer interface for communication to device. The goal of effective UI is to make the user's experience easy and intuitive requiring minimum effort on the user's part to receive the maximum desired outcome.

History: Back in the 1970's, if you wanted to use a computer, you had to use the command line interface and it is requiring infinite lines of code to complete a simple task. The user interface evolved with the introduction of the command line interface, which first appeared as a nearly blank display screen with a line for user input. Finally, the GUI arrived.

- Graphical UI: Elements of a GUI include such things as windows, pull-down menus, buttons, scroll bars and icons.
- Mobile UI: Interfaces on the smaller screens of smartphones and tablets and improving special features, like touch controls
- Web UI Interfaces on screens laptop and computer and you can deal with the interface by mouse and keyboard.

1.1.2 User experience (UX):

The user experience (UX) refers to any interaction a user has with a product. UX considers each and every element that shapes this experience, how it makes the user feel, and how easy it is for the user to end tasks**.**

History: in the early 90s, cognitive scientist Don Norman joined the team at Apple as their User Experience Architect, making him the first person to have UX in his job title. He came up with the term because he wanted to cover all aspects of the person experience with a system, including industrial design, graphics, the interface, the physical interaction.

1.2.0 Benefits of Good System Design

Better understanding by the developer of the system. A properly designed system helps in elaborating and understanding refined aspects of system. System development becomes efficient. An error free designed system saves development time since it helps in understanding without discussion. A well-designed system also helps in early detection of problems in critical areas, if any. Better quality of reporting formats. Modifications can be made easily in the system and little maintenance is required in the operations phase.

1.3.0 Motivation for creating UI/UX design in the project

We have created a mobile application to help the blind in practicing their lives better and the facilities are supervised by the blind remotely and easily through that application that has two interfaces and some features have been added that link that application to the glasses to be an integrated project to meet the needs of the blind.

Then we created a web application also for utilities with a different interface so that the facilities could communicate with the blind on an ongoing basis

1.4.0 Process

It's an iterative design process in which designers focus on the users and their needs in each phase of the design process.

1.4.1 User-Centered design (UCD)

UCD, design teams involve users throughout the design process via a variety of research and design techniques to create highly usable and accessible products for them.

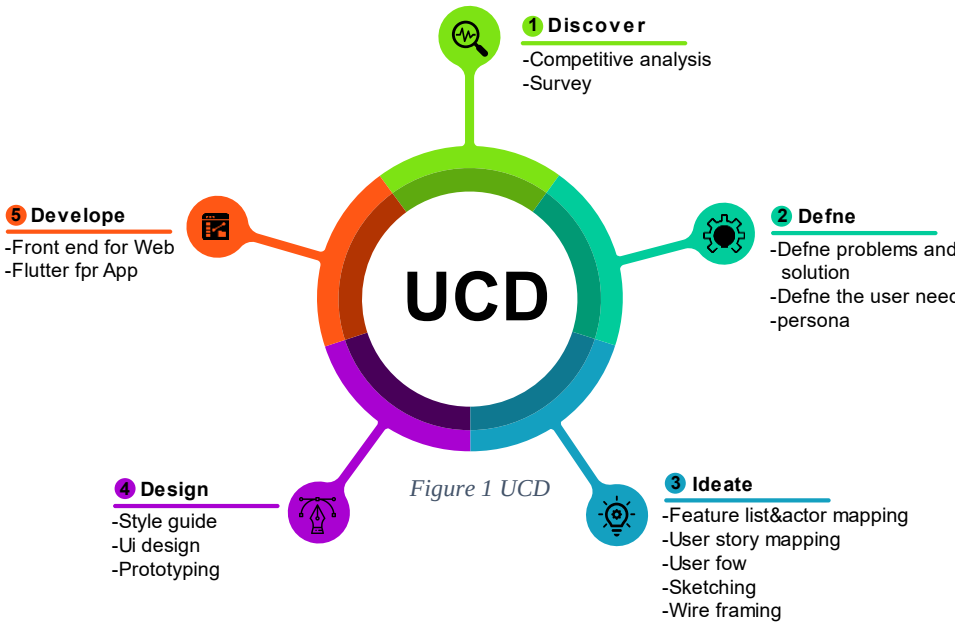


Figure 1 UCD

1.5.0 Discover:

1.5.1 Competitive analysis:

It is a way to collect and compare data about products in order to make more informed decisions about strategy, its types are Direct competitor and Indirect competitor.

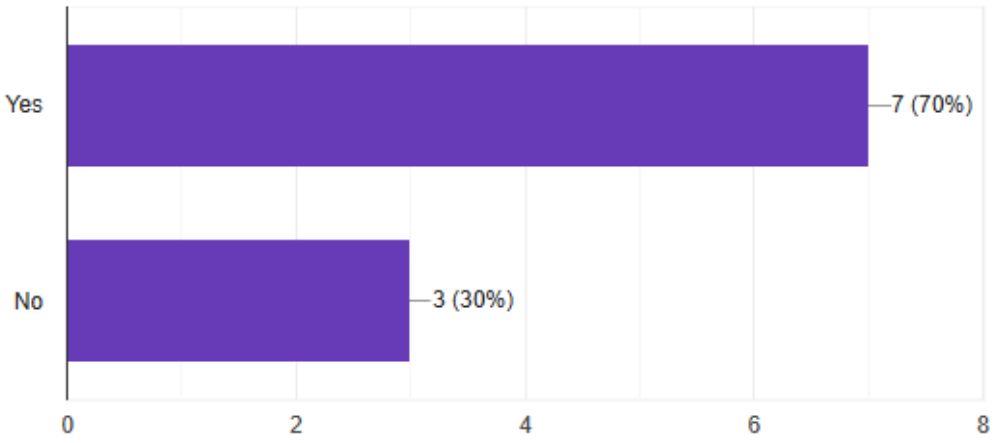
When we searched for apps related to our app, we couldn't find any directly competitor that gives the same features that we have, such as linking the features of smart glasses to an application that also has some features to help the blind person practice his life better on the other hand, we found some features in some applications separately we analysis them.

Feature name	competitor type	strengths	Weakness
Be my eye	Direct competitor	Large user base, available in multiple languages	Relies on volunteers who may not always be available
Seeing Ai	Direct competitor	Accurate and reliable, offers a range of features	Only available on iOS devices
Aira	Direct competitor	Professional assistance, remote guidance through live video streaming	Requires a subscription fee
BlindSquare	Direct competitor	Provides location-based information, supports multiple languages	Only available on iOS devices
Access Able	Indirect competitor	Provides accessibility information for public places and spaces	Relies on user-generated data for information
TapTapSee	Indirect competitor	Provides object recognition	Limited accuracy for some objects

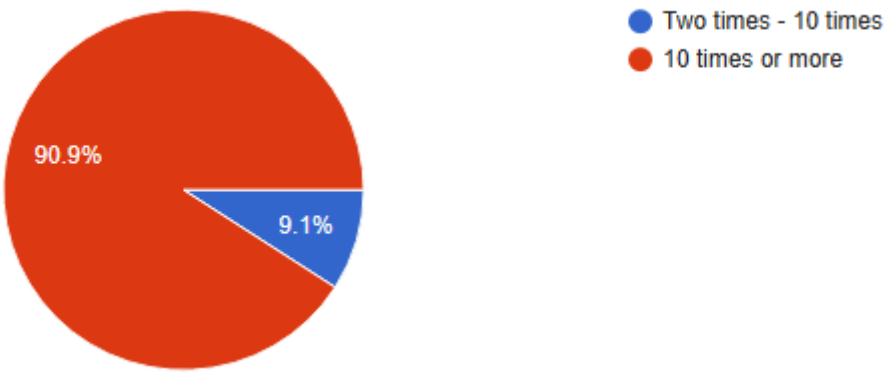
1.5.2 Survey:

I did an online survey using google forms to observe any pattern and similarity in what the potential blind user may want.

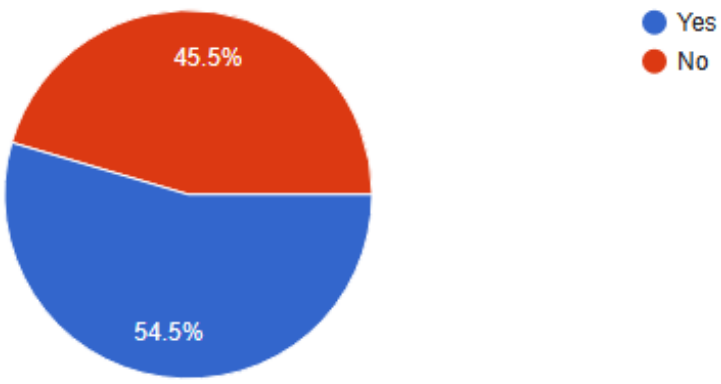
1. Are you having difficulties using mobile applications due to access issues?



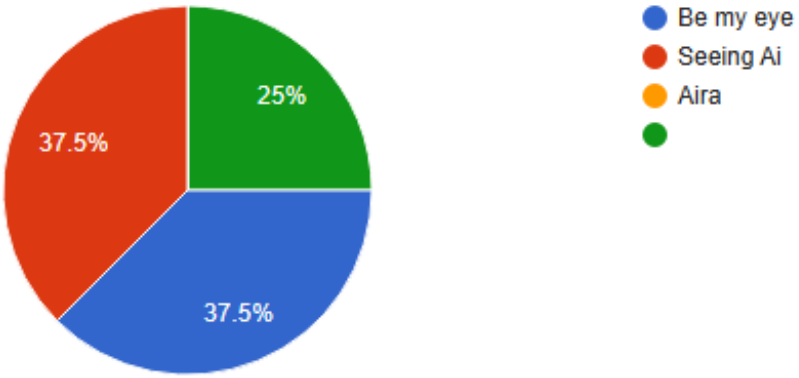
2. How often do you use mobile apps for daily activities?



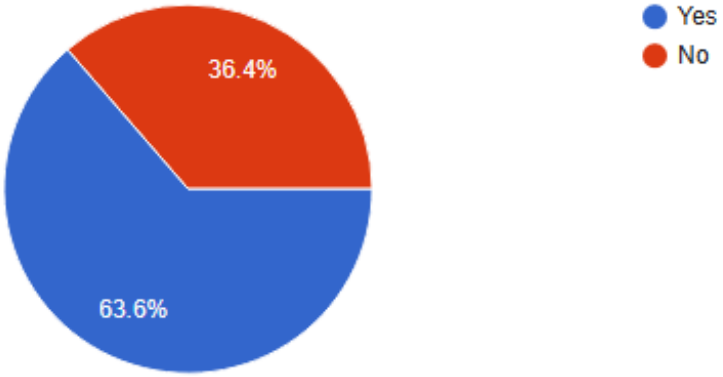
3. Have you ever used any of the mobile apps designed specifically for visually impaired people?



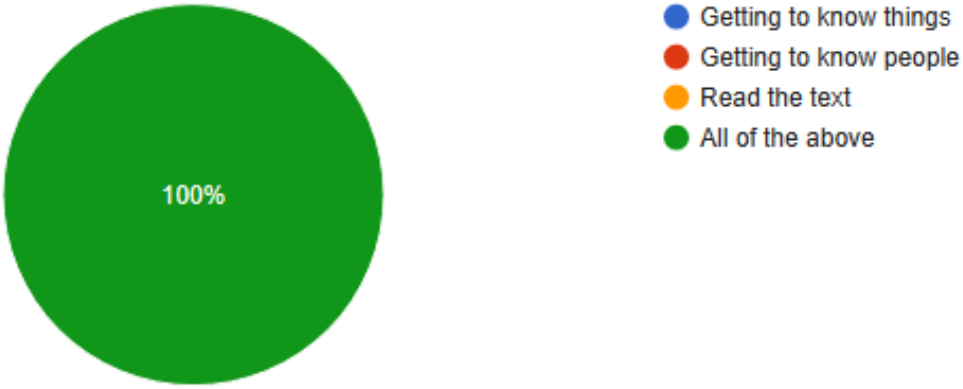
4. If yes, which mobile app did you use?



5. Would you be willing to pay for a mobile app designed specifically for visually impaired people?



6. What is the most important feature you want in your use of the phone?



1.6.0 Define:

1.6.1 Define problem and solution:

problem:

Blind people face difficulty accessing information and features in apps due to their primarily visual design

Images or videos in apps may be inaccessible to blind users without alternative descriptions or audio descriptions.

Lack of all the features that help the blind in one application

Solution:

Simple UI designed specifically for the blind

providing alt text for images, proper text labeling, and designing interfaces compatible with assistive technologies like screen readers.

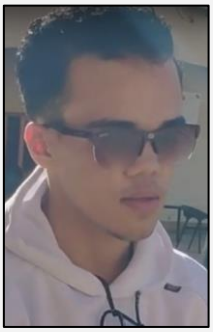
Our application combines all the features that the blind need

1.6.2 Define the user need:

1. The blind person connected with the facilities by chat and video call.
2. Linking glasses features with the application.
3. Application be free.
4. Ease of use of the application

1.6.3 Persona:

In UX (User Experience) design, a persona is a fictional character that represents a group of users who share common goals, needs, and behaviors. Personas are created based on user research and data to help designers understand their target audience and design products or services that meet their needs.



Name: Ibrahem

Old:19

- **Ibrahim is a blind person and he is a student at the Faculty of Commerce, Alexandria University.**
- **He is always using phone apps that make his life easier.**
- **He hopes that the phone apps be smoother to use and he hopes to convert any command into voice.**
- **He wants GPS to be easily identified.**

Figure 2 Persona

1.7.0 Ideate:

Ideation is a crucial part of the UX (User Experience) design process, and it refers to the phase in which designers generate and explore a wide range of ideas for solving a particular design problem. During the ideation phase, designers aim to generate as many ideas as possible, without judging or evaluating them.

The goal of ideation is to generate a large quantity of ideas, which can then be narrowed down and refined in later stages of the design process.

1.7.1 Actor mapping:

By mapping out the actors involved in a design problem, designers can better understand the needs, goals, and motivations of each actor, as well as their relationships and interactions with other actors. This can help designers identify potential conflicts or areas of overlap, and can also help them prioritize design solutions that meet the needs of all actors involved.

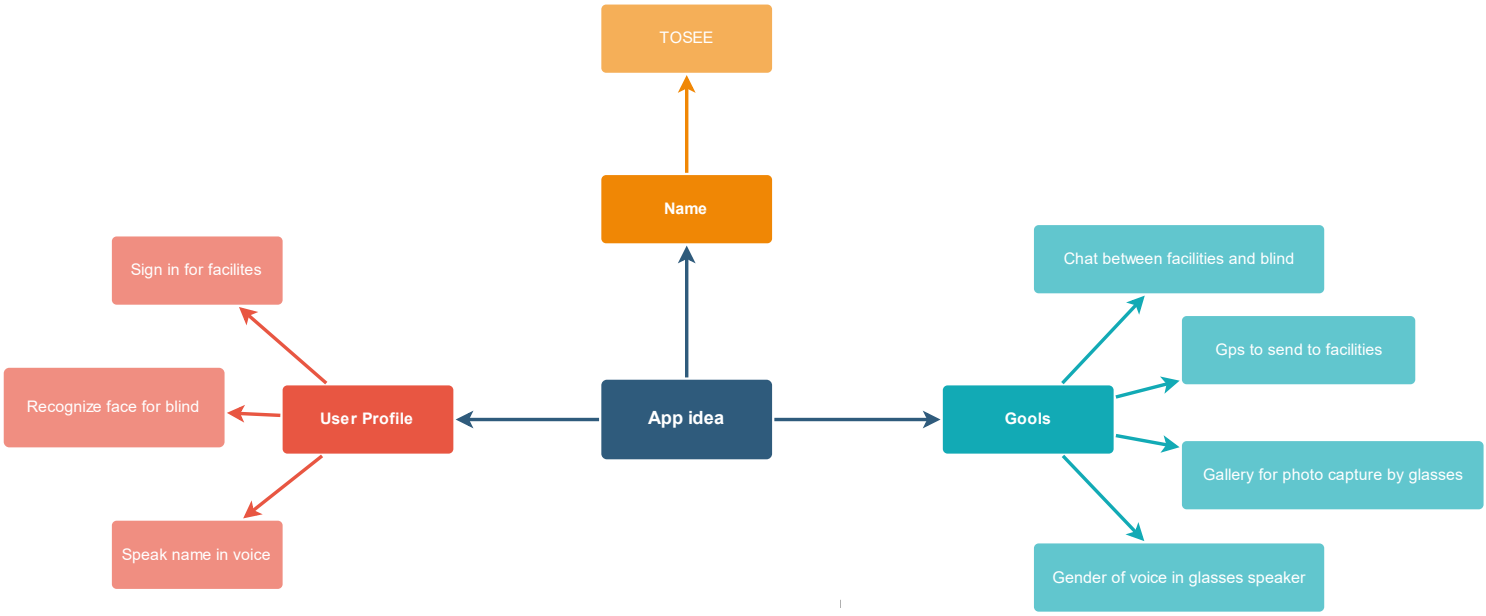


Figure 3 Actor Mapping for App

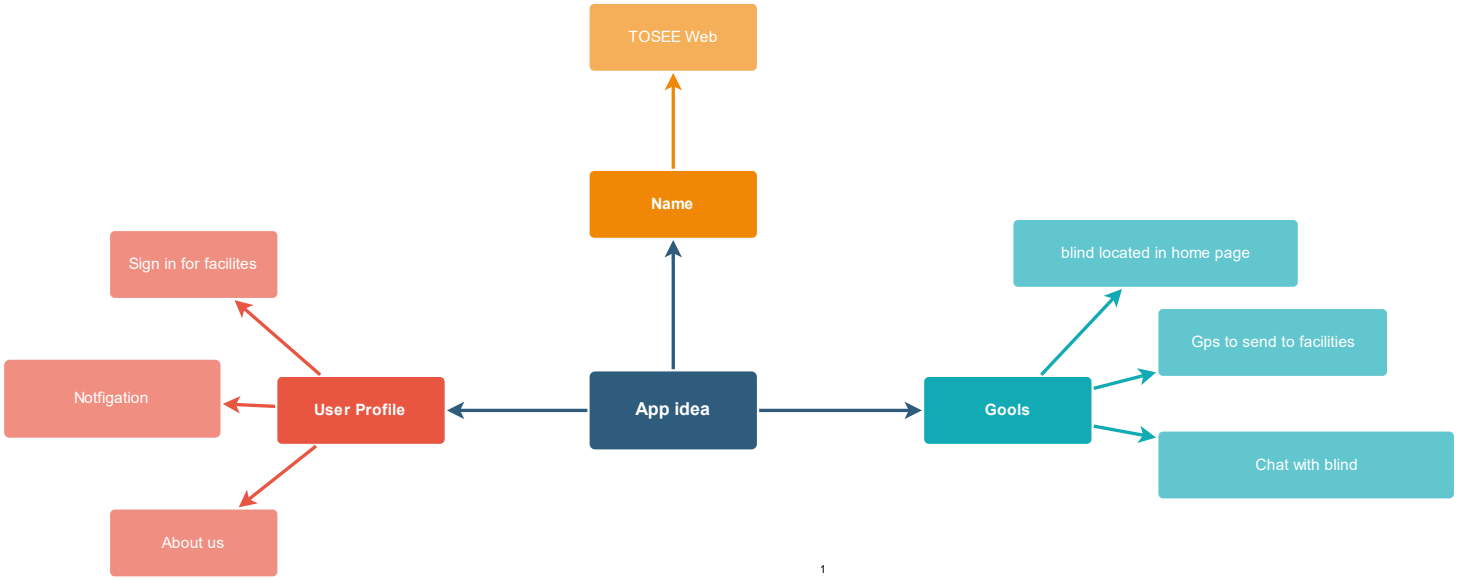


Figure 4 Actor Mapping for web

1.7.2 User flow:

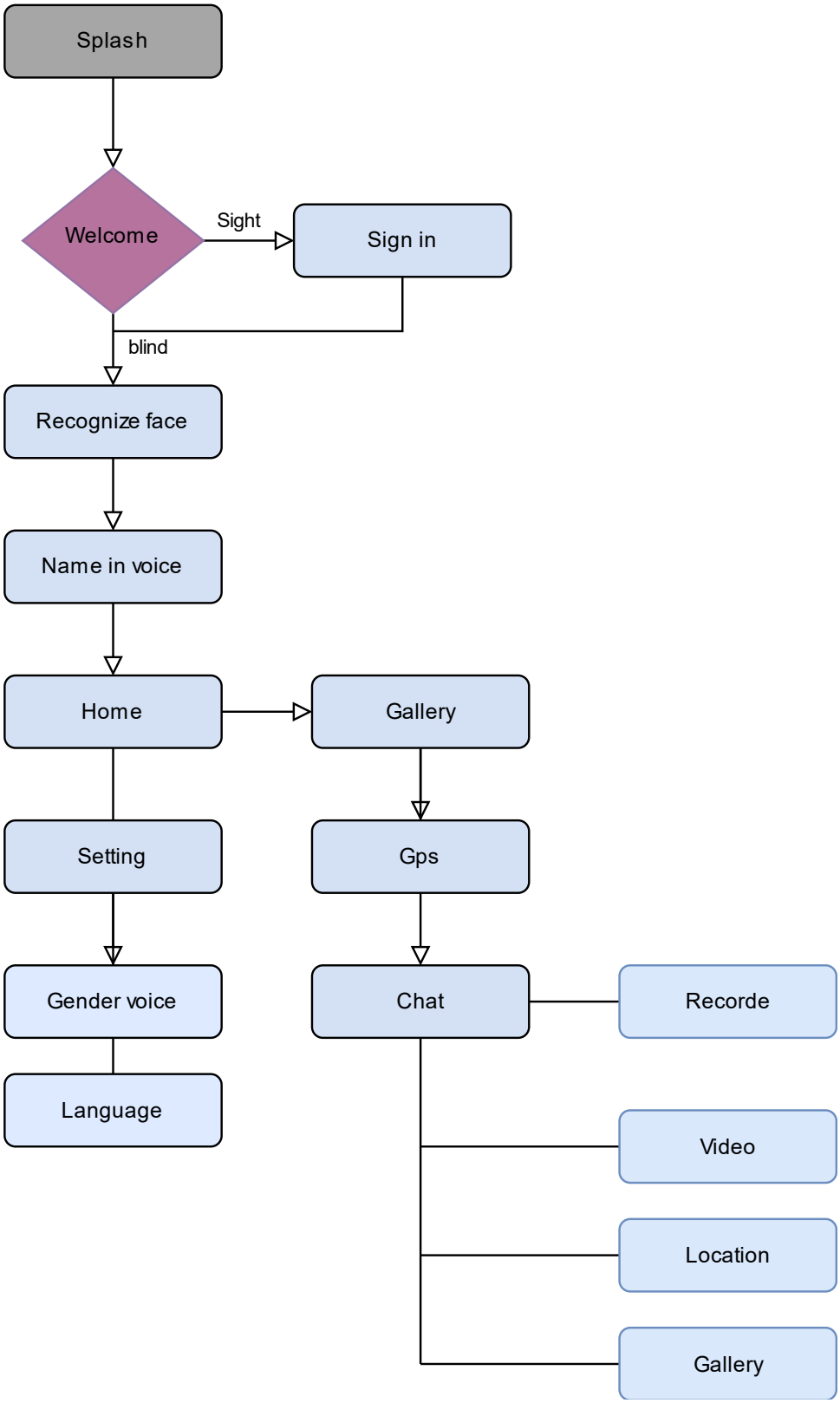


Figure 5 App user flow

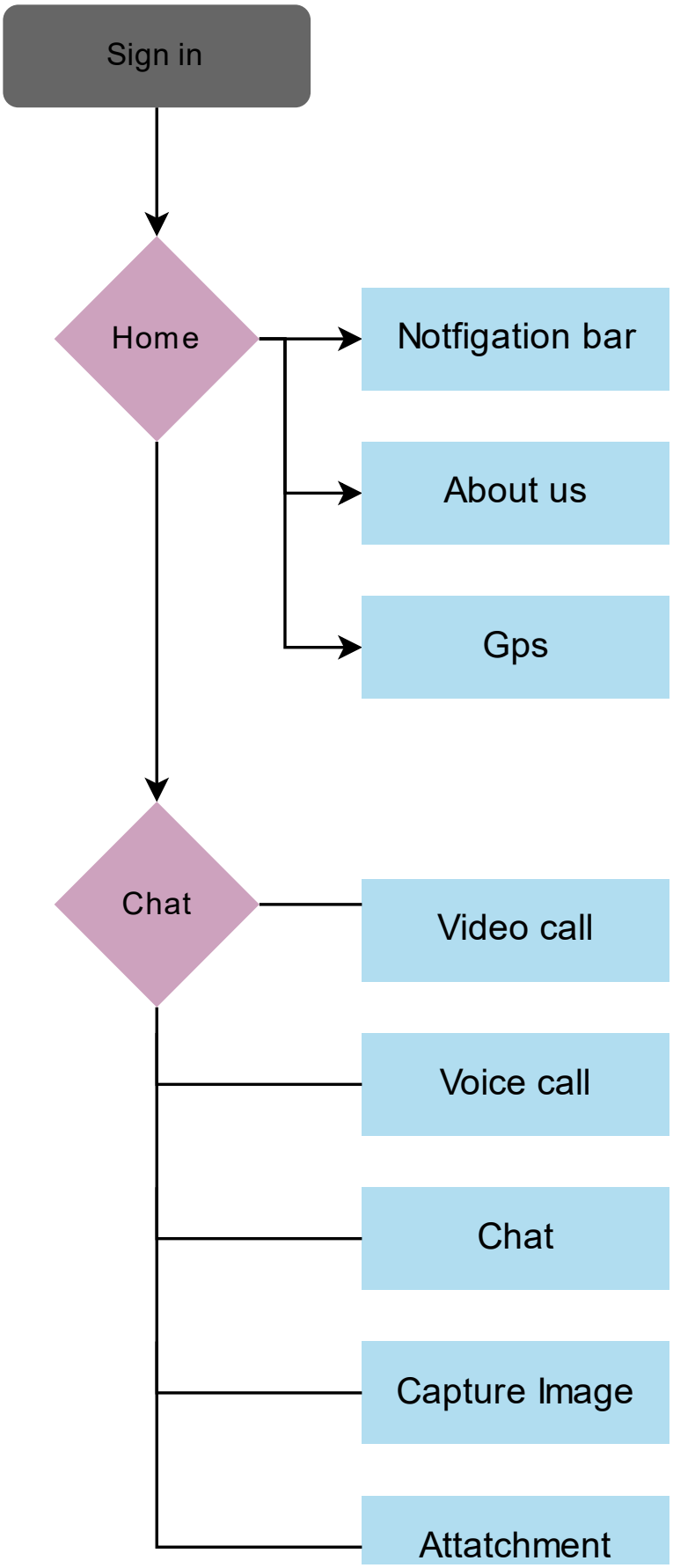


Figure 6 User flow for web

1.7.3 Sketching: “Low fidelity”

Paper wireframing is a widely used method in the UCD process that helps designers create software that meets the user’s expectations and needs.

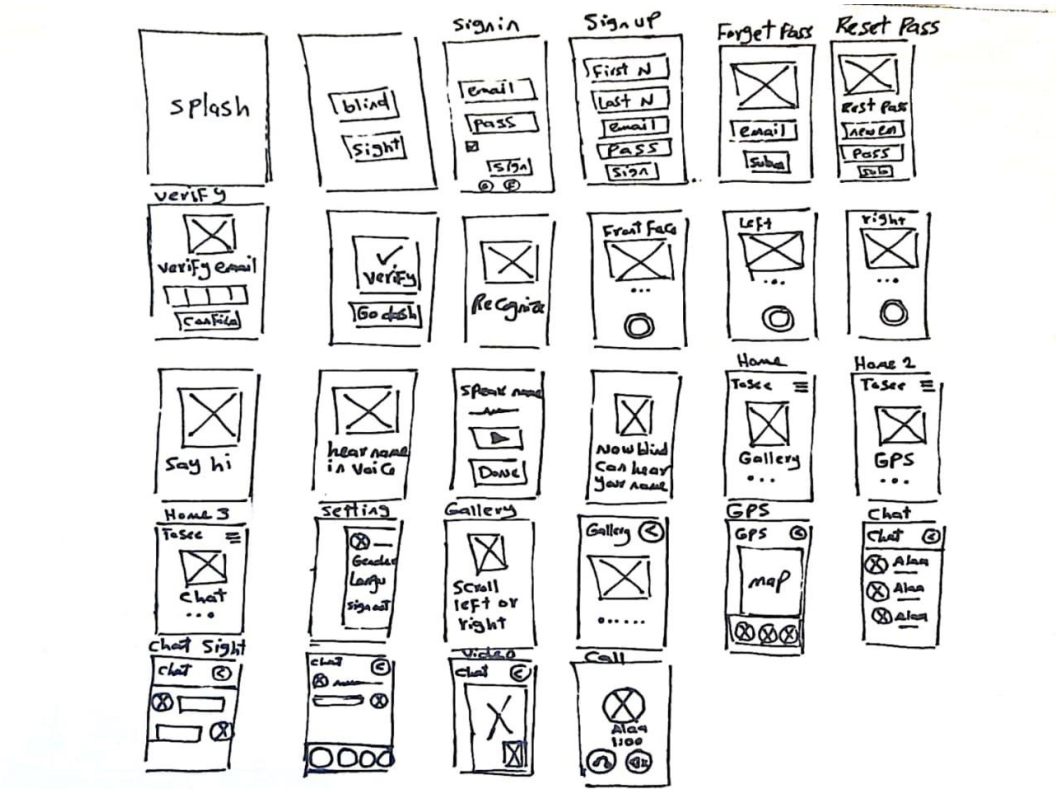


Figure 8 Sketching for App

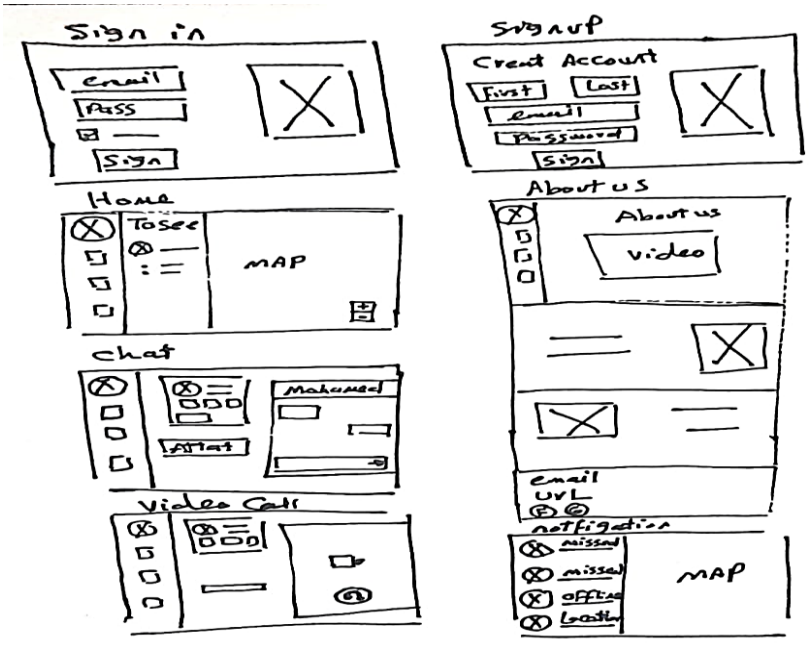


Figure 7 Sketching for Web

1.8.0 Design:

1.8.1 Ui design:” High fidelity”:

once the initial flow was completed and wireframes were ready, we started choosing a typeface and set of colors. We created a simple, clean UI style guide to maintain consistency, then created a typography scale to ensure that the hierarchy throughout the project is preserved

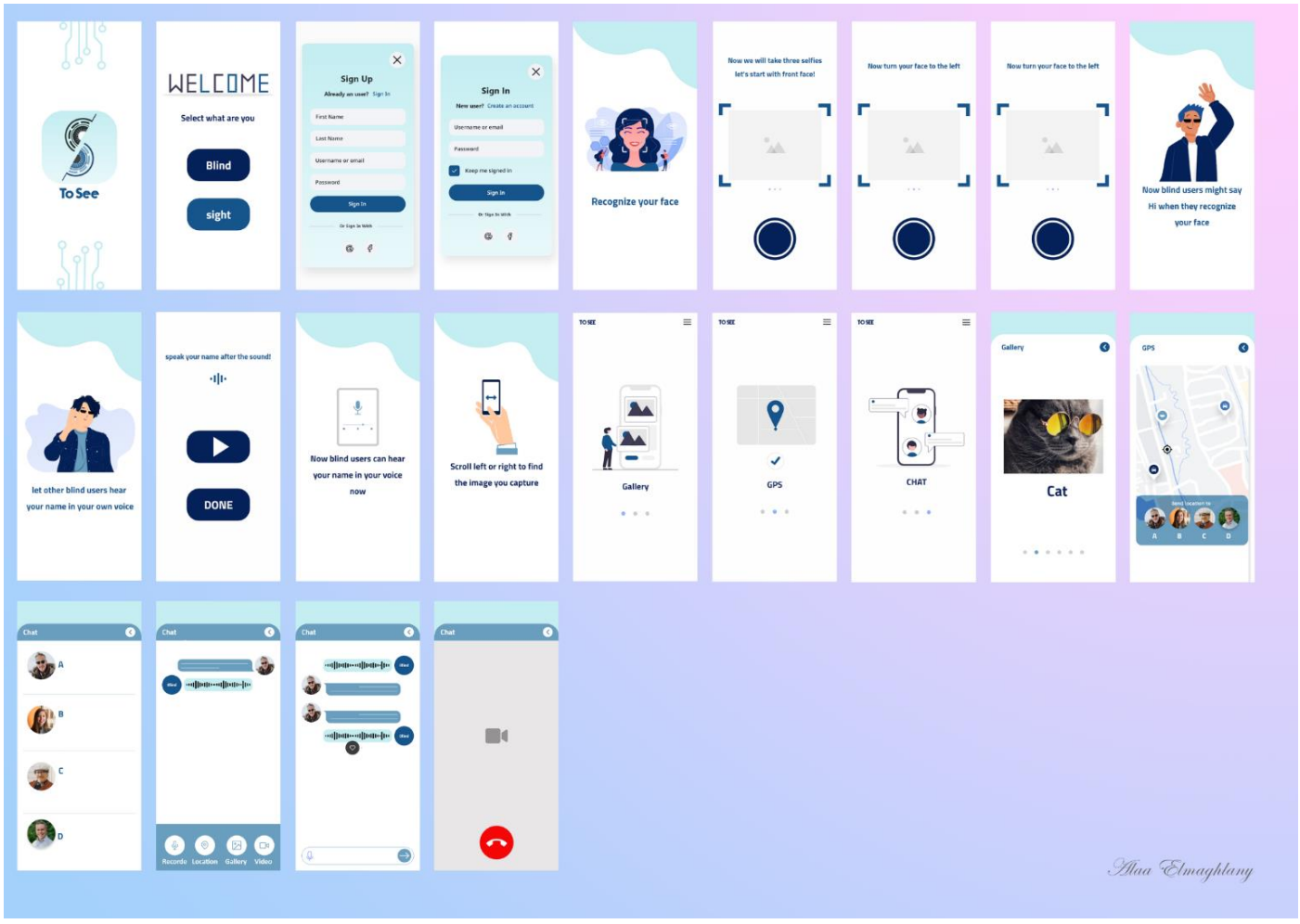


Figure 9 App High fidelity

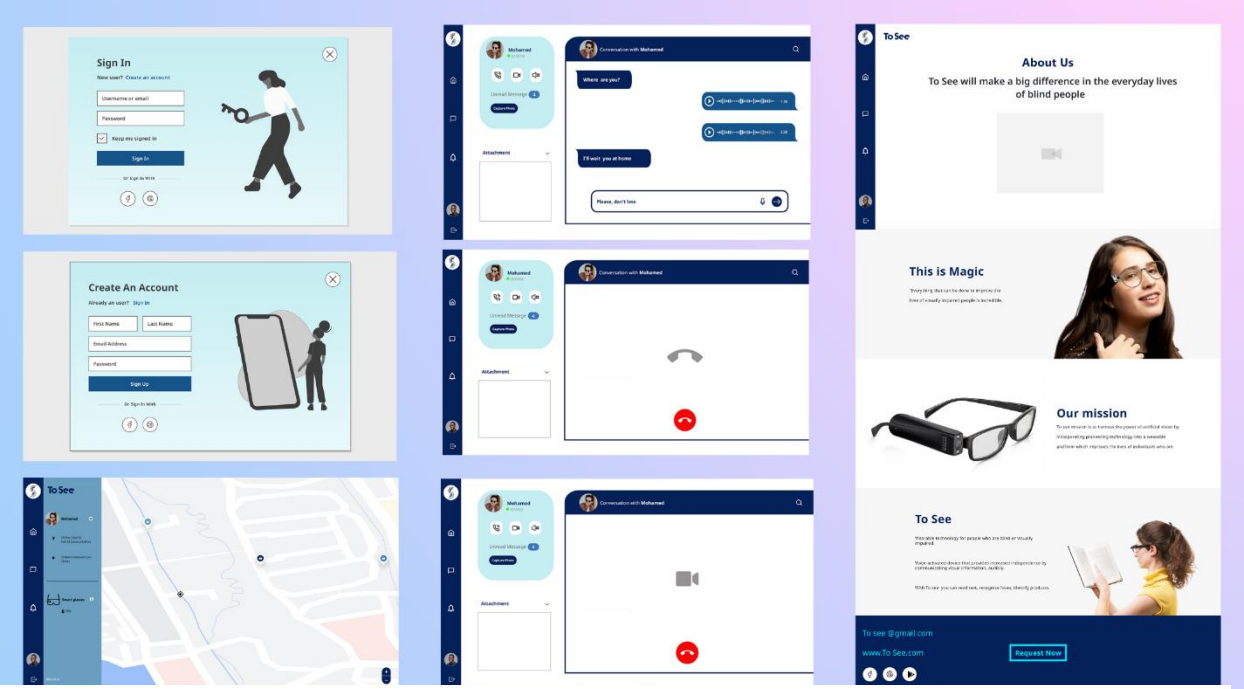


Figure 10 High fidelity web

1.8.1.1Typography

Cairo font, Noto Sanse

1.8.1.2Color:

#031740 **#052159** **#16558C** **#699EBF** **#C4EEF2**

These colors were chosen in accordance with the **WCAG** standard.

WCAG is stands for Web Content Accessibility Guidelines. It is a set of guidelines published by the World Wide Web Consortium (W3C) that provide a framework for making web content more accessible to people with disabilities.

We have done a test on the selection of those colors used in web design and application through a WCAG website**, and this is result**

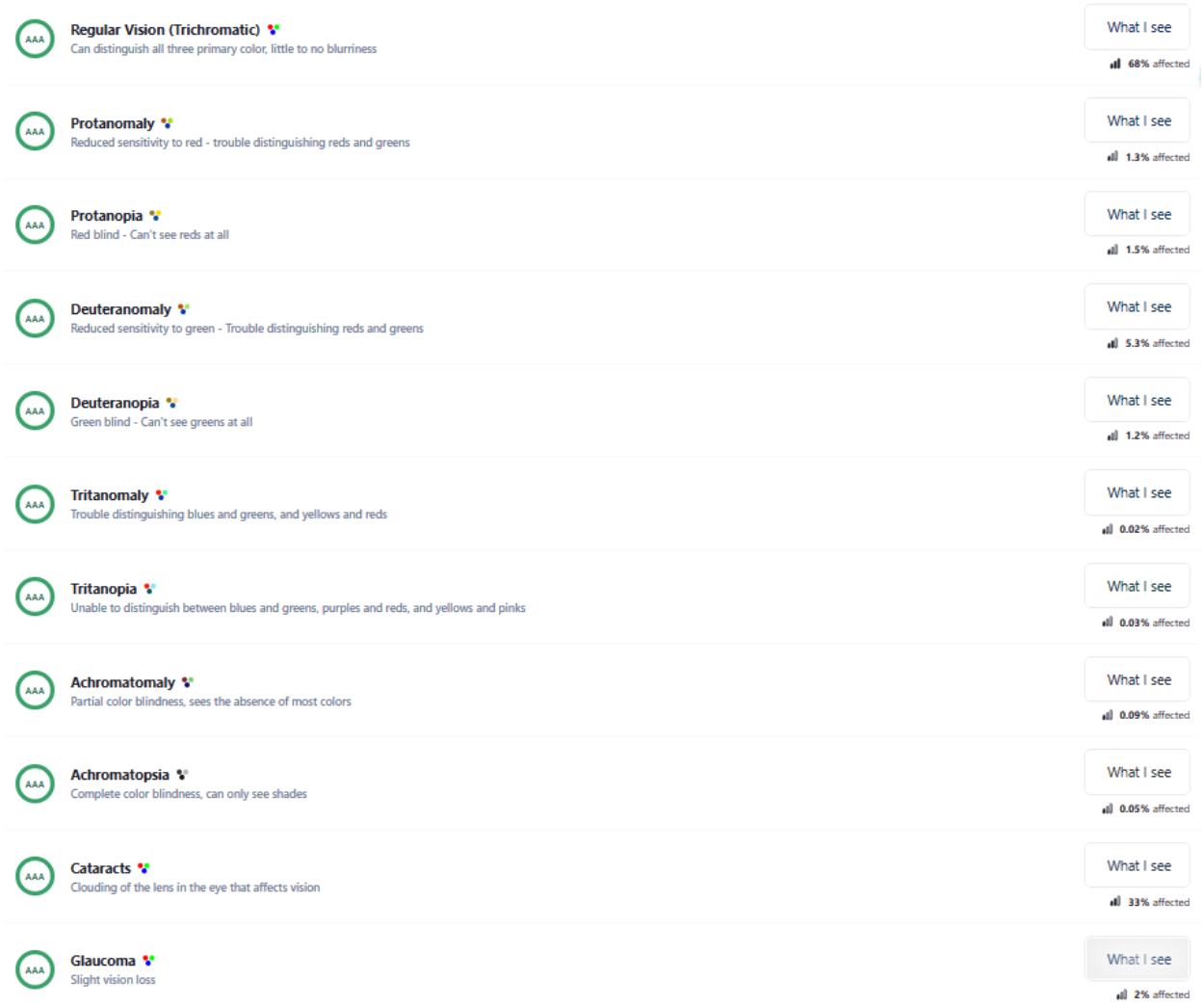


Figure 11 WCAG color result

1.8.1.3Logo:

From concept about half of eye for the blind and half eye for the sight person A symbol of integration between them and that's what our app wants.

The logo is in the shape of a S letter and this letter is the first letter of the name of our application.



1.9.0 Prototype:

A prototype is a preliminary version of a product or service that is used to test and validate design ideas before the final product is developed

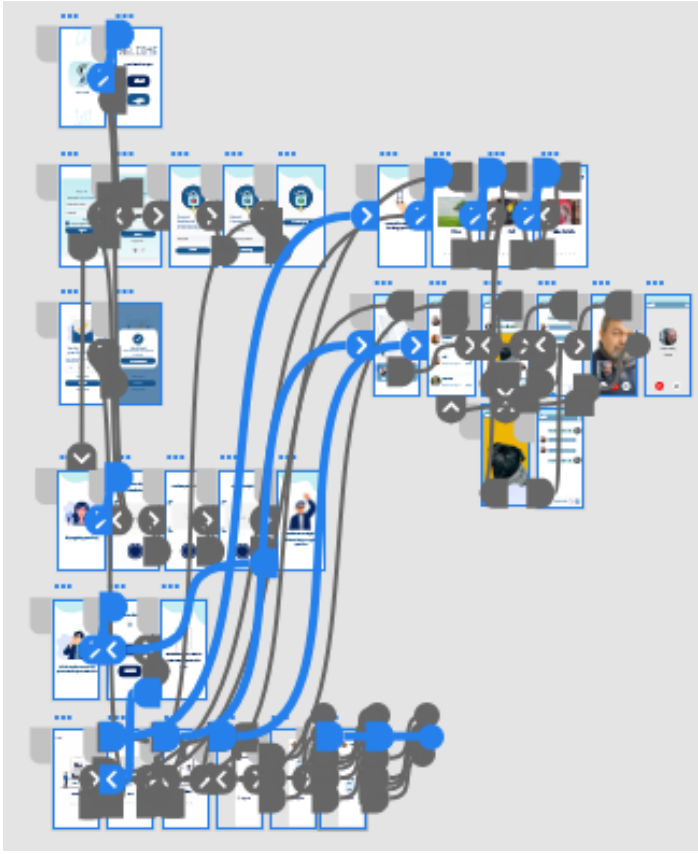


Figure 12 App prototype

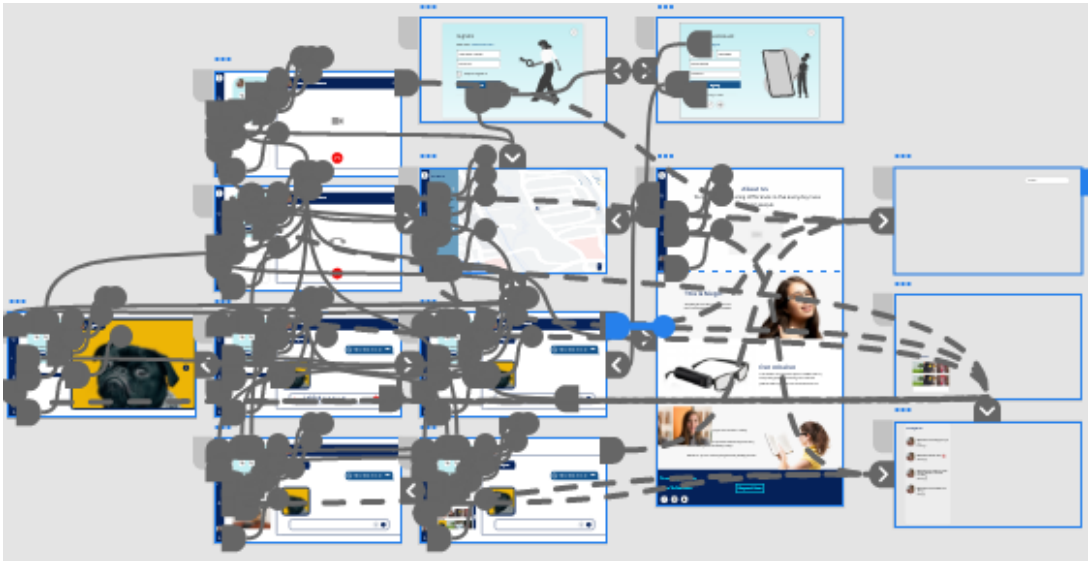


Figure 13 web prototype

Chapter 2: Machine learning

2.1.0 Introduction:

Machine learning (ML) modules are critical components of smart glasses designed to assist visually impaired individuals. These modules can perform a variety of tasks, including Optical Character Recognition (OCR), Text-to-Speech (TTS) conversion, face recognition, and object detection. OCR allows the glasses to recognize and read text in the user's environment, while TTS enables the glasses to speak the text aloud for the user. Face recognition technology can help users identify people they know, while object detection can provide information about the user's surroundings, such as the location of objects and obstacles. By leveraging the power of ML, these smart glasses can provide enhanced accessibility and independence for visually impaired individuals, enabling them to navigate their surroundings more effectively and efficiently.

2.1.1 Artificial Intelligence:

Artificial Intelligence (AI) involves creating machines that can simulate human intelligence, allowing them to perform tasks without explicit instructions and to think and act in a way that is rational and human-like. Developing an AI system is a complex and deliberate process that involves reverse-engineering the traits and capabilities that make us human and incorporating them into a machine. By leveraging the computational power of AI, we can push beyond the limits of human ability and achieve feats that were previously thought impossible.

2.2.1 Machine Learning:

Machine Learning is a facet of AI that enables machines to learn from past experiences by analyzing input data, and then use that knowledge to make predictions about the future. The goal of a Machine Learning system is to perform at a level that is at least comparable to human intelligence. By leveraging the power of data, these systems can make highly accurate predictions and automate complex decision-making processes. Machine Learning is generally categorized into three types: Supervised Learning, Unsupervised Learning, Reinforcement learning.

2.2.1.1 Supervised: Type of ML that involves supervision, where machines are trained on labelled datasets.

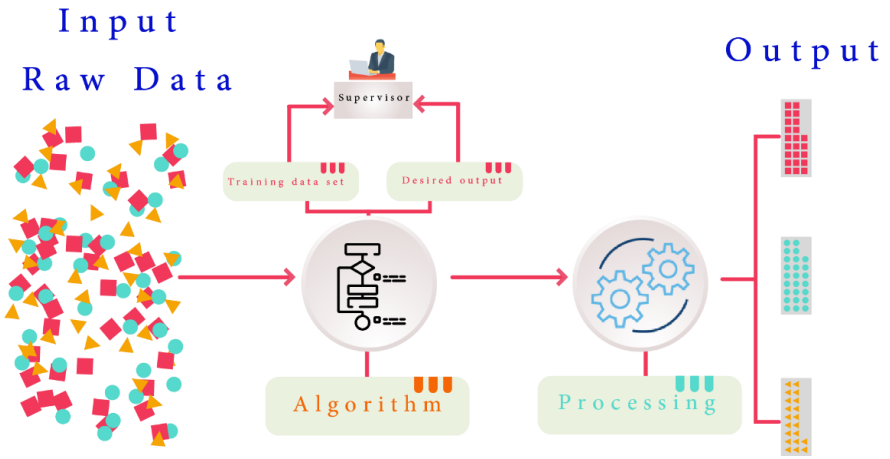


Figure 14(Supervised ML)

2.2.1.2 Unsupervised: Type of ML that devoid of supervision, where machines are trained using an unlabeled dataset.

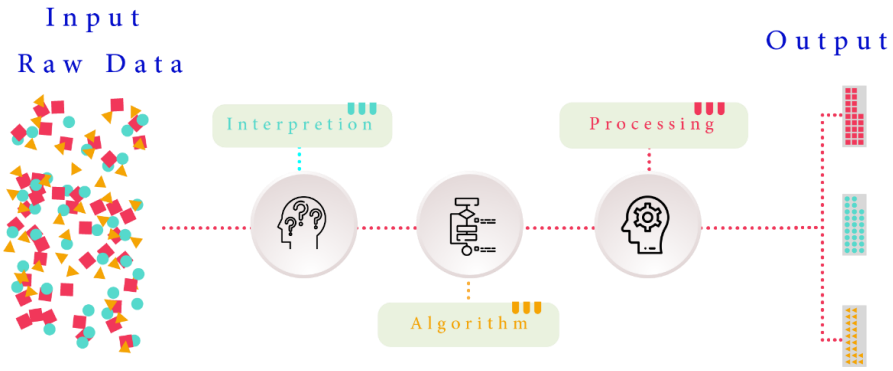


Figure 15 (Unsupervised ML)

2.2.1.3 Reinforcement Learning: When the training dataset is set of rules. It uses hit and trial method, takes action, learns from experiences.

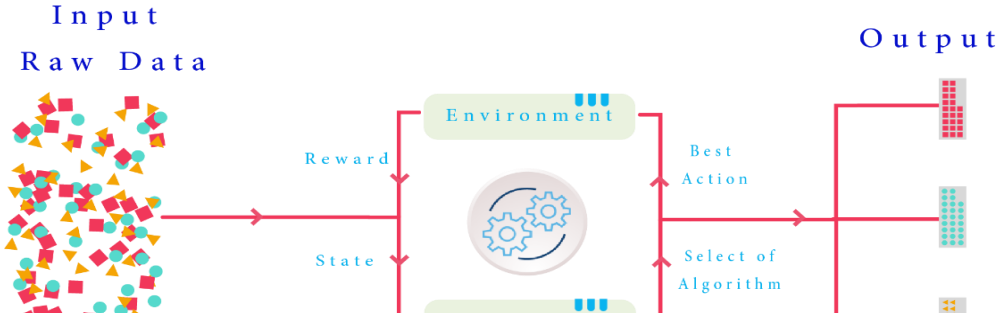


Figure 16 (Reinforcement Learning)

The objective of ML algorithms is to estimate a predictive model that best generalizes to a set of data. For ML to be super-efficient, one needs to supply a large amount of data for the learning algorithm to understand the system's behavior and generate similar predictions when supplied with new data.

2.1.3 Deep Learning:

Deep Learning is a type of Machine Learning that utilizes artificial neural networks to mimic the structure of the human brain. These networks consist of multiple layers that progressively extract higher-level features from the raw input data. One of the key advantages of Deep Learning is that it can learn which features to optimally place in each layer without human intervention, resulting in more efficient and effective processing. Deep Learning is suitable for both supervised and unsupervised learning tasks, making it a versatile tool for a wide range of applications. Ultimately, the goal of Deep Learning is to achieve performance levels that are at least comparable to human intelligence.

Thanks to recent advancements in Deep Learning, machines can now outperform humans in certain tasks, such as identifying objects in images. Unlike traditional programming, Deep Learning operates without strict rules or pre-defined instructions. Instead, Deep Learning algorithms extract patterns and trends from vast sets of unstructured data through either supervised or unsupervised learning. By leveraging the power of artificial neural networks that mimic the structure of the human brain, these algorithms can recognize increasingly complex features and patterns with each layer they traverse. Ultimately, the goal of Deep Learning is to achieve performance levels that are at least comparable to human intelligence, if not surpassing it in certain areas.

Although the concept of deep learning was first introduced in the 1980s, it has only become practical in recent years for two main reasons.

- Firstly, deep learning algorithms require large amounts of labeled data to train effectively. For example, developing a driverless car requires millions of images and thousands of hours of video footage.
- Secondly, deep learning requires significant computational power to process and analyze this data efficiently. High-performance GPUs with a parallel architecture are particularly well-suited for deep learning, and when combined with cloud computing or clusters, can significantly reduce the time required to train a deep learning network from weeks to a matter of hours.

These technological advancements have made it possible for deep learning to be applied in a wide range of fields, from image and speech recognition to natural language processing and autonomous vehicles.

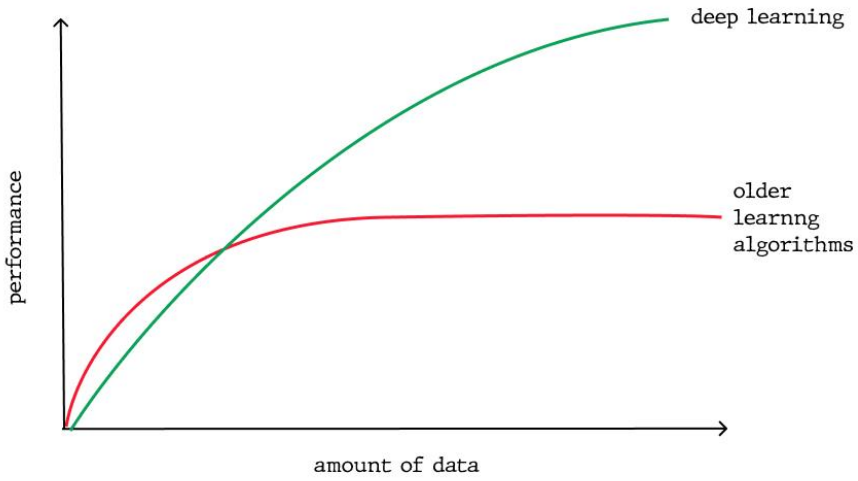


Figure 17(DL vs ML)

2.2.0 Neural Networks:

Neural networks are a type of model that is constructed with layers of interconnected nodes, or neurons, which are arranged in a structure that resembles the human brain. There are several types of neural networks that are commonly used, including Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs). These networks are designed to process and analyze large volumes of complex data, such as images, video, and natural language, and extract meaningful patterns and features from them. By leveraging the power of neural networks, we can develop highly sophisticated models that can perform a wide range of tasks, from image and speech recognition to predictive analytics and decision-making.

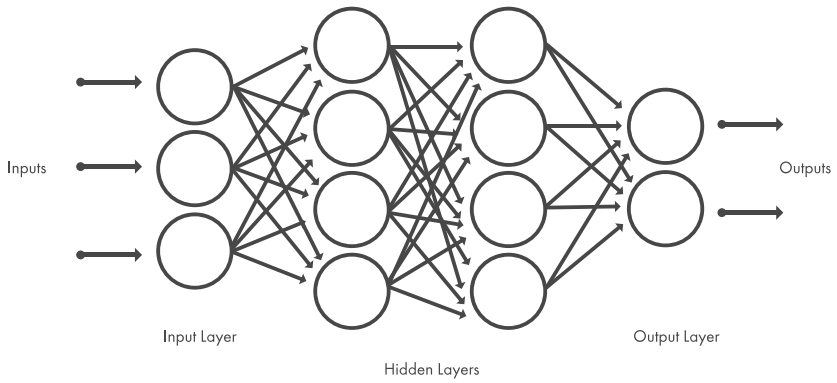


Figure 18 (Single Neuron)

2.2.1 Activation function: are used at the end of a hidden layer neurons or output layer neurons to introduce non-linear complexities to the model.

Most common activation functions:

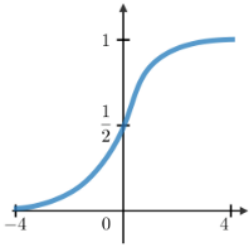
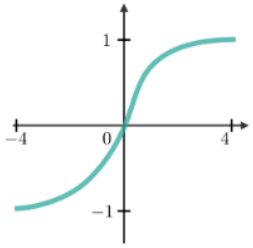
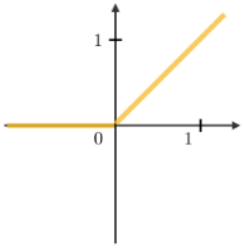
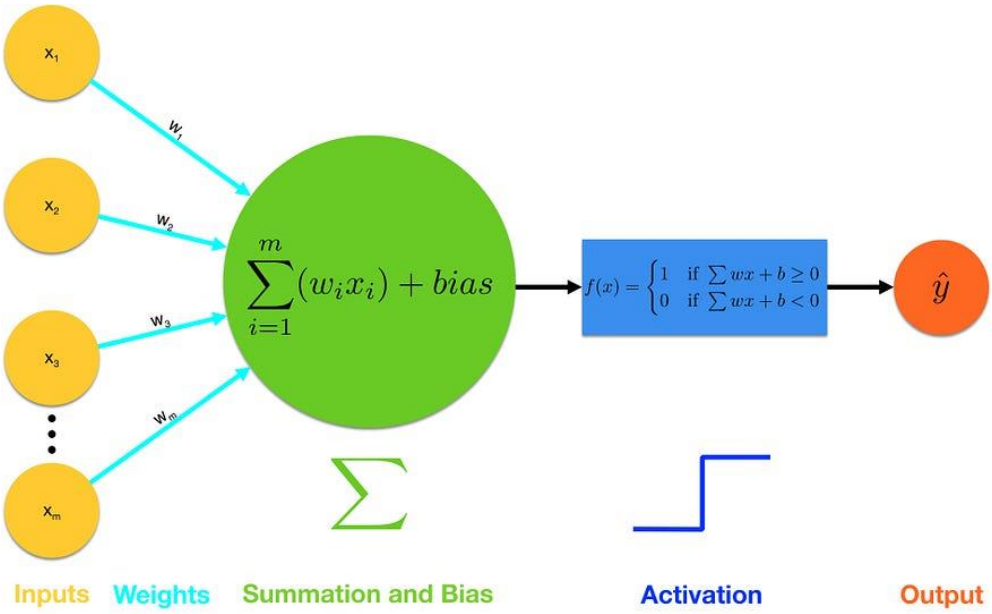
Sigmoid	Tanh	RELU
$g(z) = \frac{1}{1+e^{-z}}$	$g(z) = \frac{e^{z}-e^{-z}}{e^{z}+e^{-z}}$	$g(z) = \max(0, z)$
		

Figure 20(Activation functions)



2.2.2 Learning rate: noted as η, indicates at which pace the weights get updated. This can be fixed or changed. The current most popular method is called Adam, which is a method to adapts the learning rate (learning rate from 0 to 1).

3.2.3 Backpropagation: Is a method to update the weights in the neural network by taking into account the actual output and the desired output. The derivative respect to weight w is computed using chain rule.

$$\frac{\partial E}{\partial w} = \frac{\partial E}{\partial S} * \frac{\partial S}{\partial y} * \frac{\partial y}{\partial w}$$

Weight updated as: $W_{new} = W_{old} - \eta * \frac{\partial E}{\partial w}$

Updating weights:

- Take a batch of training data.
- Perform forward propagation to obtain the corresponding loss.
- Backpropagate the loss to get the gradients.
- Use the gradients to update the weights of the network.

2.2.4 Dropout: is a technique meant to prevent overfitting the training data by dropping out units in a neural network.

2.2.5 Convolutional Neural Networks (CNN):

CNN is a Kind of network architecture for deep learning algorithms and is specifically used for image recognition and tasks that involve the processing of pixel data

CNN helps a machine learning or deep learning model look by breaking images down into pixels that are given tags or labels.

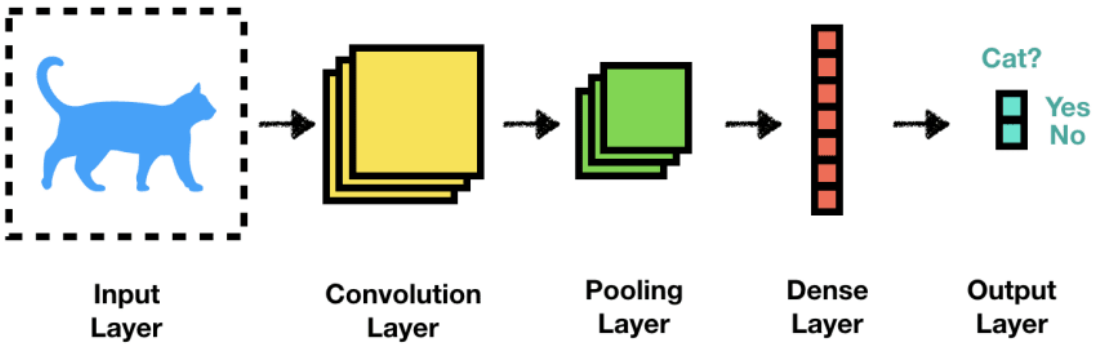


Figure 21(Convolutional Neural Networks)

2.2.5.1 Convolutional layer: Convolutional layers are the major building blocks used in convolutional neural networks

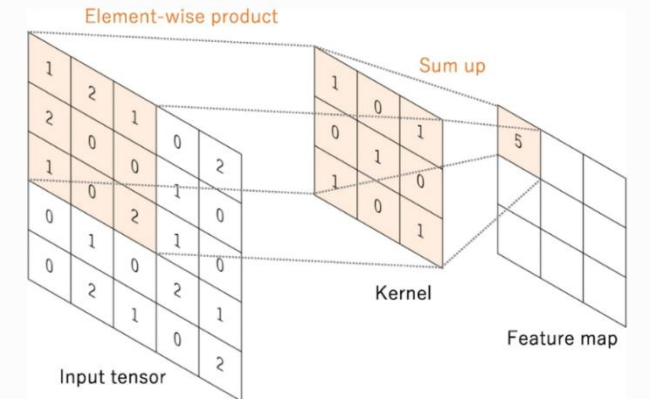


Figure 23(Convolution Layer)

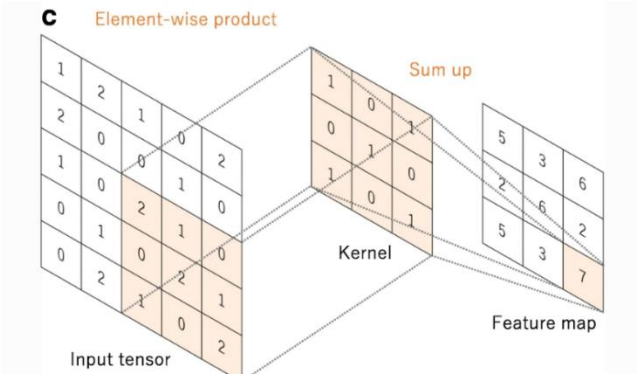


Figure 22 (Convolution Layer)

2.2.5.2 Pooling Layer*:* Down sample the feature map, Using the max operation.

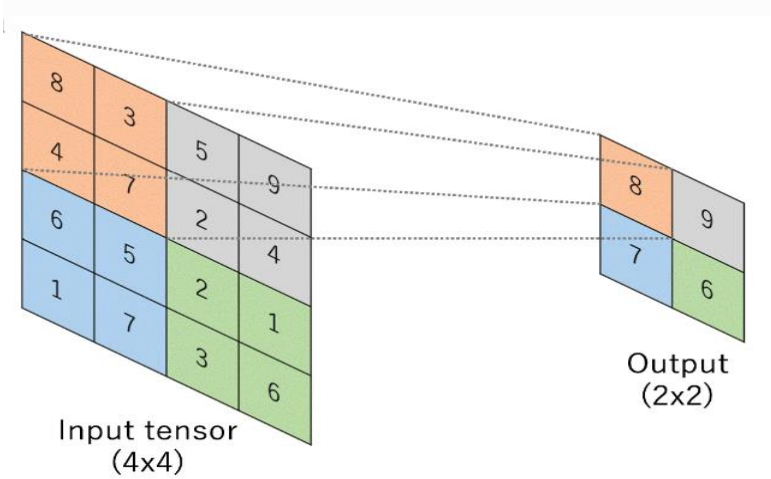


Figure 24 (Pooling Layer)

2.2.5.3 Flattening Layer: Simply rearranged the output matrices to be one dimension array

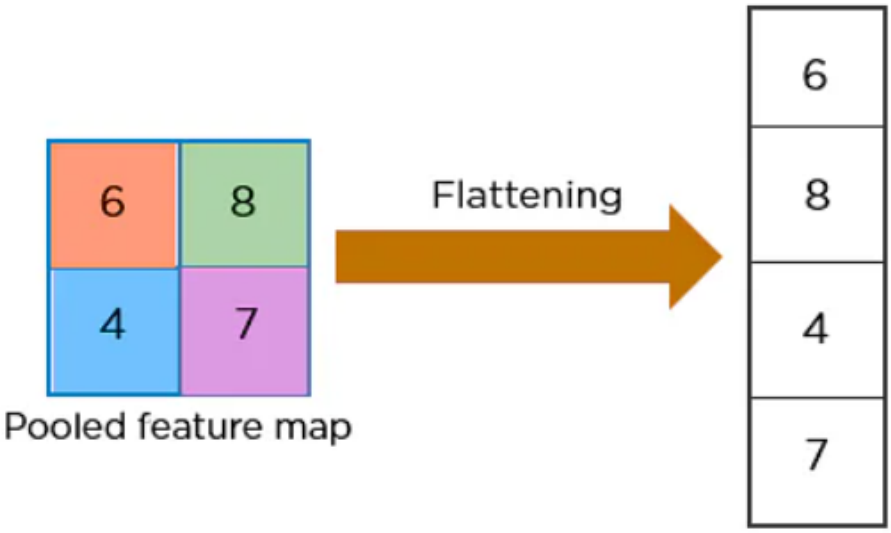


Figure 25(Flatten Layer)

2.2.6 Recurrent Neural networks (RNN):

RNN works on the principle of saving the output of a particular layer and feeding this back to the input in order to predict the output of the layer.

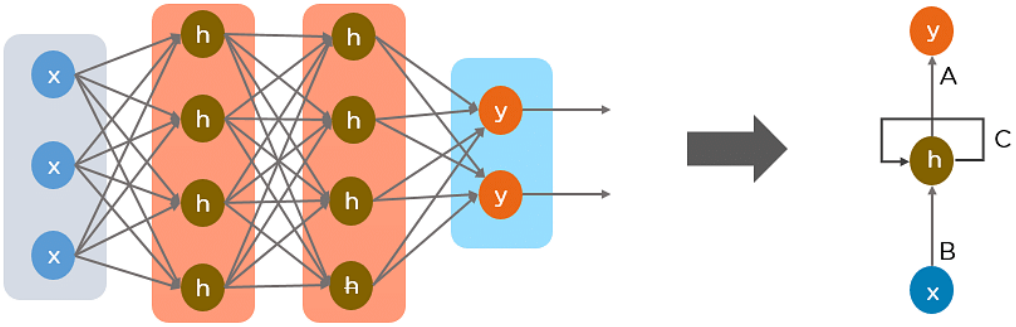


Figure 26(ANN vs RNN)

RNN were created because there were a few issues in the feed-forward neural network:

- Cannot handle sequential data
- Considers only the current input
- Cannot memorize previous inputs

2.2.6.1 There are four types of Recurrent Neural Networks:

One to One RNN: This type of neural network is known as the Vanilla Neural Network. It's used for general machine learning problems, which has a single input and a single output.

One to Many RNN: This type of neural network has a single input and multiple outputs. An example of this is the image caption.

Many to One RNN: This RNN takes a sequence of inputs and generates a single output. Sentiment analysis is a good example of this kind of network where a given sentence can be classified as expressing positive or negative sentiments**.**

Many to Many RNN: This RNN takes a sequence of inputs and generates a sequence of outputs. Machine translation is one of the examples.

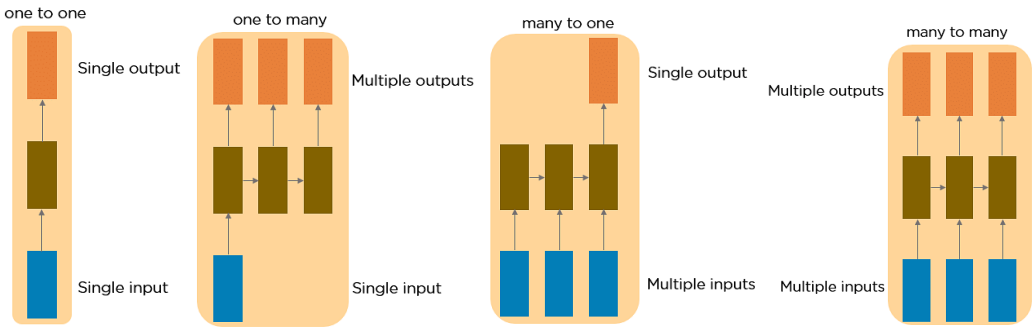


Figure 27(RNN types)

2.3.0 OCR:

OCR (Optical Character Recognition) is a key field in computer vision. Traditional OCR is usually used to scan documents. Now it often refers to scene text recognition (STR), mainly for natural scenes such as signs shown in the figure below and other texts in various natural scenes.



Figure 28(OCR applications(scenes))

There are two kinds of technical challenges: the algorithmic challenges and application challenges.



Figure 29(algorithmic challenges)

- Massive data requires OCR to achieve real-time processing.
- The end-side application requires that the OCR model is light enough and its recognition speed is fast enough.

2.3.1 OCR Cutting-edge Algorithms: Although OCR is relatively specific, it involves many aspects of technologies, including text detection, text recognition, end-to-end text recognition, document analysis, and so on.

2.3.2 Text Detection: The text detection task is to locate text regions of the input image.

Text detection is to find out the position of text in an image or video. Different from object detection, object detection is intended to solve not only the positioning problem, but also the problem of target classification.

Object detection: finds out the box of the target of the given image or video, and make classification.

Text detection: finds out the text area of the given image or video, which can be a single character or a whole text line.

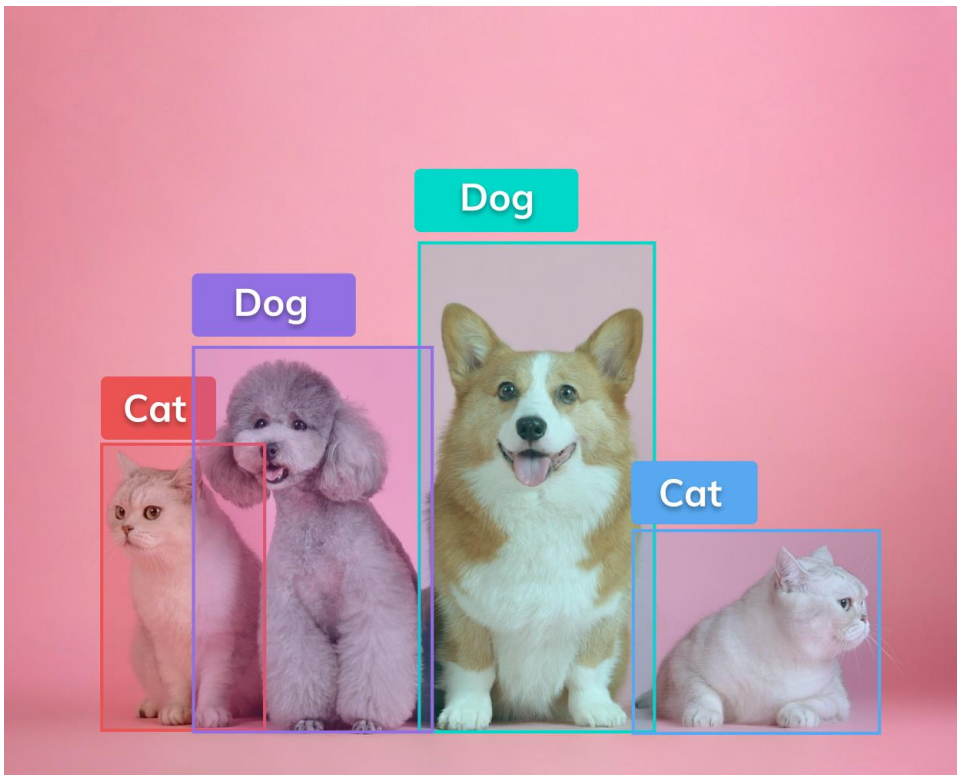


Figure 30(Object Detection)

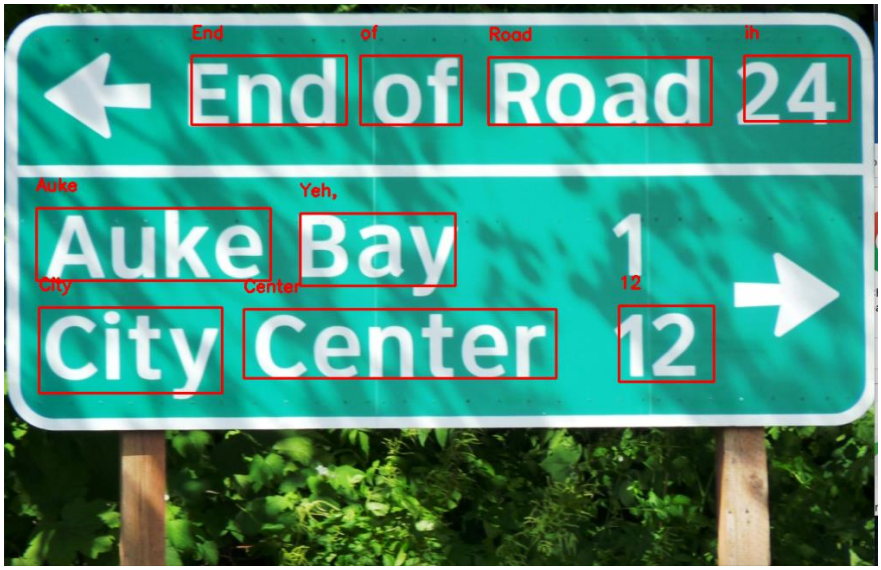


Figure 31(Text detection)

2.3.3 Text Recognition: Text recognition is to recognize the text content in the image, and the input generally comes from the textual area of the image cut out by the text box generated by text detection.



Figure 32(Text types)

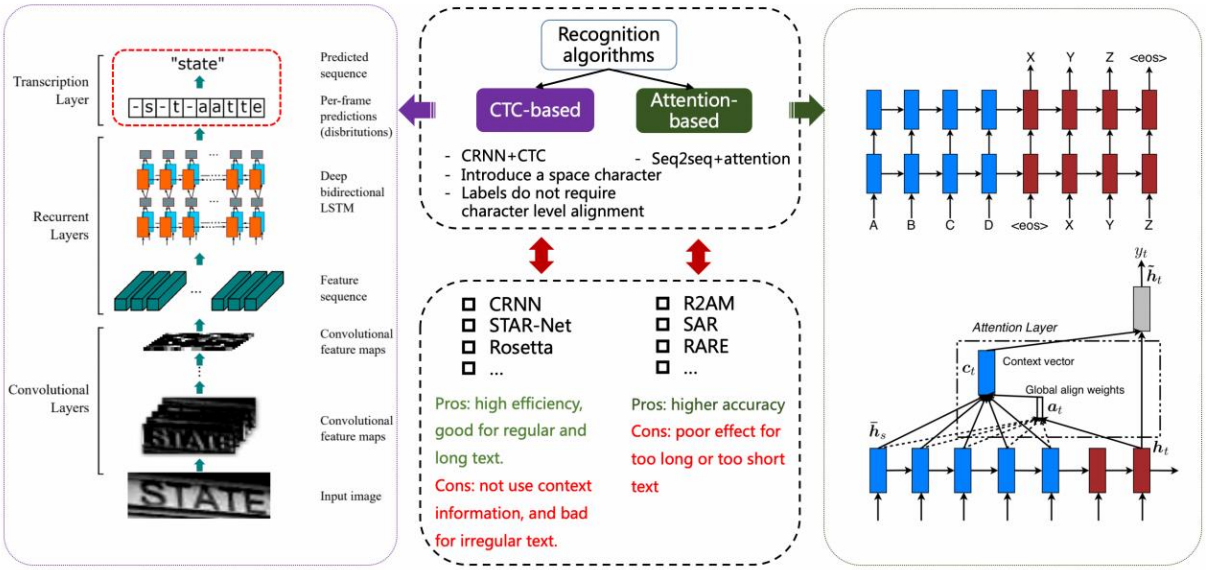


Figure 33(OCR model)

2.3.4 PaddleOCR:

The construction of PaddleOCR starts from user portraits and needs, selects and reproduces diverse cutting-edge algorithms with the core framework of PaddlePaddle, develops PP models that are more suitable for industrialization based on recurred algorithms, and integrates training and inference to provide various inference and deployment methods and meet different demand scenarios in application.

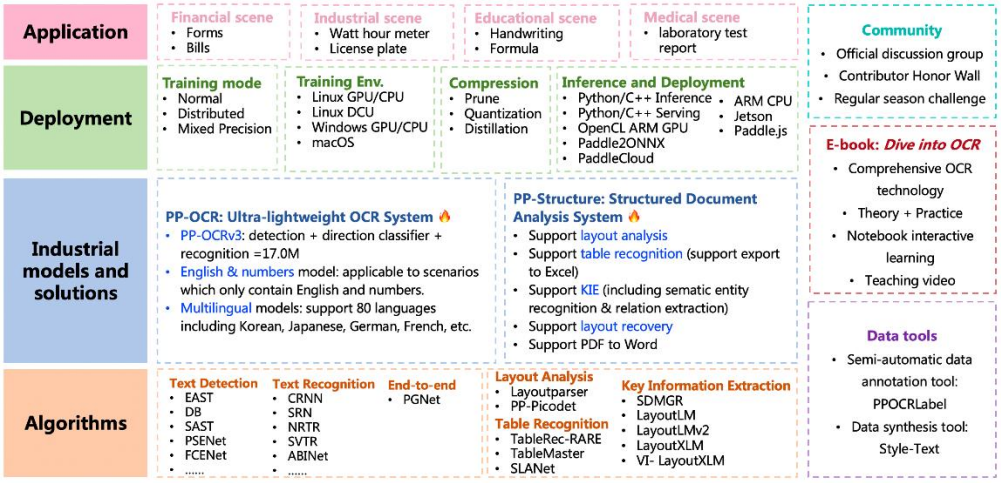


Figure 34 (PaddleOCR Features)

It can be seen from the panorama that PaddleOCR provides abundant solutions in model algorithms, pre-training model libraries, and industry-level deployment with the help of the core framework of PaddlePaddle.

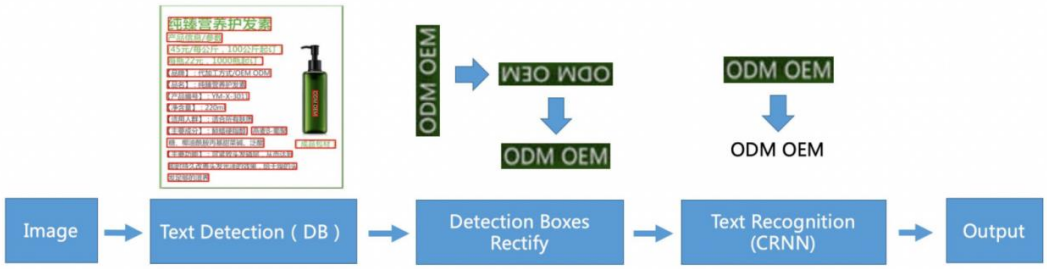


Figure 35(PaddleOCR deployment)

Paddle models can be compressed, quantified, and distilled by using Paddle Slim tools. They support five deployment schemes, including Paddle Serving (service-based), Paddle Inference (server-side or cloud-side), Paddle Lite (mobile or edge-port), Paddle.js (front end of the web). can be converted into a third-party framework supportive to ONNX with the help of Paddle2ONNX.

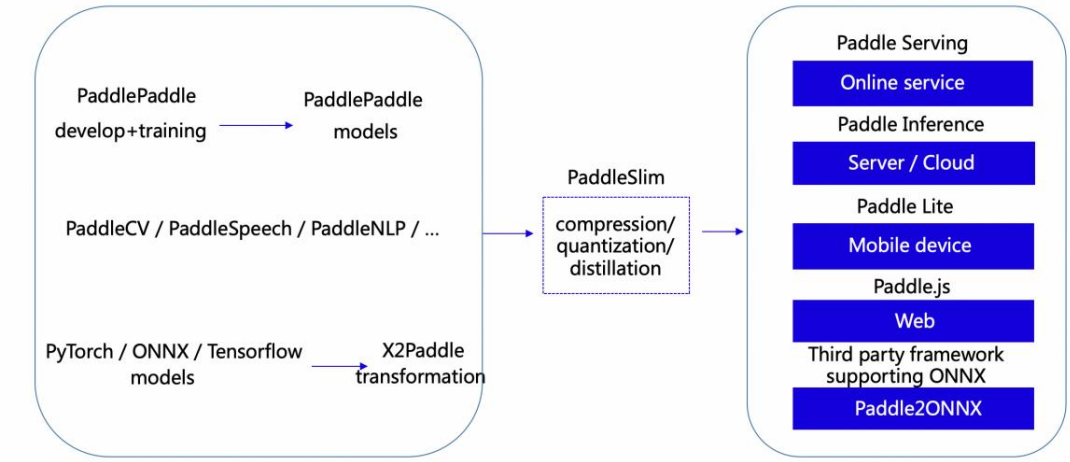


Figure 36(Paddle Support)

2.4.0 Object detection

2.4.1 What is Object detection

Object detection is a powerful computer vision technique that enables machines to identify and locate objects within images or video footage. By accurately recognizing and localizing objects, object detection can be used to count objects in a scene and track their precise locations,

while also labeling them correctly. Essentially, object detection involves detecting the presence of an object, assigning it a label to identify it, and then using boundary boxes to precisely locate it within the image or video. This technique has a wide range of applications, from security and surveillance to autonomous vehicles and robotics.

Object detection and image recognition are two distinct computer vision techniques. Image recognition involves assigning a label to an entire image, such as "dog" for a picture of a dog. However, in image recognition, a picture of multiple dogs would still be labeled as simply "dog". Object detection, on the other hand, goes a step further by drawing a box around each object within an image and labeling each box with the appropriate label, such as "dog" for each individual dog. By precisely identifying and labeling each object within an image, object detection provides more detailed information than image recognition alone. The technique can be useful for a wide range of applications, from counting and tracking objects to improving the accuracy of autonomous vehicles and robotics.

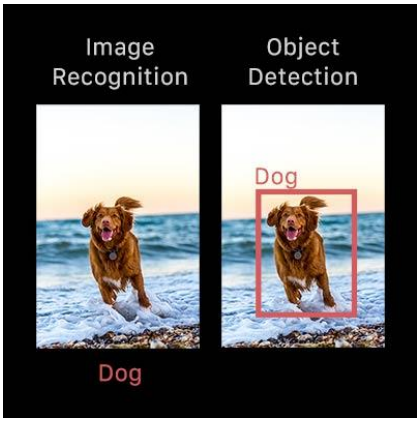


Figure 37 Example of Object detection

2.4.1.1 How to program Object detection:

There are three steps:

- **Image classification:** Assigning an image to one of several different classes (e.g., car, dog, cat, human, etc.).
- **Object localization:** Specifying the location of an object in the image
- **Object detection:** Finding all the objects in an image and drawing the bounding boxes around them.

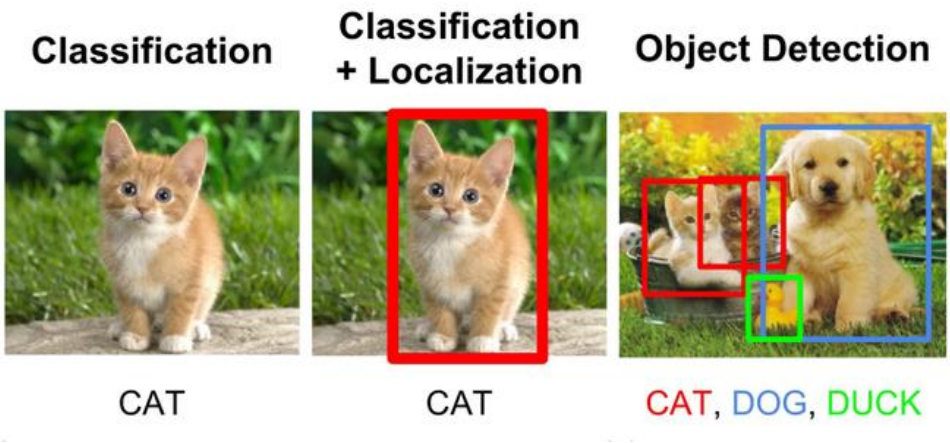


Figure 38 Steps for object detection

2.4.1.2 Possible Algorithm:

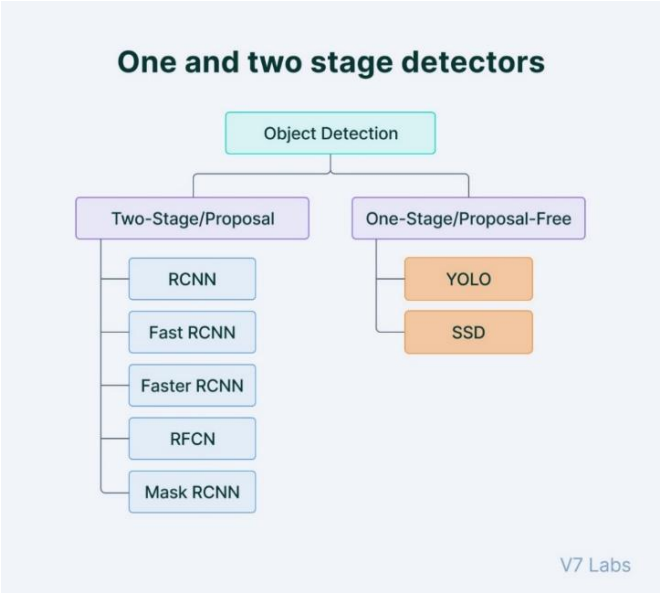


Figure 39 ALGORITHMS

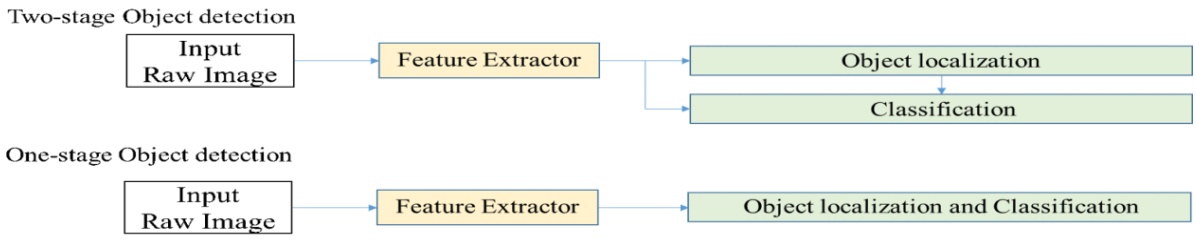


Figure 40 One stage and two stage object detection

Object detection models performance evaluation metrics:

- To determine and compare the predictive performance of different object detection models, we need standard quantitative metrics.
- The two most common evaluation metrics are Intersection over Union (IoU) and the Average Precision (AP) metrics.

2.4.2 YOLO: "You Only Look Once"

The one we used in our project.

2.4.2.1 Introduction to yolo

YOLO is an abbreviation for the term 'You Only Look Once'. This is an algorithm that detects and recognizes various objects in a picture (in real-time). Object detection in YOLO is done as a regression problem and provides the class probabilities of the detected images.

YOLO algorithm employs convolutional neural networks (CNN) to detect objects in real-time. It requires only a single forward propagation through a neural network to detect objects.

This means that prediction in the entire image is done in a single algorithm run. The CNN is used to predict various class probabilities and bounding boxes simultaneously.

Compared to the approach taken by object detection algorithms before YOLO, which repurpose classifiers to perform detection, YOLO proposes the use of an end-to-end neural network that makes predictions of bounding boxes and class probabilities all at once.

Following a fundamentally different approach to object detection, YOLO achieves state-of-the-art results beating other real-time object detection algorithms by a large margin.

2.4.2.2 Before YOLO

Sliding window

In the context of computer vision (and as the name suggests), a sliding window is a rectangular region of fixed size that "slides" across an image, such as in the following figure:

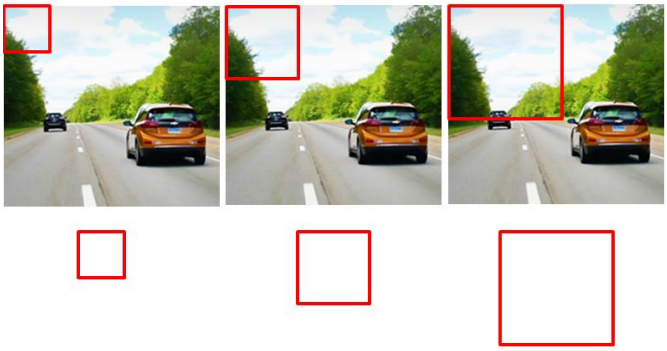


Figure 41 Sliding window

For each of these windows, we would normally take the window region and apply an image classifier to determine if the window has an object that interests us — in this case, a face.

Combined with **image pyramids** we can create image classifiers that can **recognize objects at varying scales and locations in the image.**

These techniques, while simple, play an absolutely critical role in object detection and image classification.

2.4.3 How does YOLO work?

Our system models detection as a regression problem. It divides the image into an S × S grid and for each grid cell predicts B bounding boxes, confidence for those boxes, and C class probabilities. These predictions are encoded as an S × S × (B ∗ 5 + C) tensor.

YOLO algorithm works using the following three techniques:

- Residual blocks
- Bounding box regression
- Intersection Over Union (IOU)

1. Residual blocks

First the image is divided into grid cells of fixed dimension, each cell is responsible of detecting objects in it.



Figure 42 image is divided into grid cells

2. Bounding box regression

A bounding box is an outline that highlights an object in an image.

Every bounding box in the image consists of the following attributes:

- Width
- Height
- Class
- Bounding box center

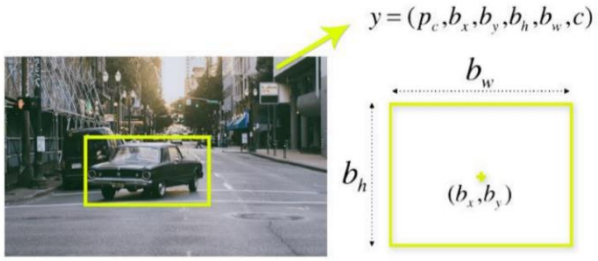


Figure 43 Bounding box regression

YOLO uses a single bounding box regression to predict the height, width, center, and class of objects. In the image above, represents the probability of an object appearing in the bounding box.

3. Intersection Over Union (IOU)

- It's an evaluation metric that describes the overlap between the real box and the predicted box.
- YOLO uses IOU to provide an output box that surrounds the objects perfectly.
- Each grid cell is responsible for predicting the bounding boxes and their confidence scores. The IOU is equal to 1 if the predicted bounding box is the same as the real box. This mechanism eliminates bounding boxes that are not equal to the real box.

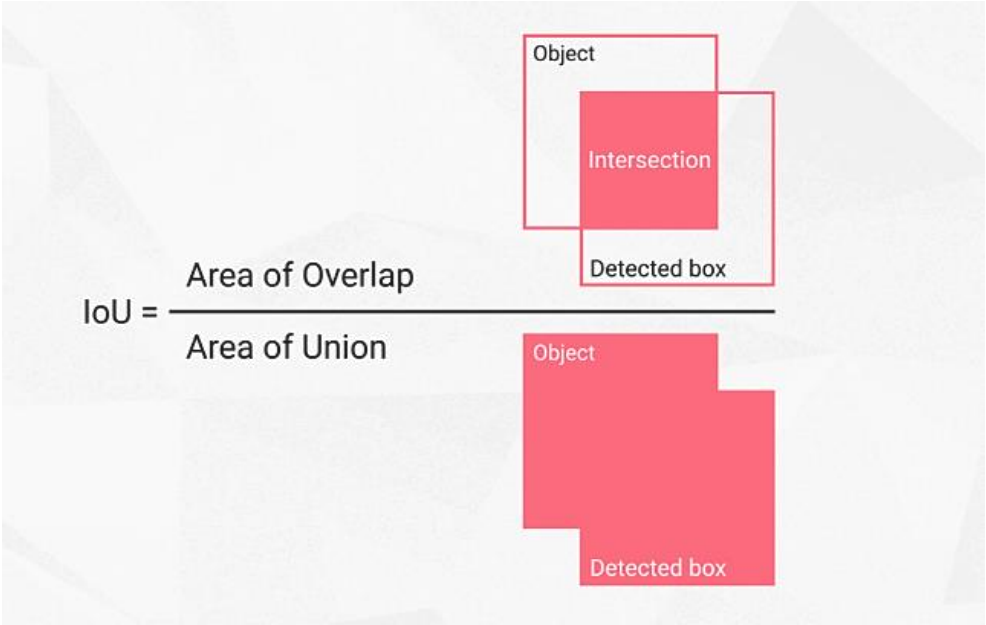




Figure 44 Intersection Over Union

Anchor Box

The idea of anchor box adds one more "dimension" to the output labels by pre-defining a number of anchor boxes. So, we'll be able to assign one object to each anchor box.

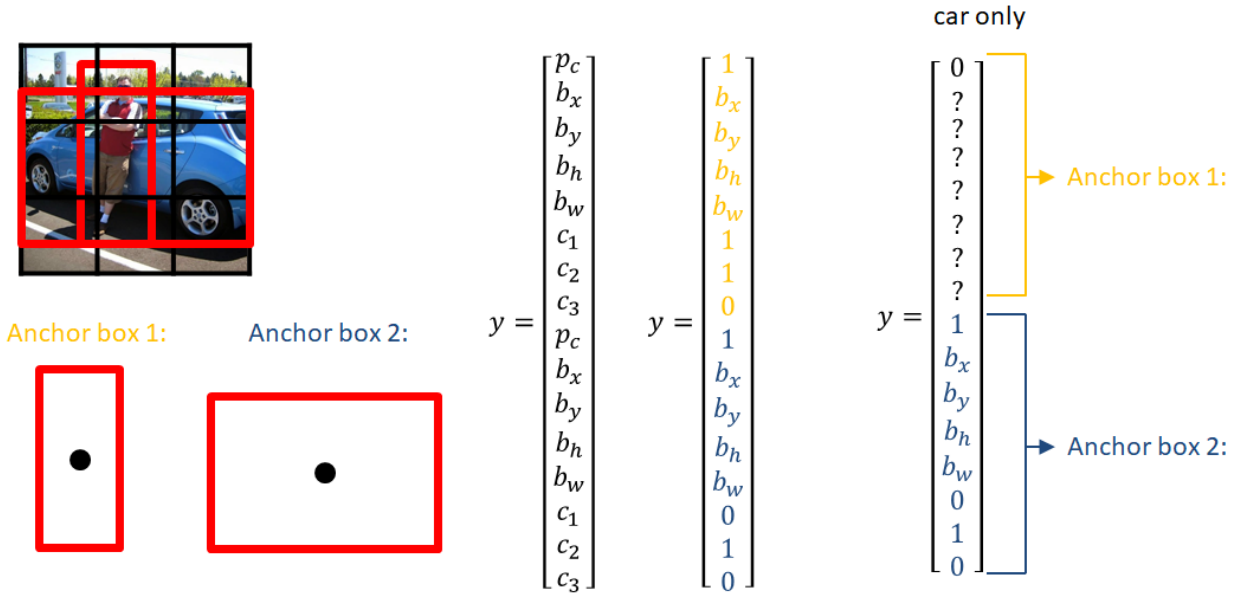


Figure 45 Anchor boxes algorithm

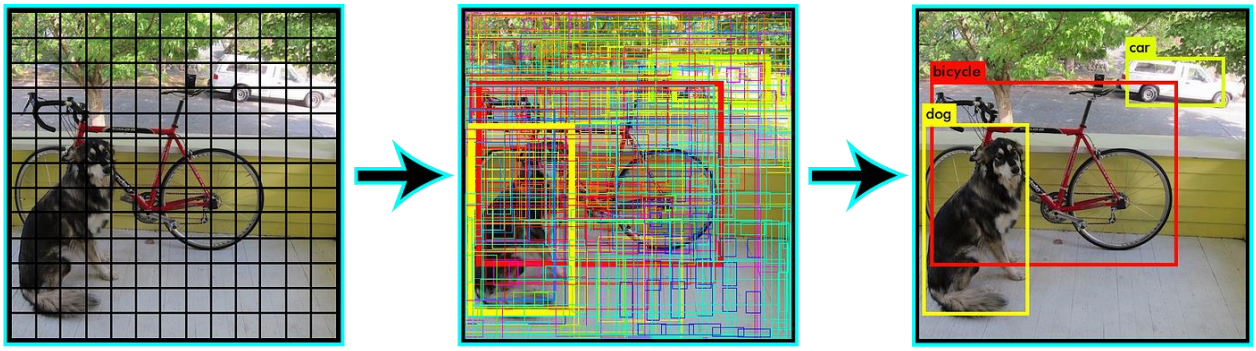


Figure 46 Operation summary

2.4.4 Operation summary

The YOLO algorithm works by dividing an image into a set of N grids, each of which has a fixed size of SxS. Each grid is responsible for detecting and localizing any objects within its bounds. To achieve this, the grids predict B bounding box coordinates relative to their cell coordinates, along with the object label and the probability of the object being present within the cell. This approach greatly reduces computation time, as both detection and recognition are handled by cells within the image. However, it can result in duplicate

predictions due to multiple cells identifying the same object with different bounding box predictions. To address this issue, YOLO uses a technique called Non-Maximal Suppression. In this process, YOLO first examines the probability scores associated with each decision and selects the highest one. It then suppresses any bounding boxes that have a lower probability score or overlap significantly with the currently selected high-probability bounding box. This process is repeated until the final set of bounding boxes is obtained, ensuring that each object is detected and localized accurately within the



Figure 47 The final bounding boxes

2.4.5 MS COCO DATASET

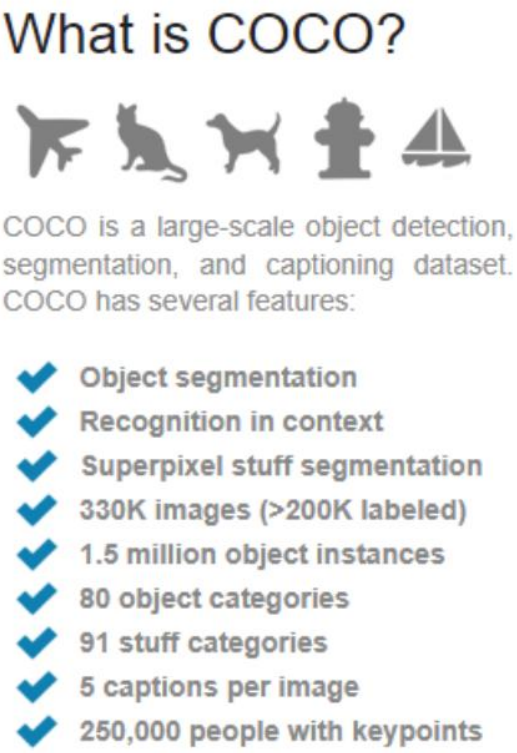


Figure 48 COCO DATASET

COCO stands for **Common Objects in Contex**t, as the image dataset was created with the goal of advancing image recognition. The COCO dataset contains challenging, high-quality visual datasets for computer vision, mostly state-of-the-art neural networks.

For example, COCO is often used to benchmark algorithms to compare the performance of real-time object detection. The format of the COCO dataset is automatically interpreted by advanced neural network libraries.

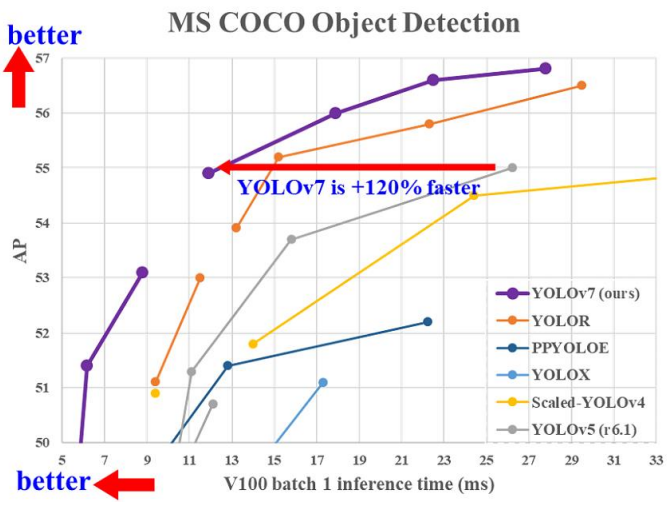


Figure 49 MS COCO

2.4.5.1 List of the COCO Object Classes

The COCO dataset classes for object detection and tracking include the following pre-trained 80 objects:

```
'person', 'bicycle', 'car', 'motorcycle', 'airplane', 'bus', 'train', 'truck',
'boat', 'traffic light', 'fire hydrant', 'stop sign', 'parking meter', 'bench',
'bird', 'cat', 'dog', 'horse', 'sheep', 'cow', 'elephant', 'bear', 'zebra',
'giraffe', 'backpack', 'umbrella', 'handbag', 'tie', 'suitcase', 'frisbee',
'skis','snowboard', 'sports ball', 'kite', 'baseball bat', 'baseball glove',
'skateboard', 'surfboard', 'tennis racket', 'bottle', 'wine glass', 'cup', 'fork',
'knife', 'spoon', 'bowl', 'banana', 'apple', 'sandwich', 'orange', 'broccoli',
'carrot', 'hot dog', 'pizza', 'donut', 'cake', 'chair', 'couch', 'potted plant',
'bed', 'dining table', 'toilet', 'tv', 'laptop', 'mouse', 'remote', 'keyboard',
'cell phone', 'microwave', 'oven', 'toaster', 'sink', 'refrigerator', 'book',
'clock', 'vase', 'scissors', 'teddy bear', 'hair drier', 'toothbrush'
```

Figure 50 Classes of coco dataset

2.4.5.2 Annotated COCO images

The MS Coco dataset contains a vast collection of annotated photos that depict everyday scenes with common objects in their natural context. The objects within these photos are labeled using pre-defined classes such as "chair" or "banana," a process known as image annotation. Unlike other object recognition datasets that focus on image classification, object bounding-box localization, or semantic pixel-level segmentation, the MS Coco dataset focuses on segmenting individual object instances. This means that the dataset provides detailed information on the precise boundaries of each object within an image, enabling more accurate object detection and recognition. By leveraging the power of the MS Coco dataset, computer vision researchers can create sophisticated models that can accurately identify and classify objects within a wide range of real-world contexts.

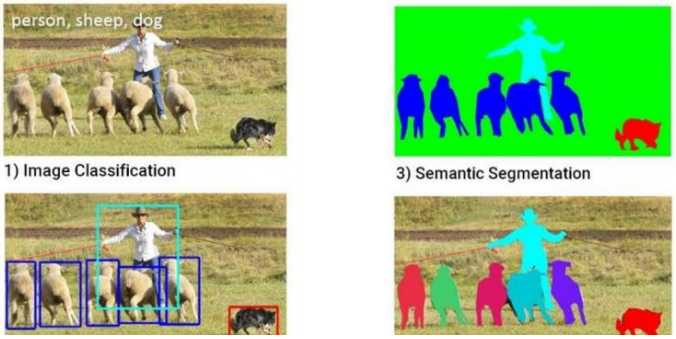


Figure 51 Annotated COCO images

2.4.5.3 Common objects in natural context

For certain categories of objects, there are well-established and easily recognizable views. For example, when conducting a web-based search for a specific object category like "chair," the top-ranked images often feature unobstructed views of chairs that are centered and organized within the photo. While image recognition systems can typically perform well on these "iconic views," they often struggle to recognize objects in real-life scenes that are more complex or partially obscured. As a result, the MS Coco dataset includes a wide range of natural images that contain multiple objects and are representative of real-world scenarios. By including these types of images, the MS Coco dataset provides a more comprehensive and realistic training set for computer vision models, enabling them to perform more accurately in a wider range of contexts.

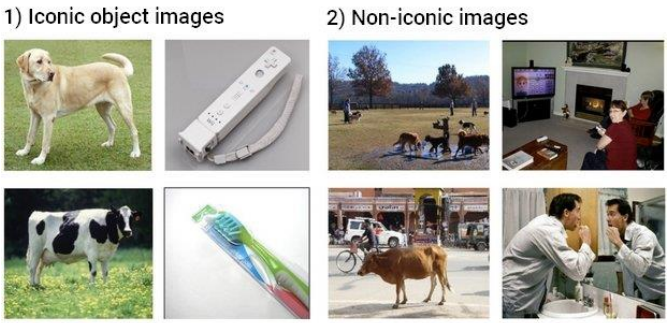


Figure 52 iconic and Non iconic Object

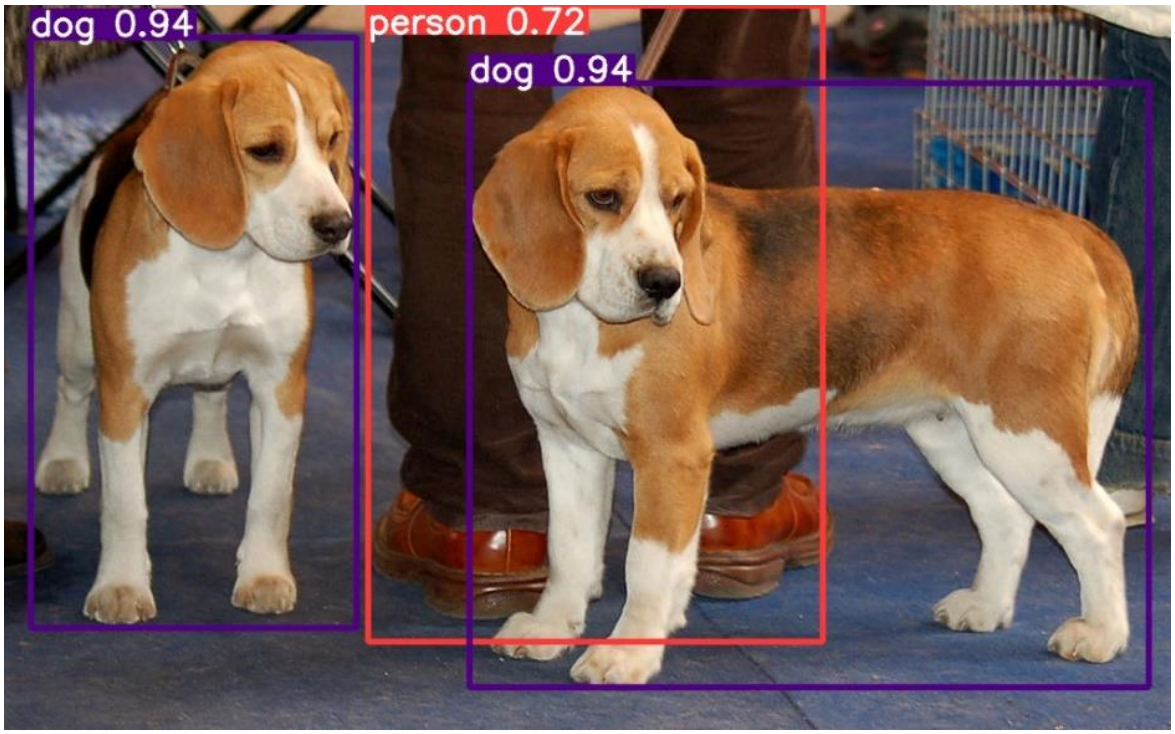
2.4.6 Output

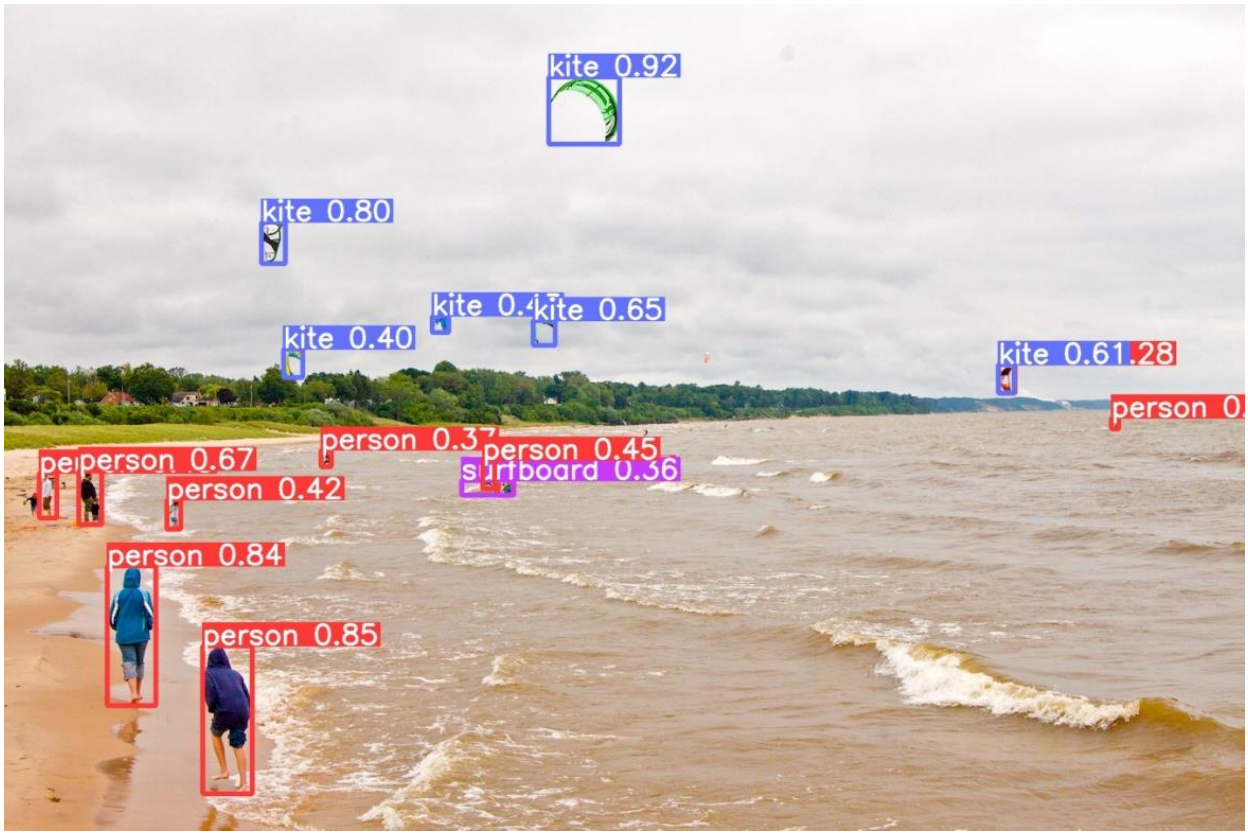

person 0.85
person 0.79
person 0.83
person 0.83
chair 0.69
book 0.75


dog 0.96


laptop 0.92
book 0.77
cell phone 0.81


traffic light 0.66
traffic light 0.49
traffic light 0.29
person 0.80
person 0.87
person 0.47
car 0.80
person 0.78
person 0.74
bicycle 0.38
bicycle 0.77


dog 0.94
person 0.72
dog 0.94


kite 0.92
kite 0.80
kite 0.65
kite 0.40
kite 0.61
person 0.67
person 0.42
person 0.45
person 0.84
person 0.85

2.5.0 Face Recognition:

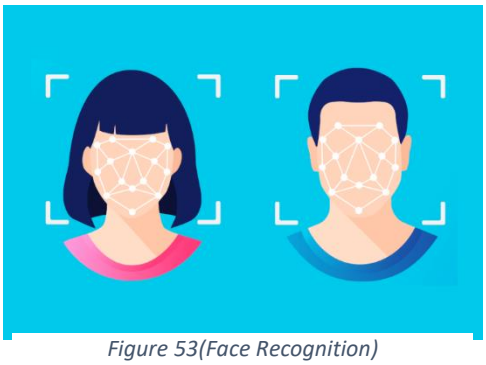


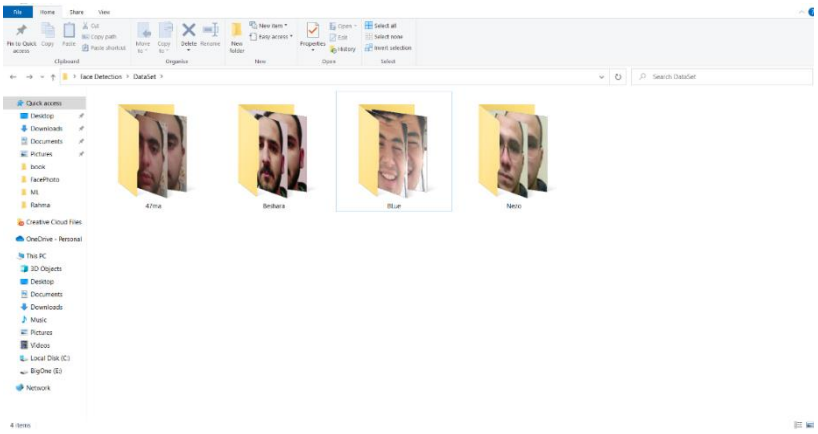
Figure 53(Face Recognition)

Facial recognition is a technology that enables the identification and confirmation of an individual's identity using their face. This technique can be applied to photos, videos, or real-time scenarios, and is commonly used in security and surveillance systems, as well as in mobile devices for user authentication. Facial recognition systems work by analyzing and comparing unique facial features, such as the distance between the eyes, the shape of the jawline, and the contours of the face, to a database of known individuals. By leveraging the power of machine learning and computer vision techniques, facial recognition systems can accurately identify and verify individuals with a high degree of accuracy, even across large datasets. This technology has a wide range of applications, from law enforcement and security to personal device authentication and social media platforms.

2.5.1 Step 1: Face detection:

We are using Haar-Cascade, Object Detection using Haar feature-based cascade classifiers is an effective method.

Haar-Cascade: Haar Cascade is a feature-based object detection algorithm to detect objects from images. A cascade function is trained on lots of positive and negative images for detection. We make a dataset for our faces using the face detection method.



(Figure 54) Custom Dataset

2.5.2 Step 2: Reshape and resize the Data:

We reshape the size of the all images to be all the same resolution as example 128*128 pixel.

```
face_img = cv2.resize(face_img, (128, 128))
face_img = face_img.reshape(1, 128, 128, 1)
```

Figure 55(Resize and Reshape)

2.5.3 Step 3: Converting the image to data:

The face capture process transforms analog information (a face) into a set of digital information (data) based on the person's facial features. Your face's analysis is essentially turned into a mathematical formula. The numerical code is called a faceprint. In the same way that thumbprints are unique, each person has their own faceprint.

2.5.4 Step 4: Finding a match:

Your faceprint is then compared against a database of other known faces, and return the name of face owner from the database.

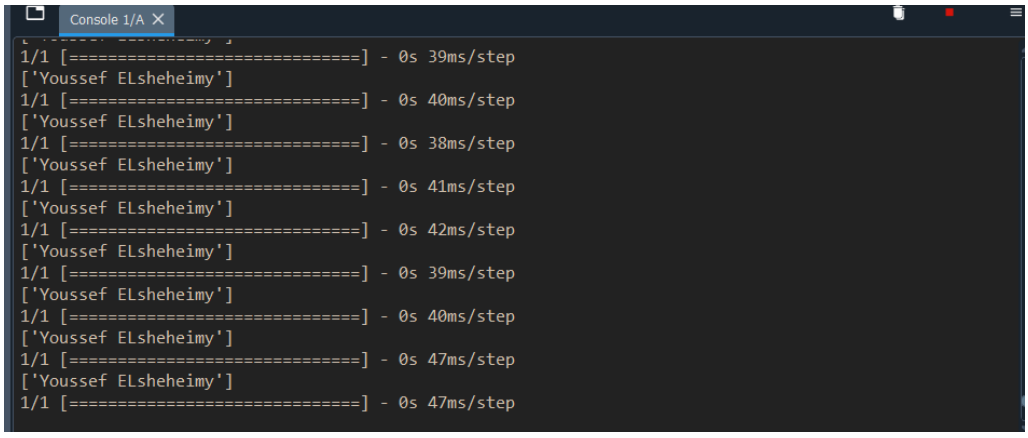


Figure 56 (Output)

2.6.0 Text-to-Speech:

The goal of TTS is the automatic conversion of written text into corresponding speech. Text To Speech (TTS) is technology that changes the medium of information. It enables digital texts into a voice speech format. TTS acts as an assistive tool for applications which need to convert text into voice format. The advancement of artificial technology enhances Text to Speech more human-like voice in real time conversion. Optical character recognition (OCR) is often used in conjunction with text-to-speech programs. This type of technology converts printed text into digital text by scanning it into a computer or handheld device.

- Text To Speech (TTS) technology reads digital text on computers, smartphones, and tablets.
- Reading difficulties can be overcome with TTS.
- Text messages turn into voice messages if the person is busy with other tasks, like driving.

2.6.1 Google Text-to-Speech:

Text-to-Speech (TTS) technology aims to automatically convert written text into corresponding speech. This technology enables digital texts to be transformed into a voice format, serving as an assistive tool for applications that require text-to-voice conversion. With the advancement of artificial intelligence, TTS systems have become more sophisticated, producing more human-like voices in real-time conversion. Additionally, TTS technology is often used in conjunction with Optical Character Recognition (OCR) programs, which convert printed text into digital text by scanning it into a computer or handheld device. Ultimately, TTS technology offers a range of benefits, including improved accessibility for individuals with visual or reading impairments, as well as enhanced convenience and efficiency for a wide range of applications, from personal assistants to public announcement systems. Google Text-to-Speech allows you to choose another default language and then download your language of preference.

Google Translate has a text-to-speech feature that converts written content to speech. It’s a useful app for situations where pronunciation is important. Google’s tool has moved from

being a text-only web function to being a versatile app that can process images, text, speech and even conversations in real-time, to make it easier to understand and converse with people who talk and write another language.

This text-to-voice service is available in up to 30 languages. You can choose to listen to the original written content or its translation.

Google's Text-to-Speech (TTS) service provides a valuable tool for individuals with disabilities, limited mobility, or dexterity problems to access the same features as everyone else. The lifelike and natural-sounding voices used in the service make it easier for those who may struggle with reading on-screen or using voice recognition software. By converting written text into speech, Google's TTS service enables individuals to access information and content more easily and efficiently. This technology has the potential to enhance accessibility and improve the quality of life for individuals with disabilities, ensuring that they are able to enjoy the same benefits as others in the digital world.

2.6.2 Festival Text-to-Speech:

Festival is an open-source Text-to-Speech (TTS) tool developed by The Centre for Speech Technology Research in the UK. This tool provides a framework for building speech synthesis systems and currently supports British English, American English, and Spanish. While the voice produced by Festival may sound rough and robotic, its sound quality is still superior to that of other TTS tools, such as speak. By offering a flexible and customizable platform for TTS development, Festival enables researchers and developers to create sophisticated speech synthesis systems that can be tailored to specific languages or dialects. This technology has a wide range of applications, from assistive devices for individuals with visual or reading impairments to automated customer service systems and language learning tools.

2.6.3 Pico Text-to-Speech:

The Pico Text-to-Speech (TTS) service utilizes the TTS binary from SVOX to generate spoken text. To use this service, the pico TTS library must be installed on the system. This library is

compatible with Debian, a popular Linux distribution, and can be easily installed through the package manager. Once installed, the Pico TTS service can be used to convert written text into speech, providing a valuable tool for individuals with visual or reading impairments, as well as for a wide range of applications, from personal assistants to automated customer service systems. By leveraging the power of TTS technology, the Pico TTS service offers a convenient and accessible way for individuals to access information and content through speech.

2.6.4 H5 file:

H5 is a file format designed for storing structured data, rather than a specific model or algorithm. In the context of machine learning, Keras utilizes the H5 format to save trained models, as it allows for the efficient storage of both model configuration and weights in a single file. By saving a trained model in H5 format, it can be easily reloaded and used again without the need to retrain the model from scratch. This approach offers numerous benefits, including improved efficiency and convenience in deploying machine learning models for a wide range of applications. Ultimately, the H5 format and Keras integration provide a powerful tool for machine learning researchers and practitioners, enabling them to save, share, and reuse their models with ease.

Chapter 3: Software

3.1 Backend

3.1.0 Introduction:

The backend of a software project is responsible for the server-side processing and data management that makes the application function. It serves as the bridge between the frontend and the database, processing requests from the frontend and returning the appropriate response. In our project, we are using .NET Core, a cross-platform, open-source framework for building scalable and high-performance applications. With its robust features and libraries, .NET Core enables us to efficiently handle data processing, security, and integration with other systems. This allows us to deliver a reliable and responsive application that meets the needs of our users.

In addition to its core responsibilities, the backend of our project is also responsible for managing the upload and storage of media files to a cloud storage provider. This includes handling the validation and processing of media files uploaded by users, as well as storing the files securely and efficiently in the cloud. By offloading the storage of media files to the cloud, we can reduce the load on our servers and ensure that our application remains responsive and scalable. Additionally, by leveraging cloud storage solutions, we can take advantage of built-in features such as redundancy and disaster recovery, ensuring that our users' data is always safe and available.

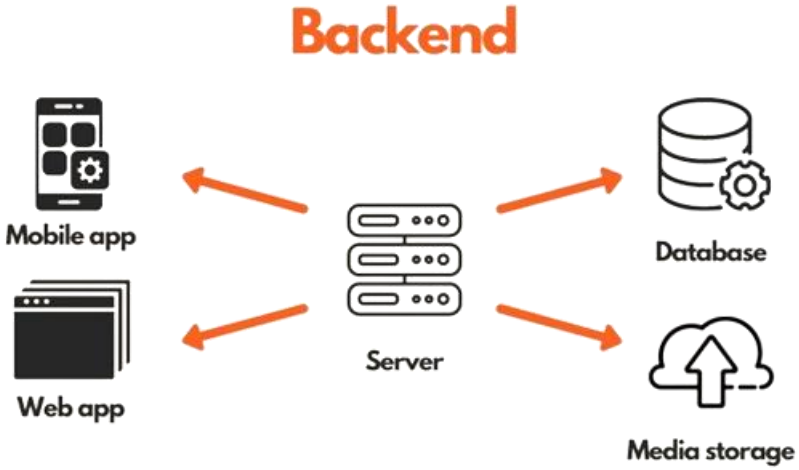


Figure 57 (Backend)

3.1.1. REST API:

API stands for Application Programming Interface. In software development, an API is a set of protocols, routines, and tools for building software applications. It provides a standard way for different software components to interact with each other.

In the context of a project, an API typically refers to a web API, which is a type of API that uses HTTP requests to perform various operations. A web API enables the backend of a project to provide services to other applications or clients, such as mobile apps or web frontends.

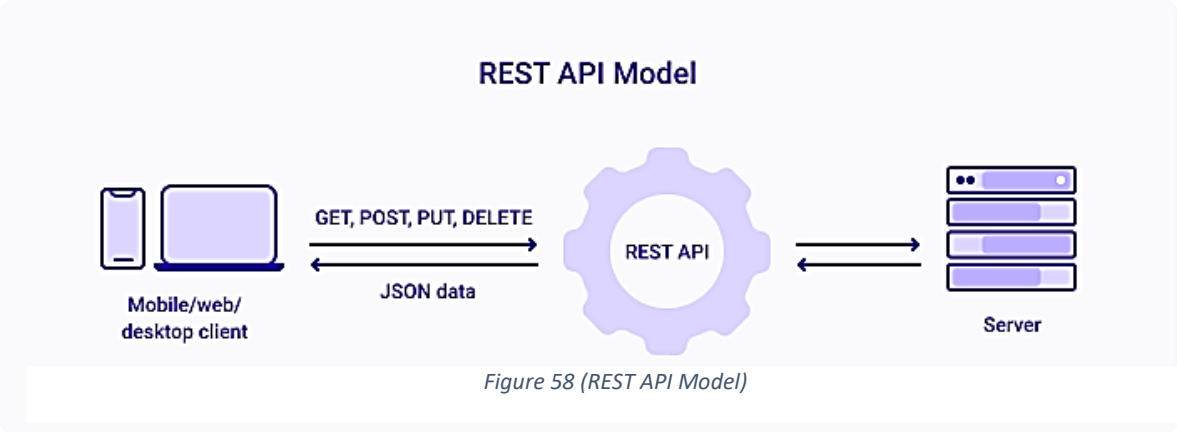


Figure 58 (REST API Model)

The backend of a project is responsible for implementing the API, which involves defining the endpoints, handling requests, and returning responses. This can include tasks such as querying a database, processing data, and returning the results in a format that can be easily consumed by the client.

APIs can serve many different purposes in a project. For example, an API can be used to facilitate communication between different parts of a system or between different systems entirely. This could involve receiving requests for data or other resources and sending back appropriate responses.

APIs can also be used to expose certain functionality or services to third-party developers, allowing them to build applications or services that interact with your system. In this way, APIs can facilitate integration with other systems and allow for the creation of new applications that extend the functionality of your system.

3.1.2.SQL Server Database: (Relational Database)

A relational database is a type of database that uses a structured approach to organizing and storing data. It is based on the relational model, which was first proposed by Edgar Codd in 1970. In a relational database, data is stored in tables that are related to each other based on common fields. This allows for easy querying and manipulation of data, as well as efficient storage and retrieval of information.

SQL Server is a relational database management system (RDBMS) developed by Microsoft. It is one of the most widely used database systems in the world, and is known for its robust features and scalability. SQL Server supports a variety of data types, including numeric, character, and binary data. It also includes support for structured and semi-structured data, such as XML and JSON.

SQL Server uses a query language called Transact-SQL (T-SQL) to manage and manipulate data. T-SQL is a powerful language that allows for complex queries and operations, including joins, subqueries, and aggregations. It also includes support for stored procedures, triggers, and views, which can be used to automate common tasks and improve performance.

SQL Server is known for its reliability and scalability, with support for high availability and disaster recovery scenarios. It also includes advanced security features, such as encryption and auditing, to protect sensitive data. SQL Server can be deployed on-premises or in the cloud, with support for hybrid scenarios that span both environments.

3.1.3. Entity Framework:

Entity Framework is a popular Object-Relational Mapping (ORM) framework developed by Microsoft. It enables developers to work with data in the form of .NET objects, abstracting away the complexity of working directly with relational databases. By using Entity Framework, developers can focus on writing code in C# or VB.NET, rather than SQL queries.

Entity Framework provides a rich set of features, including:

3.1.3.1. Code-First Approach: Entity Framework enables developers to create database schema from code, making it easier to work with databases without writing SQL queries.

3.1.3.2. LINQ support: Entity Framework allows developers to use LINQ (Language-Integrated Query) for querying databases, providing an easy-to-use syntax for retrieving data.

3.1.3.3. Lazy loading: Entity Framework supports lazy loading of related objects, allowing developers to retrieve only the necessary data when needed.

3.1.3.4. Transaction management: Entity Framework provides support for transaction management, allowing developers to perform multiple operations as a single transaction.

3.1.3.5. Database migration: Entity Framework enables developers to easily manage database schema changes, providing tools to migrate databases to new versions.

3.1.3.6. Cross-platform support: Entity Framework Core, a version of Entity Framework that is designed to work with .NET Core, enables developers to build applications that run on Windows, Linux, and macOS.

Entity Framework has become a popular choice for building applications, thanks to its ease-of-use, extensive feature set, and cross-platform support. With its rich set of features and tools, Entity Framework can help developers write efficient, scalable, and maintainable applications that interact with databases.

With Entity Framework, developers can define a data model using classes and relationships, which is then used to generate the database schema and SQL queries. This approach reduces the amount of boilerplate code that developers need to write, and allows for more flexible and maintainable code.

One of the key benefits of Entity Framework is its ability to provide a high level of abstraction between the application code and the database, allowing developers to focus on the business logic of their application, rather than the intricacies of database management. It also offers a range of features such as automatic change tracking, query optimization, and caching, that can help improve the performance of database operations.

Entity Framework also supports a variety of database providers, including SQL Server, MySQL, Oracle, PostgreSQL, and SQLite, which makes it a flexible solution for a wide range of projects. It also provides support for a range of advanced scenarios such as database migrations, code-first development, and stored procedures.

3.1.4. Database Schema:

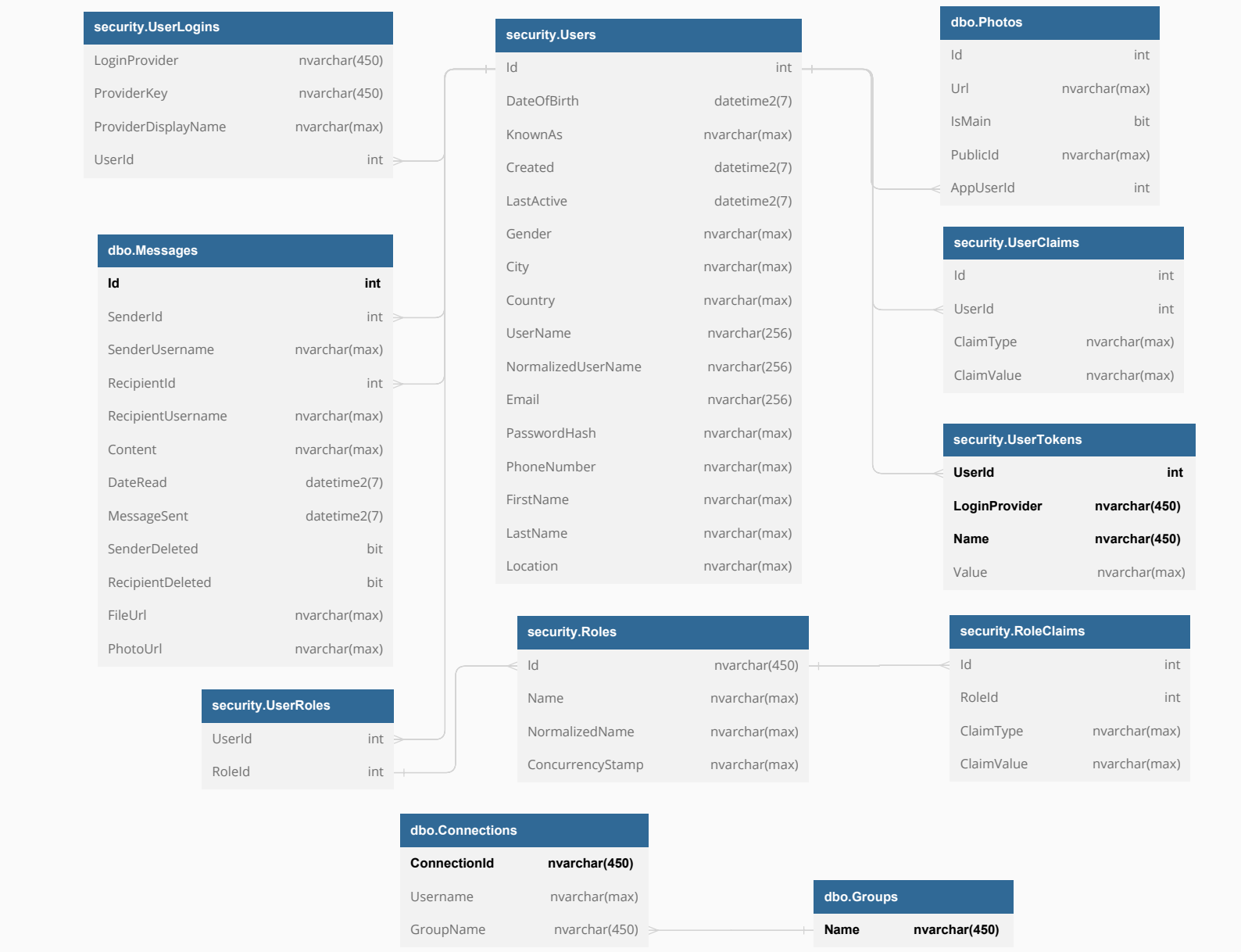


Figure 59 Database Schema

3.1.5. Identity framework:

Identity framework is an authentication and authorization framework developed by Microsoft for

use in building ASP.NET Core applications. It is a complete membership system that

provides robust support for user registration, login, and account management.

The framework provides a set of APIs and tools for managing users, roles, and permissions. It also includes features for password management, account confirmation, and two-factor authentication. Identity framework also integrates with external authentication providers, such as Google, Facebook, and Twitter, to enable users to log in using their social media accounts.

Identity framework is built on top of the ASP.NET Core Identity system, which provides a set of abstractions for user authentication and authorization. It uses a default data store that stores user information, such as usernames, passwords, and roles, in a SQL Server database. However, it also provides the flexibility to use other data stores, such as MongoDB, Redis, or Azure Table Storage.

One of the benefits of using Identity framework is that it provides a secure and extensible authentication and authorization system out of the box. It eliminates the need for developers to write their own authentication and authorization logic, which can be time-consuming and error-prone.

In addition, Identity framework is highly customizable, allowing developers to easily extend and modify its features to meet the specific requirements of their application. It also integrates seamlessly with ASP.NET Core, making it easy to build secure and scalable applications that meet the needs of modern web users.

3.1.6.JWT (JSON Web Tokens):

Identity Framework and JSON Web Tokens (JWT) are two different technologies that are often used together in modern web applications to provide secure authentication and authorization.

Identity Framework is a framework that is used to manage user authentication and authorization in ASP.NET applications. It provides a set of libraries and templates that make it easy to

implement user management functionality such as user registration, login, and password reset. Identity Framework is built on top of ASP.NET Core and uses the ASP.NET Core Identity system to provide a robust set of features for managing user authentication and authorization.

JWT, on the other hand, is a standard for creating and verifying JSON-based tokens that can be used for authentication and authorization purposes. A JWT contains a set of claims that can be used to represent information about a user or other entity, such as their identity, role, and permissions. JWTs are often used as a lightweight alternative to traditional session-based authentication mechanisms, as they can be easily transmitted over the wire and do not require server-side storage.

In an ASP.NET Core application that uses Identity Framework, JWTs can be used as a means of authentication and authorization for API endpoints. The user logs in using their credentials, and if the login is successful, a JWT is issued by the server. This token can then be included in subsequent API requests to authenticate the user and authorize them to perform certain actions.

4.1.6.1. Difference between authentication and authorization:

Authentication is the process of verifying the identity of a user or system. It involves the validation of a user's credentials, such as a username and password, to ensure that the user is who they claim to be.

Authorization, on the other hand, is the process of determining what actions a user is allowed to perform once they have been authenticated.

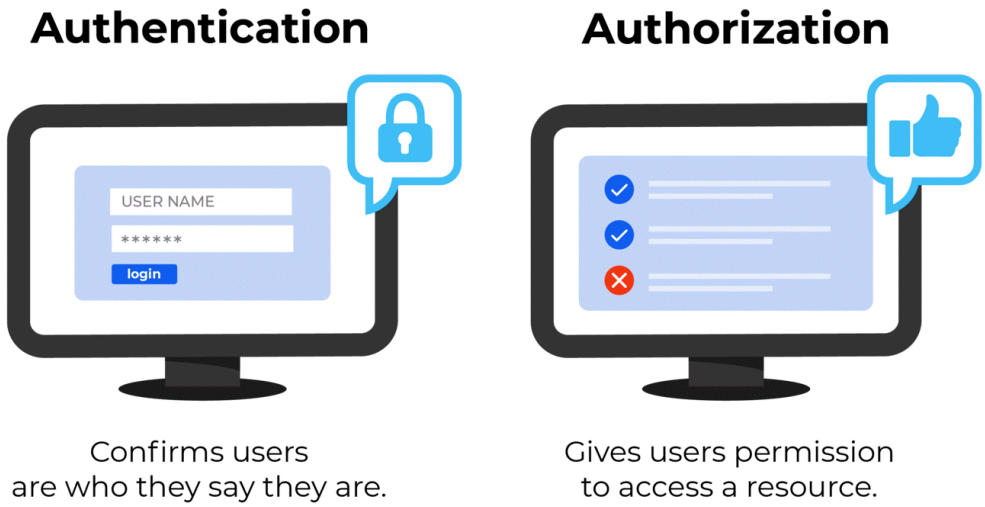


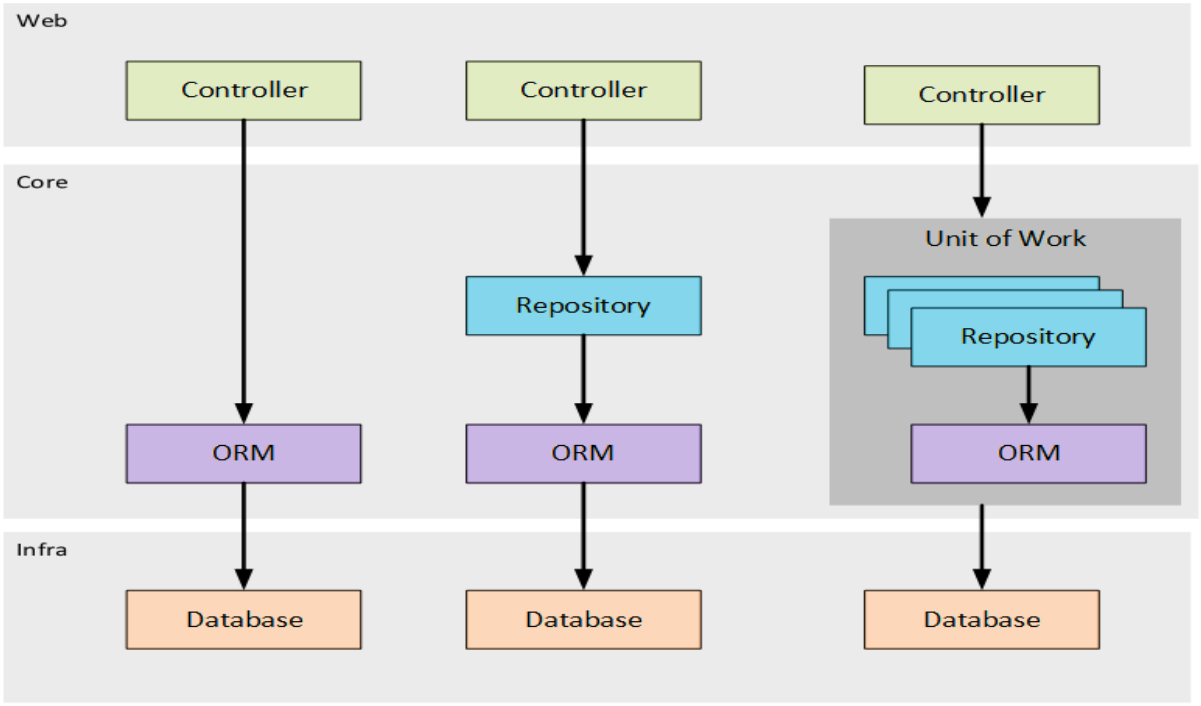
Figure 60 (Authentication VS Authorization)

3.1.7. The repository and unit of work patterns:

The Repository Pattern is a common design pattern used in software development to abstract the data access layer of an application. It involves creating a separate class, called a repository, for each entity in the application's data model. The repository acts as an intermediary between the application's business logic and the data source, providing a clean and consistent interface for the application to interact with the data. The Repository Pattern also helps to decouple the application's business logic from the underlying data storage technology, making it easier to switch to a different database or data source in the future.

The Unit of Work pattern is often used in conjunction with the Repository Pattern. It is a design pattern that helps to manage transactions and ensure that changes to the data source are handled atomically. The Unit of Work pattern defines a single object, called the Unit of Work, that is responsible for coordinating changes across multiple repositories. The Unit of Work tracks changes made to the repositories and ensures that they are all saved or rolled back as a single transaction.

Together, the Repository Pattern and Unit of Work pattern provide a powerful toolset for managing the data access layer of an application. By abstracting the data access layer into repositories and managing transactions with a Unit of Work, developers can create applications that are easier to maintain and extend.



(Figure 61)The repository and unit of work patterns

3.1.8. Data Transfer Object (DTO):

DTO stands for Data Transfer Object. It is a design pattern used to transfer data between software subsystems or layers, typically between the data access layer and the business logic layer. The purpose of using DTOs is to encapsulate data and minimize the amount of data that needs to be transferred across the system, as well as to provide a layer of abstraction between the internal data structures of the application and the external systems that interact with it.

DTOs are commonly used in web development and service-oriented architectures, where data needs to be transferred across the network. They allow developers to define a simple and standardized interface for communicating data between different layers of the application, which can help reduce coupling and improve maintainability.

In practice, DTOs are often used to represent subsets of entities or domain models, where only certain fields are required for a specific use case. They can also be used to hide sensitive data or to provide additional data not present in the original model.

One common use case for DTOs is in mapping data between the application and a database.

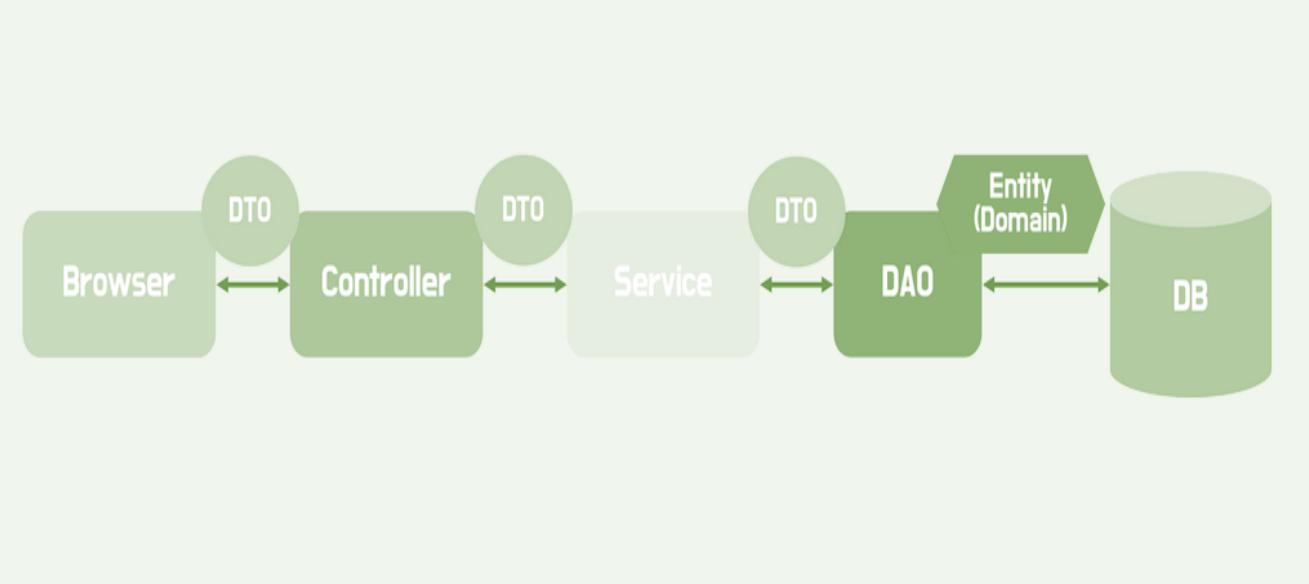


Figure 62 (Data Transfer Object)

3.1.9. SignalR:

SignalR is a library for ASP.NET that enables real-time communication between the server and clients. It allows you to build web applications with bi-directional communication channels between the server and client, where either party can initiate communication. This is achieved by using web sockets, server-sent events, or long-polling, depending on the capabilities of the client and server.

One of the most common use cases for SignalR is in building real-time chat applications. With SignalR, you can build chat applications that can instantly push messages to connected clients, without the need for manual refreshing. This makes it ideal for chat applications, where users expect to receive messages in real-time.

To use SignalR in a chat application, you would typically create a hub on the server that represents the chat room, and then clients can connect to the hub using a SignalR client library. The hub is responsible for handling incoming and outgoing messages, as well as managing connections to the clients.

With SignalR, you can easily handle various scenarios that are common in chat applications, such as joining and leaving chat rooms, sending and receiving messages, and typing indicators. You can also add additional features, such as user authentication and authorization, to ensure that only authorized users can access the chat room.

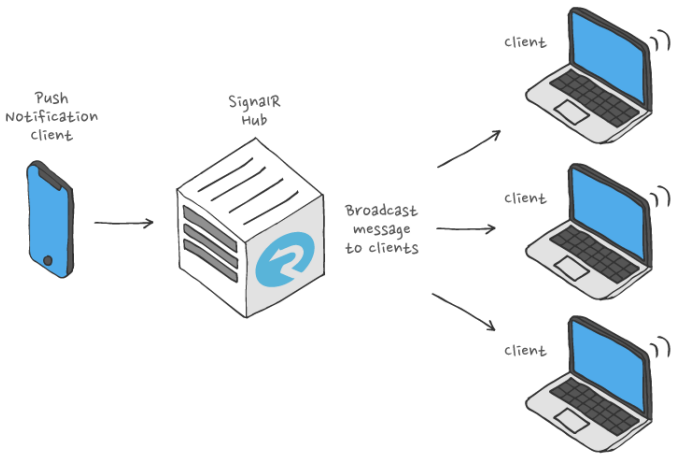
3.1.9.1. Communication types:

In SignalR, there are two main types of communication techniques: long polling and WebSockets.

- **3.1.9.1.1. Long polling** is a technique where the client sends a request to the server, and the server holds the request open until it has new data to send to the client. When the server has new data, it sends a response to the client, which then immediately sends another request to the server to continue the process. This allows for real-time communication between the client and server, but can be resource-intensive and slow.

- **3.1.9.1.2. WebSocket,** on the other hand, is a more efficient communication technique. It allows for real-time, two-way communication between the client and server, with low latency and high performance. With WebSocket, the connection between the client and server is kept open, allowing for constant communication without the need for continuous requests and responses.

SignalR can automatically choose which communication technique to use based on the capabilities of the client and server. If the client and server both support WebSockets, SignalR will use WebSockets for the communication. If not, it will fall back to long polling.



3.2.0 Flask Server for Machine Learning Modules

3.2.1 Introduction:

In our graduation project, we employed Flask, a lightweight web framework in Python, as the foundation for hosting our machine learning modules. Flask provided us with a versatile and efficient platform for creating an API that could handle incoming requests, process data, and deliver responses seamlessly.

Figure 63 Flask Logo

3.2.2 Why Flask with Machine Learning?

The primary language used in the field of machine learning is Python. Therefore, integrating our machine learning models with a Python-based web server was a natural choice. Flask's simplicity and flexibility made it an ideal candidate for hosting our machine learning modules, enabling us to build a robust and user-friendly application.

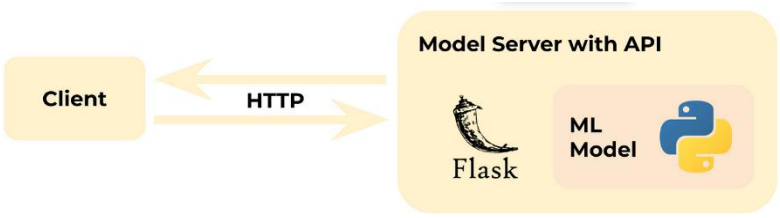


Figure 64 Flask API and machine learning

3.2.3 Server Specifications:

Server Specifications	
OS	Ubuntu 22.04 (Linux kernel)
CPU	Intel Core2 Duo E8400 @ 3.00GHz
RAM	8 GB
GPU	Nvidia GeForce GTX 1050
GPU memory	2 GB

To ensure effective hosting of our machine learning modules, we carefully selected the server specifications. Our server ran on Ubuntu 22.04, utilizing the Linux kernel for stability and performance. The CPU, an Intel Core2 Duo E8400 @ 3.00GHz, provided ample processing power to handle the computational requirements of our machine learning algorithms. With 8 GB of RAM, the server could efficiently manage the memory-intensive tasks associated with training and inference. Additionally, we incorporated a Nvidia GeForce GTX 1050 GPU with 2 GB of dedicated memory, which played a crucial role in accelerating parallel computations and optimizing the performance of our machine learning models.

3.2.4 CPU problem with TensorFlow

The server's CPU is an old Core2 Duo model that does not support the AVX (Advanced Vector Extensions) instruction set. This instruction set is required by the current version of TensorFlow installed via pip. We have two options to solve this problem: either compile TensorFlow from the source without using the AVX instruction set or find an already compiled version on the internet that does not use it. Alternatively, we can use conda to install TensorFlow, which will automatically select a compatible version for the CPU.

3.2.5 Conda

Conda is good for machine learning because it allows you to create and share environments with different packages and dependencies. Conda also provides access to a large repository of machine learning and data science tools such as TensorFlow that you can easily install and use. Conda environments can help you isolate your experiments and ensure reproducibility. Conda can also install non-Python packages such as CUDA, which are useful for GPU-accelerated machine learning. Conda is also compatible with pip, so you can use both tools to install packages in the same environment. However, you should use conda and pip carefully to avoid conflicts.

3.2.6 The Importance of GPU:

The Graphics Processing Unit (GPU) emerged as a key component in our setup. With its specialized architecture designed for parallel computations, the GPU proved to be instrumental in accelerating machine learning tasks. Our Nvidia GeForce GTX 1050 GPU efficiently processed the large volumes of data involved in our machine learning models. Leveraging the GPU's computational power and memory capacity, we were able to achieve significant speed improvements and enhance the overall efficiency of our system.

Furthermore, the GPU's prowess in handling mathematical operations provided a substantial advantage. Its ability to execute complex calculations in parallel not only expedited training and inference but also enabled us to tackle more sophisticated machine learning algorithms. We harnessed the GPU's capabilities to their fullest extent, resulting in improved accuracy, reduced training time, and enhanced overall performance.

0	**Raspberry Pi 4 B**	**Jetson Nano**	**phone, OnePlus Nord N10**	**My PC at home**
GPU	Broadcom VideoCore VI	NVIDIA Maxwell (GM20B)	Qualcomm Adreno 619L	NVIDIA GTX 1050 (75 watt)
# of cores (shading units)	64	128 (cuda)	128	640 (cuda)
FP32 of GPU (Single Precision)	32 GFLOPS	235.8 GFLOPS	536 GFLOPS	1.862 TFLOPS
FP32 of CPU (Single Precision)	3,125 MFLOPS	2,423 MFLOPS	61.5 GFLOPS	3,051 MFLOPS

Video Encode	H264(1080p30)	H.264/H.265 (4Kp30)	4K@30fps, , 1080p@120fps	-
Video Decode	H.265(4Kp60), H.264(1080p60)	H.264/H.265 (4Kp60, 2x 4Kp30)	-	3x4Kp60

3.2.7 Challenges of GPU Memory Usage:

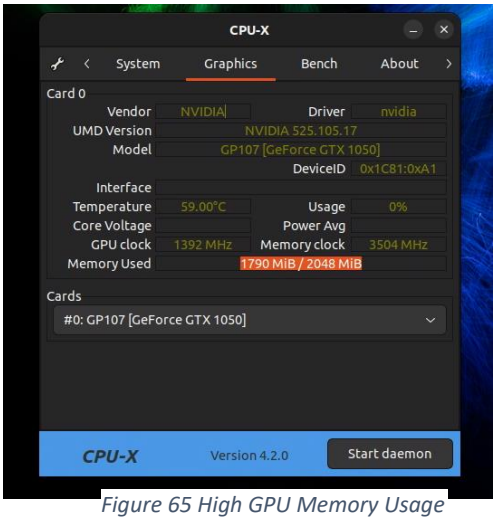


Figure 65 High GPU Memory Usage

While the GPU's computational prowess was invaluable, we encountered challenges related to memory usage. As machine learning models continue to grow in complexity, their memory requirements also increase. Our project involved storing and utilizing large machine learning models, which resulted in high GPU memory usage. At peak utilization, our GPU memory reached 87%, highlighting the necessity for careful memory management and optimization techniques. Through efficient memory allocation and utilization strategies, we were able to mitigate memory constraints and maintain smooth operation of our machine learning modules.

3.2.8 Choosing On-Premises Hosting over Cloud Services:


Figure 66 Pricey Cloud Services

In selecting our hosting approach, we deliberated between utilizing cloud services and deploying an on-premises solution. While cloud services offer convenient and scalable solutions, they can be prohibitively expensive, particularly for smaller projects. Our project aimed to keep costs down without compromising performance, prompting us to opt for on-premises hosting. By leveraging our dedicated server resources, we were able to achieve cost-effectiveness while maintaining full control over our infrastructure. This approach also provided us with greater flexibility and customization options, tailored to our specific project requirements.

3.2.9 Developing a User-Friendly Web Interface for Testing:

As part of our project, we recognized the need for a user-friendly web interface to facilitate testing and evaluation of our machine learning modules. To address this requirement, we created a simple yet intuitive web user interface. Hosted at ai.tosee.live, this interface empowered users to interact with our machine learning models and assess their effectiveness firsthand. By providing input such as images, users could receive real-time predictions or assistance based on the functionalities of our smart glasses. The web interface acted as a bridge between our machine learning algorithms and the end-users.



Figure 68 UI to test machine learning models

Scan the QR code to check the website

http://ai.tosee.live/

SCAN ME

Figure 67 QR code ai.tosee.live

3.3 Mobile Application

3.3.0 Introduction:

Mobile applications have revolutionized the way we communicate, learn, and interact with the world around us. In recent years, mobile applications have also made a significant impact on the daily lives of people who are blind or visually impaired

3.3.1. App's purpose:

- Facilitate communication between the blind user and the escort.
- Send the last location of the blind user to his escort

3.3.2. How will it help users?

- The blind can add their own trusted contact and facilitate communication with them.
- A notification is sent to the escort if the blind person is out of the internet for more than an hour.
- Facilitate communication between the escort and the blind through chat.
- Save the images taken by the blind's glasses in the gallery and make them spoken for reference when needed.
- The escort is allowed to follow up on the location of the blind person.

3.3.3. Cross platform

3.3.3.1. Cross platform software:

the creation of software applications that are compatible with multiple operating systems, such as desktop and mobiles.

The benefit of cross-platform software:

Using the same program on a Windows or Mac as if you were checking in from your laptop or smartphone is a perk of cross-platform software.

3.3.3.2. User experience:

You will be more productive and able to use software that you are familiar with regardless of the operating system you use if you have a similar user experience.

3.3.4. Flutter framework:

Flutter is Google's free and open-source UI framework for developing native mobile applications.

Flutter enabled developers to create mobile applications with a single codebase and programming language in 2017.

This capability simplifies and accelerates the development of iOS and Android apps.

The Flutter framework consists of a software development kit (SDK) and a widget-based user interface (UI) library. This library contains reusable UI elements like sliders, buttons, and text inputs.

Dart will be used by developers working with the Flutter framework to create mobile applications. Dart is a typed object programming language.

3.3.4.1 Some of the top reasons why development teams choose Flutter to include:

3.3.4.1.1 Increased productivity

Using the same codebase for iOS and Android saves both time and resources.

Flutter's native widgets also minimize testing time by ensuring there are little to no compatibility issues with different OS versions.

3.3.4.1.2. Easy to learn

Flutter enables developers to create native mobile applications without requiring access to OEM widgets or a large amount of code. This, combined with Flutter's

particularly appealing user interface, simplifies the process of developing mobile apps.

3.3.4.1.3. Great performance

Users report that distinguishing between a Flutter app and a native mobile app is difficult.

3.3.4.1.4. Cost effective

Creating iOS and Android apps from the same codebase is essentially the same as creating two apps for the price of one.

3.3.4.1.5. Great documentation& community

Flutter has a wealth of resources to answer your questions, thanks to its extensive documentation and simple use cases.

3.3.5. architectural pattern

An architectural pattern is a structure that allows you to express and define a structural schema for all software systems. It is a reusable solution that includes a predefined set of subsystems, roles, and responsibilities, as well as the rules and roadmap for defining their relationships. It assists you in addressing a variety of software engineering concerns such as performance constraints, high availability, minimizing business risk, and so on.

3.3.5.1. The importance of software architecture patterns:

Software architecture patterns hold significant importance for they can solve various problems within different domains. For instance, instead of depending on a single server, complex user requests can be easily segmented into smaller chunks and distributed across multiple servers. In another example, testing protocols can be simplified by dividing various software segments rather than testing the whole thing simultaneously. And the most used architectures are:

-MVC (Model, View, Controller)

-MVP (Model, View, Presenter)

- MVVM (Model, View, View Model)

3.3.5.2. MVVM:

In the application, we need a single architecture that communicates between the user interface and the business logic. One of them is MVVM, which can separate business logic from user interface. This appears to be a simple task, but believe me when I say that if you miss something in the architecture or do something incorrectly, you will be in big trouble later in the application development process, and you will eventually end up having some code hack in the application to do your job.

MVVM facilitates the transfer of business logic from the view to the View Model and Model.

View Model acts as a bridge between the View and the Model, handling all user events and returning the result.

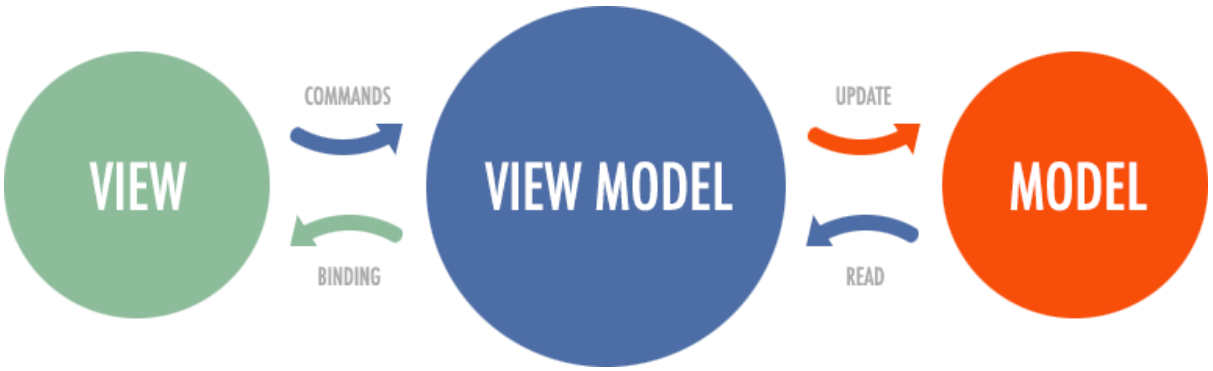


Figure 2(MVVM model)

3.3.5.2.1. Model

The model is a single source of truth for real-time fetch data or database-related queries. This layer may include business logic, code validation, and so on.

This layer communicates with ViewModel either locally or in real-time. In response to ViewModel, data is returned.

3.3.5.2.2. View Model

View Model acts as a bridge between View and Model, accepting all user events and directing them to Model for data. When the Model has data, it returns to View Model, which then notifies the data. A single View Model can provide data to multiple Views because it can be used by multiple Views.

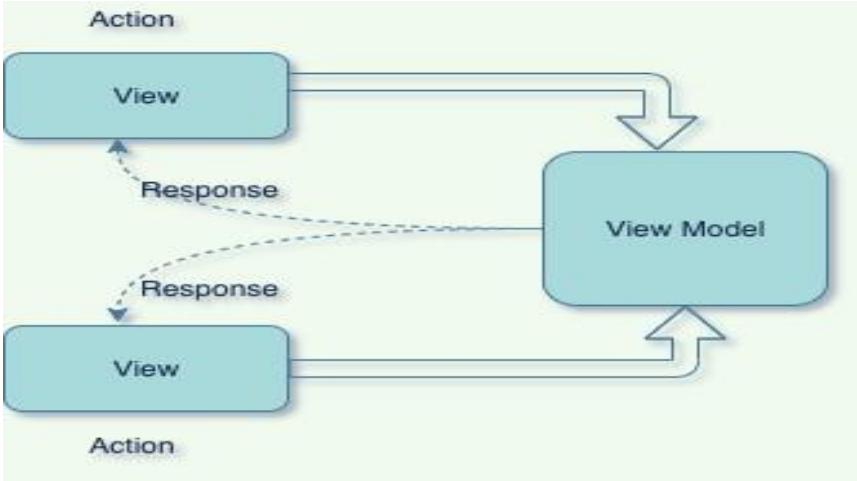


Figure 3(View Model)

3.3.5.1.3. View

The view is where the user interacts with the Widgets that are displayed on the screen. These user events trigger actions that navigate to View Model, and the rest of View Model handles the rest of the work. When View Model has the necessary data, it updates View.

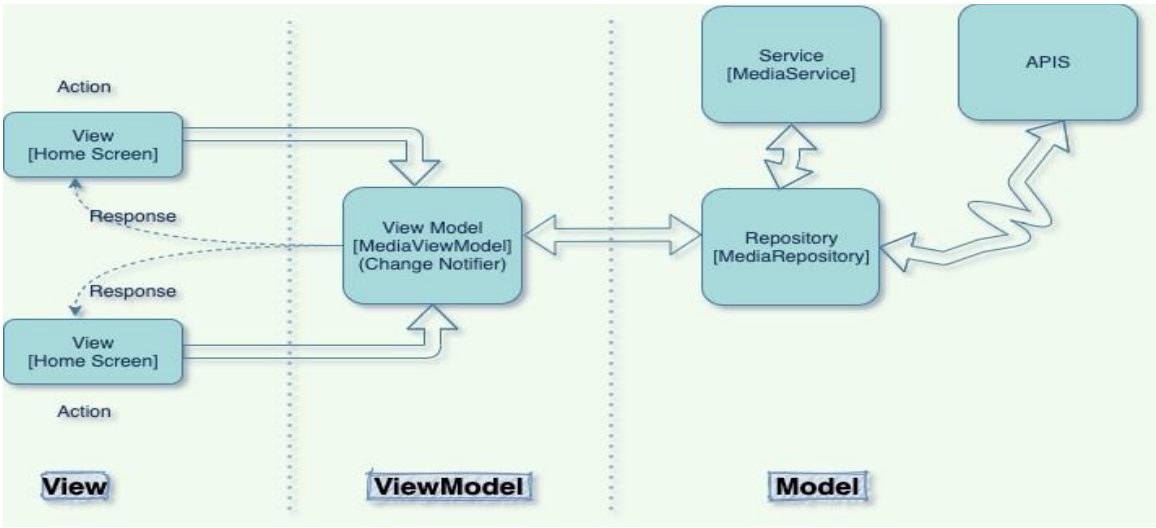


Figure 4(relation between view-view Model-Model)

3.3.6. Design pattern

To write clean and maintainable codebases, Flutter developers employ a variety of design patterns. They frequently write reusable widget implementations in separate Dart files. Decompose large and isolated widgets into private methods/classes, and separate the main app screens into different files. Data sources are used by almost all Flutter apps for data persistence and retrieval. The repository pattern can help all Flutter developers learn how to organize their code better and more manageably.

3.3.6.1. Repository pattern:

The repository pattern is a software design pattern that introduces a centralized component called a repository to decouple the data access logic from the business logic.

3.3.6.1.1. The repository benefits:

- It enables data access from a centralized location, preventing data-access-related code repetition.
- Because the data layer is decoupled from the business logic, the codebase becomes more unit-testable.
- We can easily change data sources without having to make time-consuming code changes.

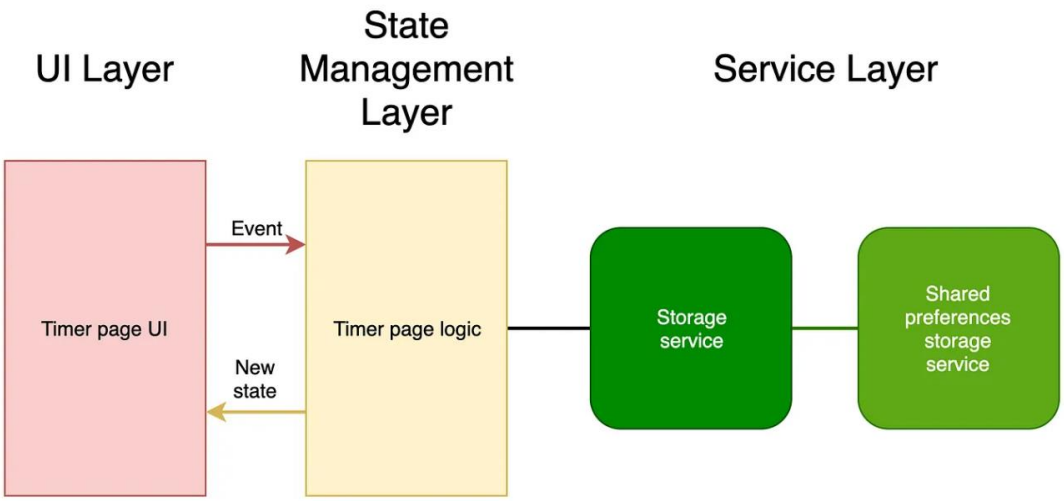
3.3.7. State Management:

When a user interacts with an application, some information must be changed, and the user interface must be rebuilt at the same time. The information required to reflect changes is referred to as the State of application at that time.

State Management is a strategic approach to managing all user interactions on an application and then reflecting those changes to the user interface, updating databases, making server requests, and so on.

Text fields, radio buttons, checkboxes, dropdowns, and other UI controls are common in any application. State management refers to dealing with the states of such UI controls because, most of the time, one or more UI controls rely on one another due to business logic requirements.

State Management is essential for managing application views whether you are developing web or mobile applications.



3.3.7.1. List of Flutter State Management Approaches/Techniques:

- Provider
- setState
- Inherited Widget
- Redux
- FishRedux
- GetIt
- MobX
- Riverport
- Getz
- Flutter Commands

Conclusion:

Our app is an excellent example of how mobile applications can improve the daily lives of people who are blind or visually impaired. The app's ability to recognize and identify objects in real-time and provide navigation assistance is essential for users who require assistance in navigating their surroundings independently. The app's development using the Flutter framework is also noteworthy, as Flutter provides several benefits for mobile application development, including highly responsive user interfaces and hot-reload features. Overall, the To See app is a testament to the power of mobile applications to improve accessibility and inclusivity for all users.

3.4. Frontend

3.4.0 Introduction to Frontend Development

Front-end development refers to the part of web development that deals with the user interface of a website or application. HTML, CSS, and JavaScript are the main technologies used in front-end development. HTML provides the structure of a web page, while CSS is used for styling and layout. JavaScript adds interactivity and dynamic functionality to a website. A strong foundation in these three technologies is essential for creating engaging and effective user experiences on the web.

HTML:	Structure
CSS:	Styling
JavaScript:	Interactivity



Figure 69 Frontend Analogy

3.4.1 Single Page Application Using React.js

3.4.1.1 Introduction to React.js

React is a widely-used JavaScript library for building user interfaces. Its popularity is due to its ability to simplify and optimize web development, making it faster and more efficient than traditional JavaScript. In this article, we'll explore five key reasons why React is better than normal JavaScript.

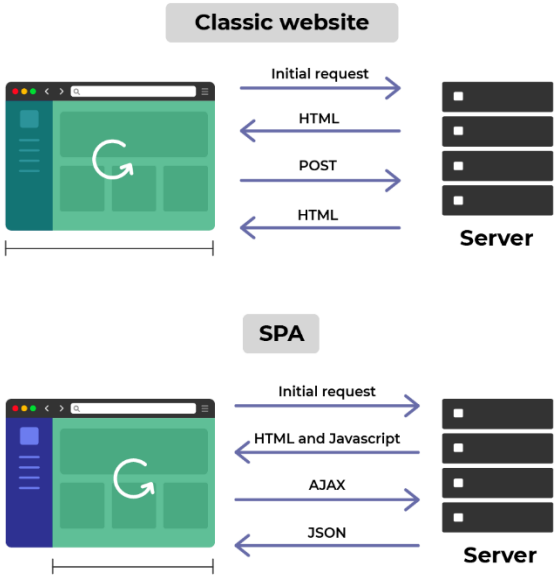


Figure 70 Classic website VS SPA

3.4.1.2 Why React?

React allows developers to build complex UIs with ease. It provides a modular structure that makes code easier to understand and maintain, making it ideal for larger projects.

3.4.1.3 React Router

React's routing capabilities make it easy to create single-page applications (SPAs). This means users can navigate through an app without reloading the page, creating a smoother experience.

3.4.1.4 React's Virtual DOM and Performance

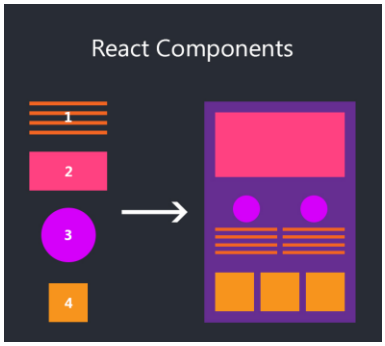


Figure 71 React Components

React's virtual DOM enables faster rendering, making it a popular choice for high-performance web applications. The virtual DOM minimizes the number of updates required to render changes in the UI. This feature makes React applications more efficient and faster than traditional JavaScript.

3.4.1.5 Components and Template

React's component-based architecture makes it easier to reuse code and ensures consistency across an application. The use of templates, such as JSX, also simplifies the creation of UI components and reduces the amount of code required. The application consists of components such as (Map, Login, Sidebar, Footer, Chat, Single Message…. and many more).

3.4.1.6 Conclusion

Overall, React's ease of use, improved performance, and flexibility make it the preferred choice for modern web development. Its ability to simplify and optimize the development process has led to its widespread adoption across the industry.

3.4.3 Testing and Debugging for Frontend Development

Testing and debugging in frontend development with React becomes easier due to the component-based architecture. React allows developers to break down the user interface into reusable components. This modularity facilitates independent testing and debugging of individual components, which can be isolated and tested in isolation. Each component's behavior can be independently tested, making it easier to identify and fix issues. Additionally, React's virtual DOM and unidirectional data flow enable efficient testing by providing a

predictable rendering and updating process. With the ability to isolate and test components, developers can ensure the stability and reliability of their frontend codebase, resulting in more efficient debugging and better overall quality assurance.

3.4.4 Styling and Responsive Design with Bootstrap

Using Bootstrap with React can greatly benefit the application's styling and layout. Bootstrap 5 provides a robust set of pre-built CSS styles, components, and utilities that can be easily integrated into the React application. With Bootstrap's responsive grid system, it's simple to create a layout that scales well on different screen sizes and devices. This means that the application will look great and be easy to use on desktops, laptops, tablets, and mobile devices. By using Bootstrap 5, developers can focus more on functionality and less on design, resulting in faster development times and increased productivity. Additionally, Bootstrap's extensive set of styling options and classes make it easy to create a visually appealing and cohesive design.

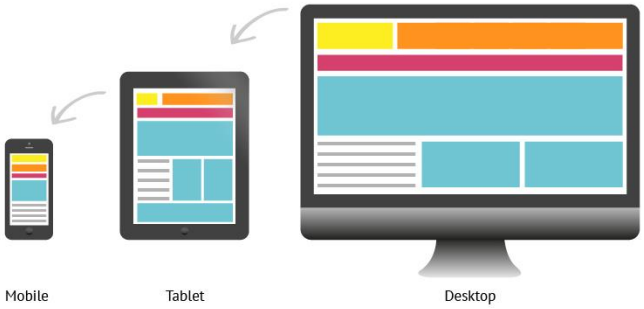


Figure 72 Responsive Design

3.4.5 Voice Notes Visualization with Wavesurfer.js

Wavesurfer.js is an open-source library that enables you to create a variety of audio visualizations. You can use it to display waveforms, spectrograms, regions, markers, and more. It also supports interactive features such as zooming, scrolling, panning, and playback. Wavesurfer.js is based on the Web Audio API and HTML5 Canvas, which means it works in modern browsers without any plugins. You can customize the appearance and behavior of the visualizations with CSS and JavaScript. Wavesurfer.js is easy to use and well-documented, making it a great choice for audio enthusiasts and developers alike.

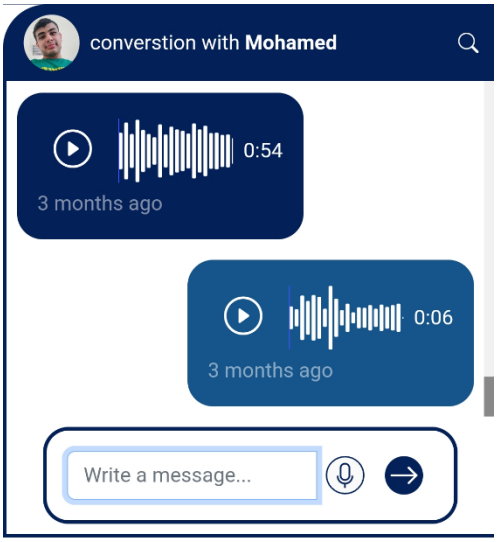


Figure 73 Voice Notes Visualization

3.4.6 Google Maps API

Our application harnesses the power of three essential endpoints from the API, enabling us to retrieve addresses, distances, and directions between users. Let's delve into each endpoint and explore their inputs and outputs:

1. Geocoding API: By providing the coordinates (latitude and longitude) of the blind person's location, this endpoint returns a formatted string representing their address. Here's an example

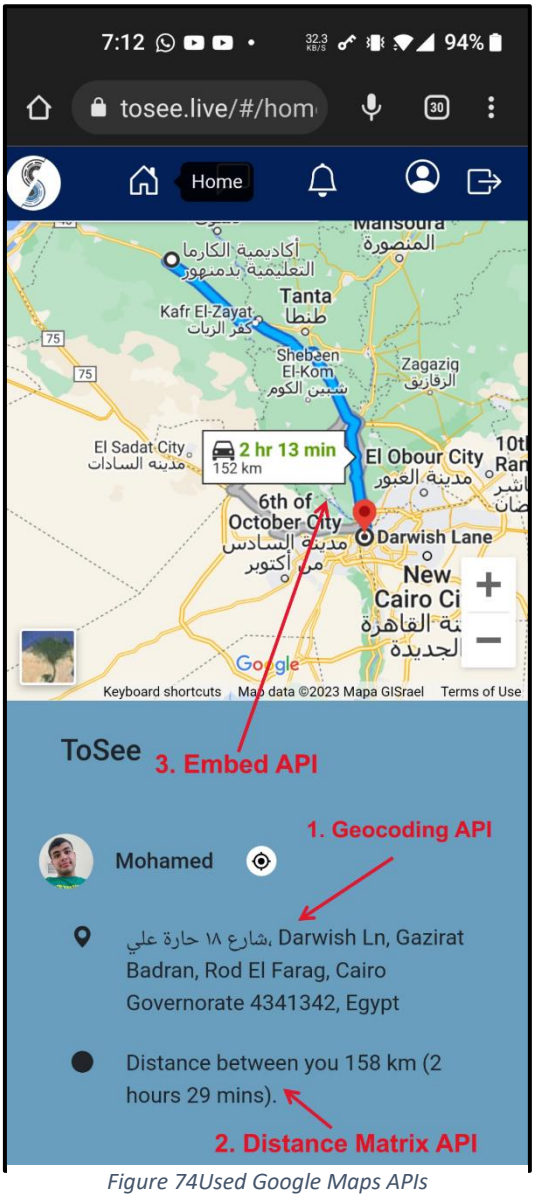


Figure 74Used Google Maps APIs

URL:`https://maps.googleapis.com/maps/api/geocode/json?latlng=${blindLocation}&key=${apiKey}`

The response will contain the precise address of the location, such as "277 Bedford Ave, Brooklyn, NY 11211, USA".

2. Distance Matrix API: This endpoint allows us to obtain the distance and duration between the blind and sighted person's locations by inputting their respective coordinates. The API provides this information in both numerical and textual formats. Take a look at the following URL:

   `https://maps.googleapis.com/maps/api/distancematrix/json?origins=${sightedlocation}&destinations=${blindLocation}&key=${apiKey}`

   The response will contain the distance, for instance, "8.5 km," and the duration, such as "25 mins," required to travel between the two locations.

Figure 75 Used Google Maps APIs

3. Embed API: This remarkable endpoint empowers us to embed a Google map on our website, illustrating the directions from the sighted person's location to the blind person's location. Here's an example URL:

   `https://www.google.com/maps/embed/v1/directions?key=${apiKey}&origin=${location}&destination=${blindLocation}`

   The embedded map will beautifully display the route and provide clear directions from the origin to the destination.

To unlock the potential of these endpoints, we will need a Google Maps API key, which can be obtained for free with a monthly credit of $200. To acquire your API key, you can sign up with a payment method like Visa or MasterCard.

3.4.7 Integrating GitHub Pages for Hosting and Version Control

Integrating GitHub Pages for Hosting and Version Control for a React project is an excellent way to showcase the project and make it accessible to a wider audience. GitHub Pages allows developers to host static websites for free using their repository. This means that once the project is uploaded to GitHub, it can be accessed from anywhere with an internet connection. Additionally, GitHub provides version control for the project, allowing developers to track changes made to the codebase over time. The integration process is straightforward by installing a package with the command "npm install gh-pages --save-dev", and once completed, the project can be deployed using the simple npm command "npm run deploy".

For more information about hosting, look up the cloud segment later in the chapter.

3.4.8 Deploying and Maintaining the Frontend Interface

Deploying and maintaining a frontend interface to GitHub Pages can be achieved using a set of commands that automate the process. First, we need to initialize a Git repository and add all the files to the staging area. Then, we commit these changes with a descriptive message. Next,

we need to set the default branch to "main" and set the remote URL to the GitHub Pages repository. Finally, we push the code to the remote repository and run the deployment script. The deployment script runs the necessary build processes and deploys the frontend interface to the specified GitHub Pages repository URL. This automation saves time and ensures a consistent deployment process.

Used Batch Script (or Bash Script for Linux Users):

```
git init
git add .
git commit -m "setup gh-pages"
git branch -M main
git remote set-url origin https://github.com/99zarka/tosee-build.git
git push -u origin main
npm run deploy && pause
```

3.4.9 Implementing Real-time Communication with WebRTC

WebRTC (Web Real-Time Communications) is a free and open-source API that allows real-time communication between browsers and mobile applications. It enables video and audio communication, as well as data sharing, without the need for plugins or additional software installations.

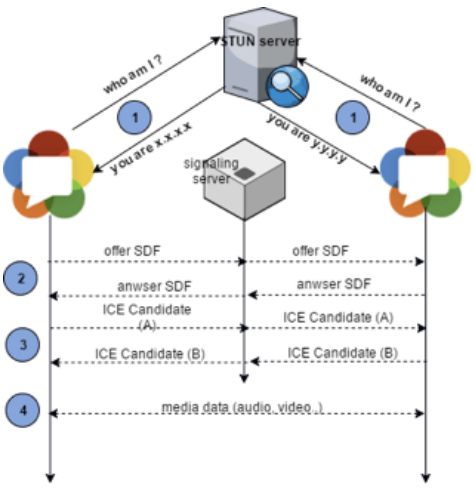


Figure 76 WevRTC Sequence Diagram

WebRTC is a powerful technology for real-time communication over the internet. For our project, implementing WebRTC for video calling is a great way to provide an intuitive and easy-to-use communication channel between the blind person and the sighted person. It allows for a direct and personalized communication experience that is especially important in situations where the blind person

may require assistance in navigating their environment. Additionally, the low-latency and high-quality video and audio make it an ideal solution for real-time interactions.

WebRTC consists of three main components:

1. MediaStream: Provides access to a device's camera and microphone to capture audio and video content.
2. RTCPeerConnection: Establishes a connection between two devices and enables secure data transfer.
3. RTCDataChannel: Enables real-time data sharing between devices.

Here are some commonly used terms related to WebRTC:

1. Peer-to-peer (P2P): A communication model where data is exchanged directly between two endpoints without the need for a central server.
2. Signaling: The process of exchanging information between two endpoints to establish a connection. It includes information such as network addresses and session descriptions.
3. Session Description Protocol (SDP): A format used to describe the parameters of a multimedia session, such as the type of media being used, codecs, and network addresses.
4. ICE (Interactive Connectivity Establishment): A protocol used to establish network connections between two endpoints, even when they are located behind firewalls and NATs.
5. STUN (Session Traversal Utilities for NAT): A protocol used to determine the public IP address and port number of a device, which is necessary for establishing a connection with other devices.
6. TURN (Traversal Using Relay NAT): A protocol used when direct P2P communication is not possible due to network restrictions. TURN servers act as relay points to facilitate communication between endpoints.

3.5 Cloud computing

3.5.0 Introduction

Cloud computing provides IT resources through the Internet on a pay-as-you-go model, eliminating the need for physical data centers and servers. Technology services offered by cloud providers include computing power, storage, networking, databases, developer tools, analytics, and security. Customers can scale their usage up or down and pay only for what they use. This makes it easier and more cost-effective for businesses to access the technology resources they need to innovate and grow without investing in expensive hardware or IT staff.

3.5.1 History of Cloud Computing

The idea of cloud computing can be traced back to the 1960s, when computer scientist J.C.R. Licklider first proposed the concept of an "intergalactic computer network." However, it wasn't until the early 2000s that cloud computing began to take shape.

In 2006 , Amazon Web Services (AWS) launched its first cloud computing service, which provided developers with access to computing resources over the internet. This was followed by the launch of Google's cloud computing service, Google App Engine, in 2008.

Since then, cloud computing has become increasingly popular, with companies of all sizes using cloud computing services to streamline their operations and reduce costs.

3.5.2 Types of Cloud Computing

There are three main types of cloud computing services: Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS), and Software-as-a-Service (SaaS).

IaaS provides companies with access to computing resources, including servers, storage, and networking, over the internet. With IaaS, companies can build and manage their applications and software, but are responsible for maintaining the underlying infrastructure.

PaaS provides developers with a platform on which to build, test, and deploy applications. With PaaS, companies can focus on developing their applications, without having to worry about the underlying infrastructure.

SaaS provides companies with access to software applications over the internet. With SaaS, companies can use software applications without having to install or maintain them on their own servers.

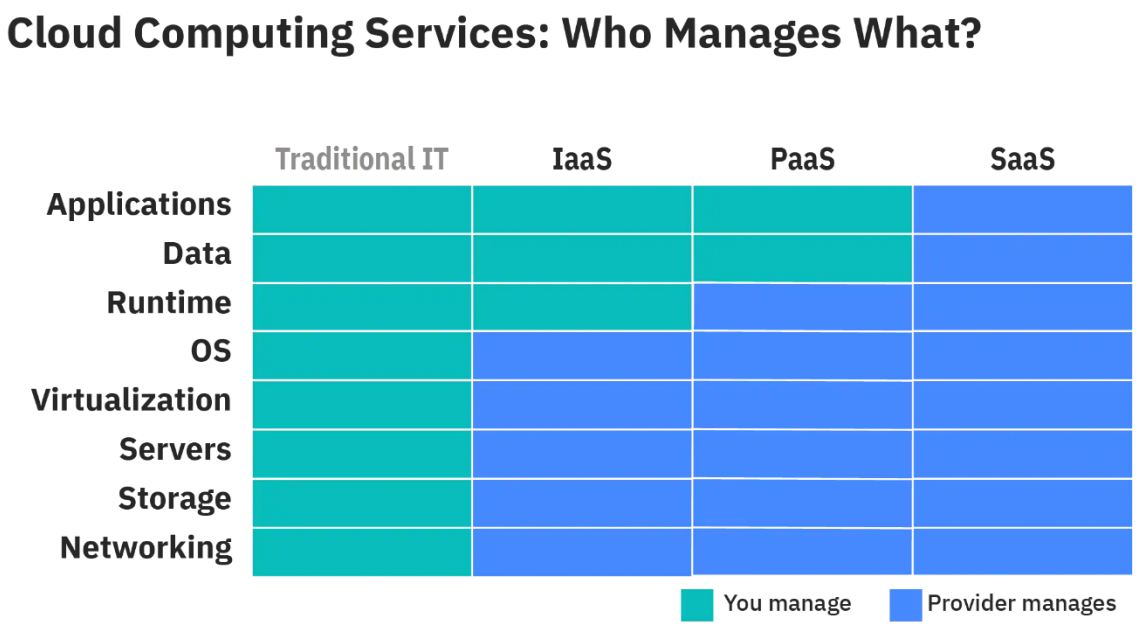


Figure 77 (managing cloud)

3.6 Hosting

3.6.0 What Does web Hosting mean?

Web hosting is a service that allows individuals or businesses to make their websites accessible to others over the internet. Web hosting providers offer the infrastructure and technologies required for a website to be viewed on the internet.

When a website is created, it is made up of files that need to be stored somewhere so that they can be accessed by people who want to visit the site. Web hosting providers offer servers, which are special computers that are designed to store these files and make them available on the internet.

When someone wants to visit a website, they enter the website's domain name (such as www.example.com) into their web browser. The browser then sends a request to the web hosting server, which retrieves the website files and sends them back to the visitor's browser. The browser then displays the website to the visitor.

3.6.1 Web hosting types

There are several types of web hosting available, each with its own features, benefits, and pricing. Here are some of the most common types of web hosting:

Shared hosting: This is the most affordable type of web hosting, where multiple websites are hosted on the same server. Since resources are shared, this type of hosting is best suited for small websites with low traffic.

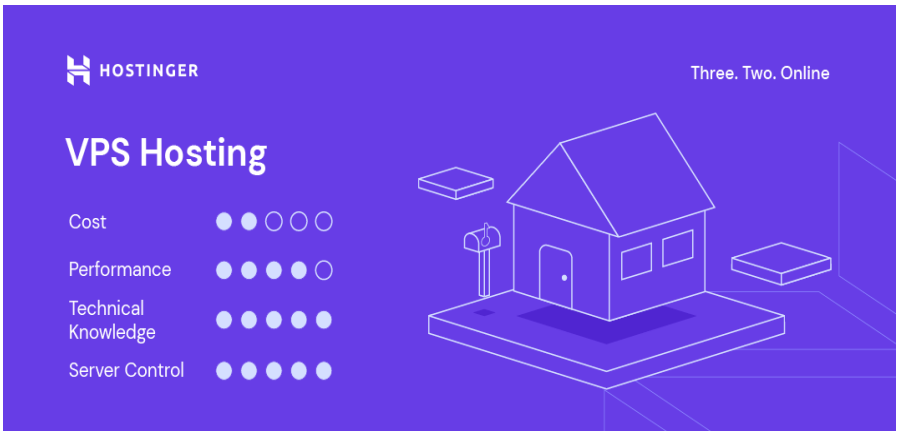


Figure 78 (shared)

Virtual Private Server (VPS) hosting: This type of hosting involves dividing a physical server into multiple virtual servers, each with its own set of resources. VPS hosting provides more control and flexibility than shared hosting, and is ideal for websites that require more resources or customization.

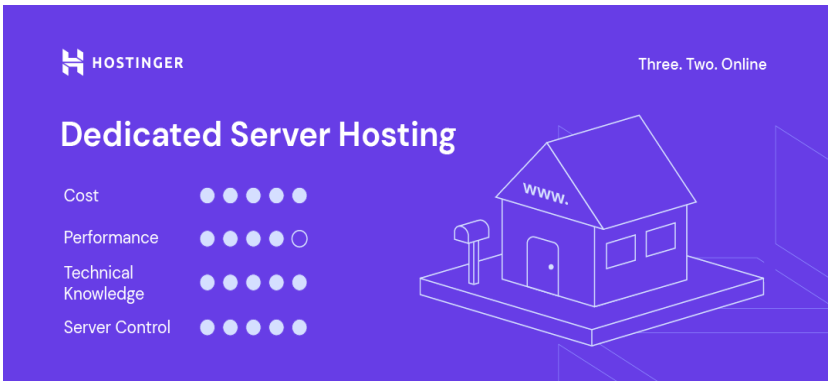


Figure 79 (vps)

Dedicated hosting: With dedicated hosting, a single server is dedicated to hosting a single website. This provides maximum control and resources, making it ideal for high-traffic websites or those with more complex requirements.



Figure 80 (Dedicated)

Cloud hosting: This type of hosting uses a network of servers to host websites, rather than a single physical server. Cloud hosting provides more scalability and flexibility than traditional hosting, making it ideal for websites with fluctuating traffic.



Figure 81 Cloud

Choosing the right type of web hosting depends on factors such as the size and complexity of the website, the expected traffic, and the level of control and customization required. It is important to carefully consider the features and pricing of each type of hosting before making a decision.

3.6.2 Why we decided to use cloud hosting.

Cloud hosting is often the best choice for web app hosting for several reasons:

Scalability: Cloud hosting enables quick scaling of resources to ensure that web apps remain available and responsive during sudden spikes in traffic, such as during peak usage periods.

Flexibility: Cloud hosting supports diverse programming languages, frameworks, and operating systems, simplifying the deployment and management of web apps with specific software configurations and dependencies.

Reliability: providers use multiple servers to host web apps, allowing for quick and seamless switching to another server in case of hardware failure or outage, ensuring uninterrupted availability and accessibility for users.

Security: Cloud hosting providers provide security measures like firewalls, intrusion detection systems, and SSL certificates to safeguard web apps against cyber threats and protect sensitive user data.

Cost-effectiveness: Cloud hosting enables web app owners to pay only for the resources they utilize, making it a cost-effective alternative to investing in expensive hardware and infrastructure for web app hosting.

3.6.3 Web app hosting

Web app hosting is the process of running a software program on a web server that is accessible over the internet through a web browser. Web app hosting providers offer tools and features for managing web applications, including monitoring, scaling, backup, recovery, and security measures. This service is crucial for businesses and individuals who want to deploy and run web applications on the internet.

3.6.4 Front-End Hosting on GitHub-pages.

GitHub Pages is a free web hosting service provided by GitHub that enables users to host static websites directly from their GitHub repositories. It supports user or organization sites hosted at the username.github.io or orgname.github.io domain and project sites hosted at the username.github.io/repository name or orgname.github.io/repository name domain. To use GitHub Pages, create a new repository, add website files to the root directory, and configure repository settings to enable GitHub Pages. It automatically builds the website and makes it available at a URL based on your username or repository name

Our URL:

- **https://99zarka.github.io/tosee-build/**

GitHub Pages provides several benefits for web hosting, including:

Free hosting: GitHub Pages is completely free to use.

Version control: GitHub provides version control for your website files, allowing you to track changes and collaborate with others.

Custom domains: can use a custom domain name for your GitHub Pages website.

HTTPS support: GitHub Pages supports HTTPS, ensuring that your website is secure.

GitHub Pages is a great option for developers who want to showcase their projects or create simple static websites for free and with version control.

3.6.5 Back-End Hosting on Microsoft Azure

Azure is Microsoft's cloud platform that offers PaaS, IaaS, and managed database services. It relies on virtualization technology and consists of a large collection of servers and networking hardware running a complex set of distributed applications. To host a back-end web application on Azure, create an account, web app, configure web app settings, deploy the application code, configure the database, and test the application.

Our API hosted on Microsoft Azure ,after we rent a web app server.

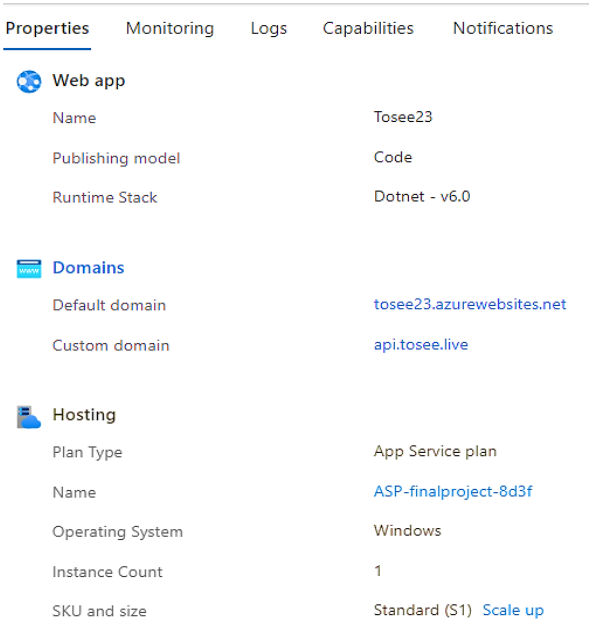


Figure 83 (api server info)

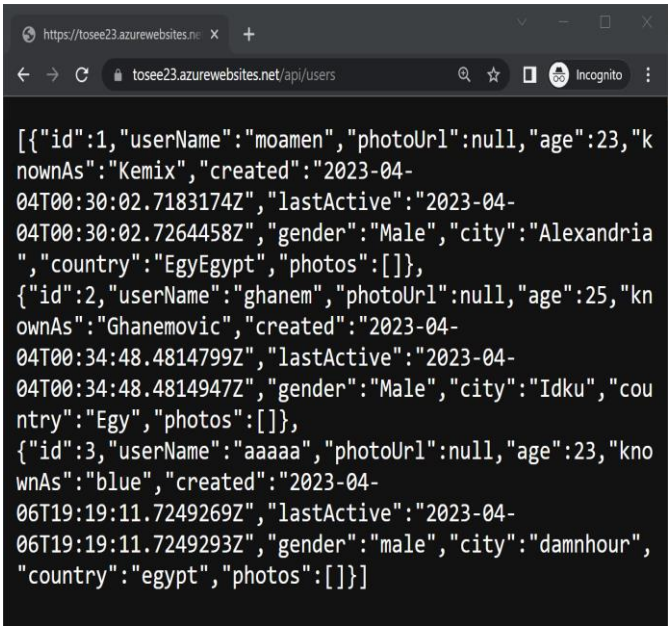


Figure 82 (data from api)

- We get our URL For API https://tosee23.azurewebsites.net/api/users

3.6.6 Database instance in Azure

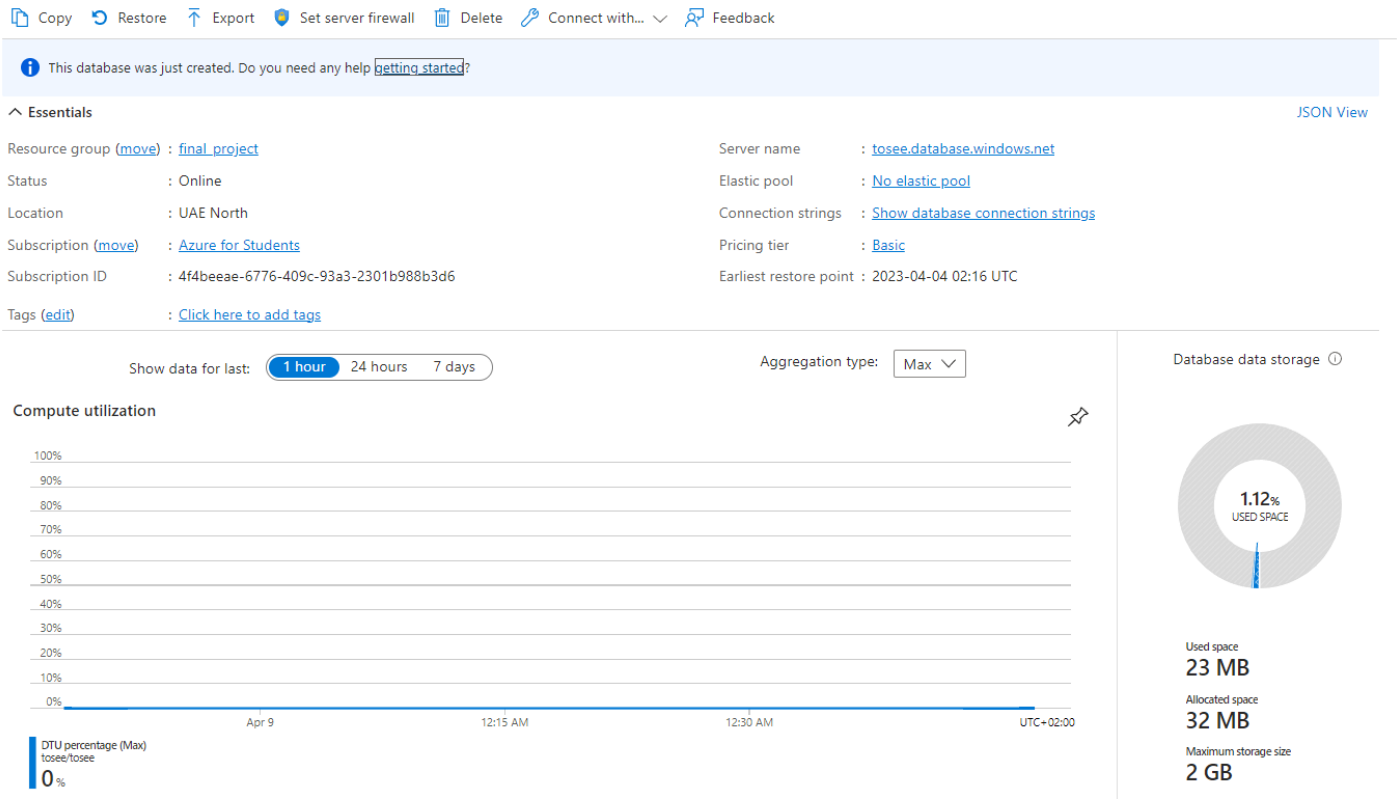


Figure 84 (database server info)

3.6.7 Linking API with Database:

Add connection strings to API from database

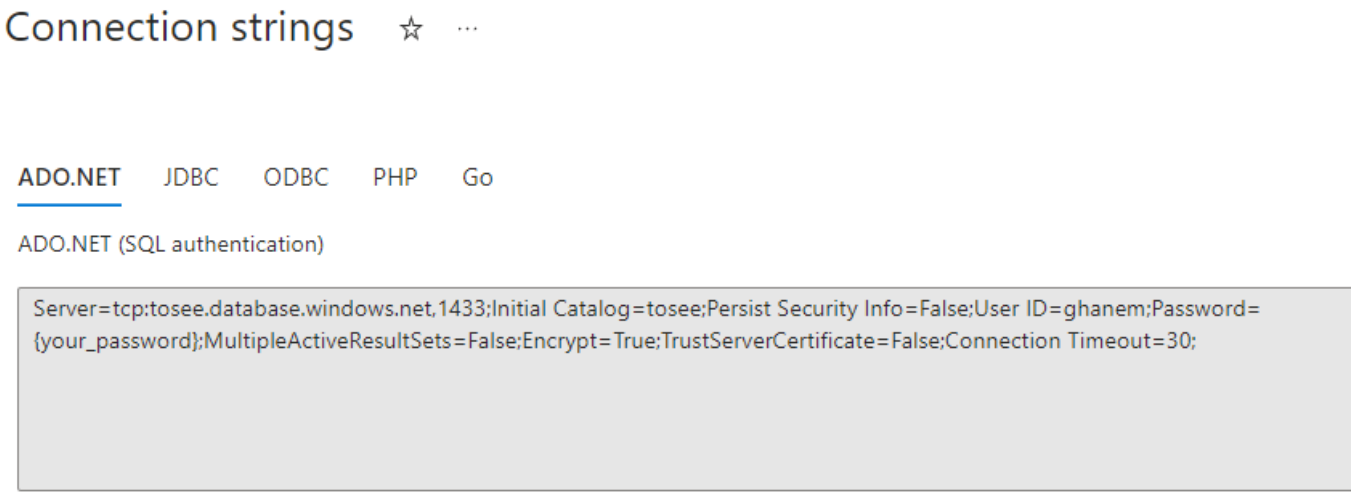


Figure 85 (connection string data)

Set the firewall in data with Ip v4 from the API server

Example of ip v4 from the server :

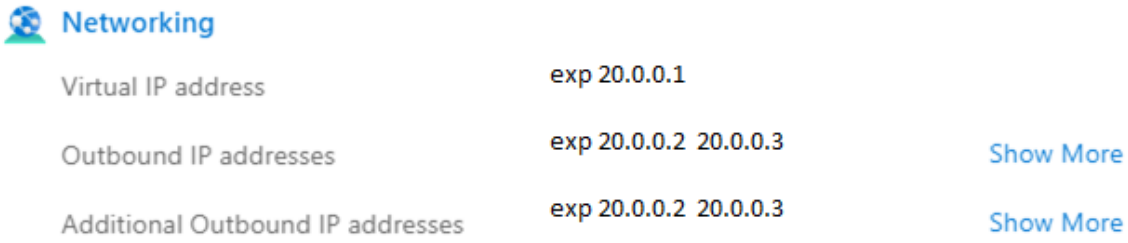


Figure 86 (api server ips)

3.6.8 Adding the IP v4 to database firewall

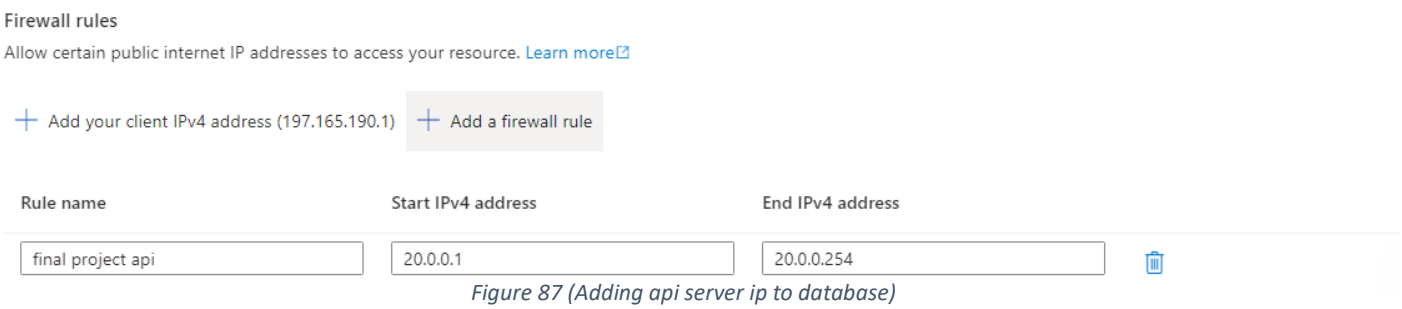


Figure 87 (Adding api server ip to database)

3.6.9 File Hosting on Cloudinary

Cloudinary is a cloud-based media management platform that offers tools and services for uploading, storing, manipulating, optimizing, and delivering images and videos on the web. It provides scalable and secure media storage, manipulation, optimization, and global content delivery network. Cloudinary integrates with third-party tools and services and offers APIs and SDKs. To host media on Cloudinary, create an account, upload media files, organize and manipulate them, optimize, and deliver them using the CDN. Cloudinary streamlines media management workflows and improves website and application user experience.

3.6.9.1 Domain Name:

A domain name is a unique character string that identifies a website on the internet, making it easy for users to remember and access a website using their web browser. Examples include "google.com" and "facebook.com".

3.6.9.1.1 Domain Name System (DNS):
The DNS translates domain names into numerical IP addresses used by computers to identify each other on the internet. When a user types in a domain name, their web browser contacts a DNS server to find the associated IP address. The server returns the IP address, enabling the website to be accessed.

3.6.9.1.2 Domain Name Registration:
Domain names are registered with a domain name registrar that manages the registration and associated IP addresses. Users pay a fee for a specific period, such as one or more years, and can renew the registration after it expires.

3.6.9.1.3 DNS Record:
DNS records are used to specify how domain names are translated into IP addresses, which are used to identify servers and other devices on the internet. There are several types of DNS records, each with its own purpose and function:

A (Address) Record: This record maps a domain name to an IP address. It is the most common type of DNS record and is used to translate domain names into IP addresses.

CNAME (Canonical Name) Record: This record is used to create an alias for a domain name. It allows multiple domain names to be associated with a single IP address.

MX (Mail Exchange) Record: This record is used to specify the mail server that is responsible for handling email for a particular domain name.

NS (Name Server) Record: This record is used to identify the name servers that are responsible for a particular domain name.

TXT (Text) Record: This record is used to store arbitrary text data associated with a domain name. It is often used to store SPF (Sender Policy Framework) records, which are used to specify which servers are authorized to send email on behalf of a particular domain name.

SRV (Service) Record: This record is used to specify the location of a specific service on a domain name. It is often used in conjunction with other records, such as MX records, to specify the location of services such as email or instant messaging.

3.6.9.5 Domain Rent:

from Name.com

Name.com is a Denver-based domain name registrar and web hosting company founded in 2003. It offers various services, including domain name registration, web hosting, website builder tools, email hosting, and SSL certificates. The company is owned by Donuts Inc.

Our URL Domain form name.com **tosee.live**

Chapter 4 : Component

4.1.0 Introduction

Raspberry Pi is a series of small single-board computers developed in the United Kingdom by the Raspberry Pi Foundation to promote the teaching of basic computer science in schools and developing countries. The original model became far more popular than anticipated, selling outside its target market for uses such as robotics. It does not include peripherals such as keyboards and mice or cases. However, some accessories have been included in several official and unofficial bundles.

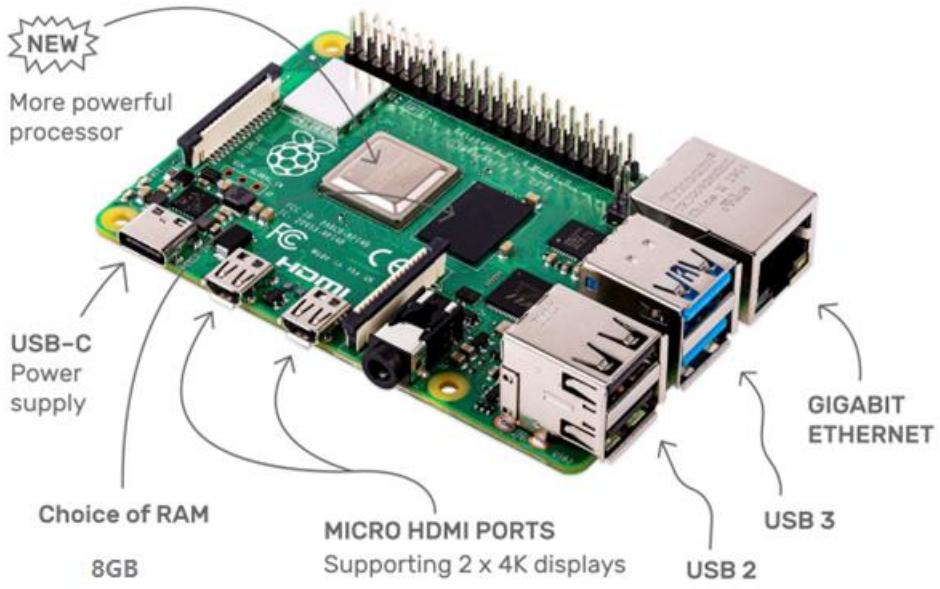


Figure 88 (Raspberry pi)

4.2.0 Specifications

- Processor: Broadcom BCM2711 64-bit SoC 1.5 GHz.
- 8GB LPDDR4-3200 SDRAM.
- Raspberry Pi standard 40 pin GPIO header
- Connectivity: Fast data transfer with USB 3.0, 2 USB 2.0 ports and Gigabit Ethernet.
- 2.4 GHz and 5.0 GHz IEEE 802.11ac wireless, Bluetooth 5.0.
- Supports two 4k HDMI displays.

- Storage: Micro-SD card slot for loading operating system and data storage.
- 2-lane MIPI DSI display port.
- 2-lane MIPI CSI camera port (for the 8 MP camera).
- 4-pole stereo audio and composite video port (3.5 mm Headphone Jack).
- 5V DC via USB-C connector (minimum 3A*).
- 5V DC via GPIO header (minimum 3A*).

4.3.0 Processor

BCM2711: this is the Broadcom chip used in the Raspberry Pi 4 Model B, the Raspberry Pi 400, and the Raspberry Pi Compute Module 4. The architecture of the BCM2711 is a considerable upgrade on that used by the SoCs in earlier Raspberry Pi models. It continues the quad-core CPU design of the BCM2837, but uses the more powerful ARM A72 core. It has a greatly improved GPU feature set with much faster input/output, due to the incorporation of a PCIe link that connects the USB 2 and USB 3 ports, and a natively attached Ethernet controller. It is also capable of addressing more memory than the SoCs used before.

The ARM cores are capable of running at up to 1.5 GHz, making the Raspberry Pi 4 about 50% faster than the Raspberry Pi 3B+. The new VideoCore VI 3D unit now runs at up to 500 MHz The ARM cores are 64-bit, and while the VideoCore is 32-bit, there is a new Memory Management Unit, which means it can access more memory than previous versions.

The BCM2711 chip continues to use the heat-spreading technology that started with the BCM2837B0, which provides better thermal management.

4.3.1. Processor: Quad-core Cortex-A72 (ARM v8) 64-bit SoC @ 1.5 GHz.

4.3.2 Memory: Accesses up to 8GB LPDDR4-2400 SDRAM (depending on model)

4.3.3 Caches: 32kB data + 48kB instruction L1 cache per core. 1MB L2 cache.

4.3.4 Multimedia: H.265 (4Kp60 decode); H.264 (1080p60 decode, 1080p30 encode); OpenGL ES, 3.0 graphics

I/O: PCIe bus, onboard Ethernet port, 2 × DSI ports (only one exposed on Raspberry Pi 4B), 2 × CSI ports (only one exposed on Raspberry Pi 4B), up to 6 × I2C, up to 6 × UART (muxed with I2C), up to 6 × SPI (only five exposed on Raspberry Pi 4B), dual HDMI video output, composite video output.\

4.3.0 Raspberry Pi GPIO

A powerful feature of the Raspberry Pi is the row of GPIO (general-purpose input/output) pins along the top edge of the board. A 40-pin GPIO header is found on all current Raspberry Pi boards (unpopulated on Raspberry Pi Zero, Raspberry Pi Zero W and Raspberry Pi Zero 2 W). Prior to the Raspberry Pi 1 Model B+ (2014), boards comprised a shorter 26-pin header. The GPIO header on all boards (including the Raspberry Pi 400) have a 0.1" (2.54mm) pin pitch.

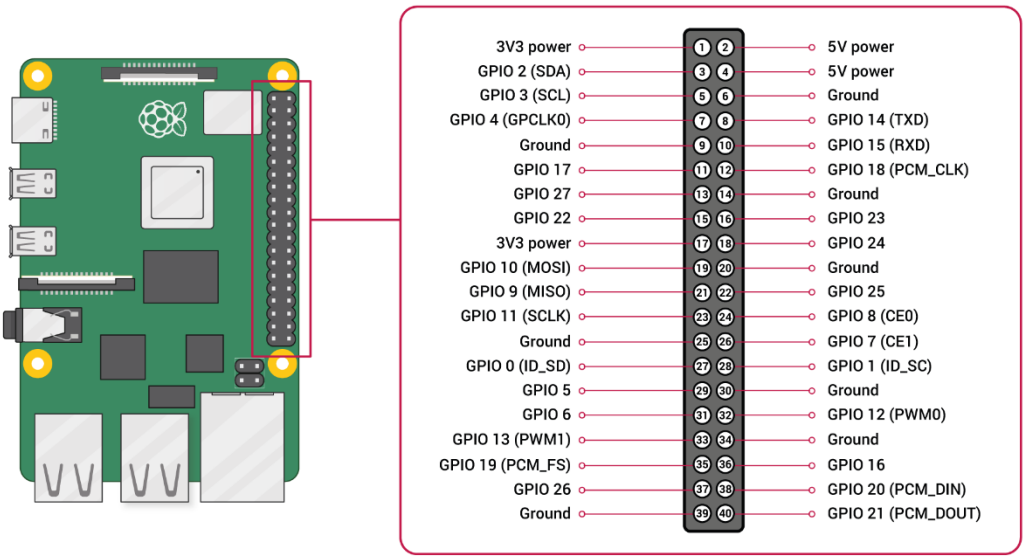


Figure 89 (Raspberry pi GPIO)

Any of the GPIO pins can be designated (in software) as an input or output pin and used for a wide range of purposes.

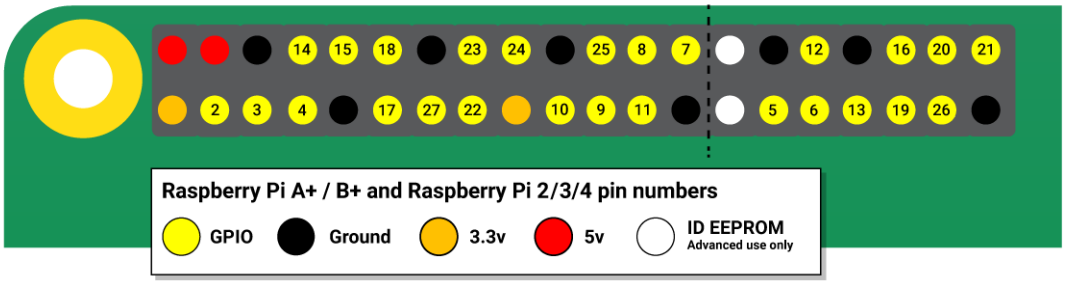


Figure 90 (Raspberry pi GPIO specification)

4.3.1 Voltages

Two 5V pins and two 3.3V pins are present on the board, as well as a number of ground pins (0V), which are unconfigurable. The remaining pins are all general purpose 3.3V pins, meaning outputs are set to 3.3V and inputs are 3.3V-tolerant.

4.3.2 Outputs

A GPIO pin designated as an output pin can be set to high (3.3V) or low (0V).

4.3.3 Inputs

A GPIO pin designated as an input pin can be read as high (3.3V) or low (0V). This is made easier with the use of internal pull-up or pull-down resistors. Pins GPIO2 and GPIO3 have fixed pull-up resistors, but for other pins this can be configured in software.

4.3.4 More

As well as simple input and output devices, the GPIO pins can be used with a variety of alternative functions, some are available on all pins, others on specific pins.

- PWM (pulse-width modulation)
  - Software PWM available on all pins
  - Hardware PWM available on GPIO12, GPIO13, GPIO18, GPIO19

- SPI
  - SPI0: MOSI (GPIO10); MISO (GPIO9); SCLK (GPIO11); CE0 (GPIO8), CE1 (GPIO7)
  - SPI1: MOSI (GPIO20); MISO (GPIO19); SCLK (GPIO21); CE0 (GPIO18); CE1 (GPIO17); CE2 (GPIO16)
- I2C
  - Data: (GPIO2); Clock (GPIO3)
  - EEPROM Data: (GPIO0); EEPROM Clock (GPIO1).

4.5.0 Transfer rate

Connection	Speed
USB 2.0	480 Mbps
USB 3.0	4.8 Gbps
GPIO (theoretically when using 8 pins)	1.9 MBps
Bluetooth 5.0	48 Mbps
2.4 GHz Wi-Fi	600 Mbps (long range)
5.0 GHz Wi-Fi	1300 Mbps (short range)
Ethernet	1 Gbps

4.6.0 Raspberry Pi Camera Module 2

4.6.1 Specification

The Raspberry Pi Camera Module 2 replaced the original Camera Module in April 2016. The v2 Camera Module has a Sony IMX219 8-megapixel sensor (compared to the 5-megapixel Omni Vision OV5647 sensor of the original camera).



Figure 91 (Raspberry pi Camera)

The Camera Module 2 can be used to take high-definition video, as well as stills photographs. It's easy to use for beginners, but has plenty to offer advanced users if you're looking to expand your knowledge. There are lots of examples online of people using it for time-lapse, slow-motion, and other video cleverness. You can also use the libraries we bundle with the camera to create effects.

You can read all the gory details about IMX219 and the Exmor R back-illuminated sensor architecture on Sony's website, but suffice to say this is more than just a resolution upgrade: it's a leap forward in image quality, colour fidelity, and low-light performance. It supports 1080p30, 720p60 and VGA90 video modes, as well as still capture. It attaches via a 15cm ribbon cable to the CSI port on the Raspberry Pi.

The camera works with all models of Raspberry Pi 1, 2, 3 and 4. It can be accessed through the MMAL and V4L APIs, and there are numerous third-party libraries built for it, including the Picamera Python library. See the Getting Started with Picamera resource to learn how to use it.

All models of Raspberry Pi Zero require a Raspberry Pi Zero camera cable; the standard cable supplied with the camera module is not compatible with the Raspberry Pi Zero camera connector.

4.6.2 GPS Sensor

The U-blox NEO-6M global positioning system (GPS) module, a very popular, cost-effective, high-performance GPS module with a ceramic patch antenna, an on-board memory chip, and a backup battery that can be conveniently integrated with a broad range of microcontrollers. It can track up to 22 satellites over 50 channels and achieve the industry's highest level of tracking sensitivity i.e. -161 dB, while consuming only 45 mA current.

Unlike other GPS modules, it can perform 5 location updates in a second with 2.5m horizontal position accuracy. The U-blox 6 positioning engine also has a Time-To-First-Fix (TTFF) of less than 1 second.

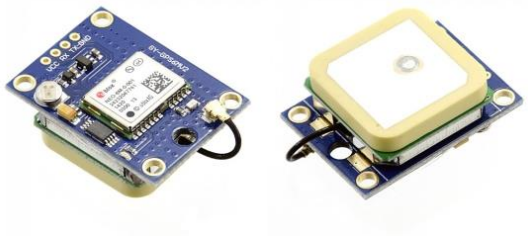


Figure 92 GPS Sensor

One of the best features offered by the chip is Power Save Mode (PSM). This allows a reduction in system power consumption by selectively switching certain parts of the receiver on and off. This dramatically reduces the power consumption of the module to just 11mA making it suitable for power sensitive applications such as GPS wristwatches.

The required data pins of the NEO-6M GPS chip are broken out to 0.1″ pitch headers. It contains the pins needed for communication with the microcontroller over the UART. The module supports baud rates from 4800bps to 230400bps with a default baud of 9600.

Specifications	feature
Receiver Type	50 channels, GPS L1(1575.42Mhz)
Horizontal Position Accuracy	2.5m
Navigation Update Rate	1HZ (5Hz maximum)
Navigation Sensitivity	161dBm
Communication Protocol	NMEA, UBX Binary, RTCM
Operating Temperature	-40°C ~ 85°C
Operating Voltage	2.7V ~ 3.6V

4.6.3 Joystick

joystick very similar to the 'analog' joysticks on PS2 (PlayStation 2) controllers. Directional movements are simply two potentiometers – one for each axis. Pots are ~10k each. This joystick also has a select button that is actuated when the joystick is pressed down. This is a Joystick module that has X,Y analog outputs and has a switch that detects button press. The joystick is made up of two passive potentio-meters (variable resistors) and a push button and therefore can be used at any standard voltage levels. Lots of robotic projects need a joystick. This module offers an affordable solution to that. The Joystick module is similar to analog joysticks found in gamepads. It is made by mounting two potentiometers at a 90 degrees angle. The potentiometers are connected to a short stick centered by springs. This module produces an output of around 2.5V from X and Y when it is in resting position. Moving the joystick will cause the output to vary from 0v to 5V depending on its direction. If you connect this module to a microcontroller, you can expect to read a value of around 512 in its resting position (expect small variations due to tiny imprecisions of the springs and mechanism) When you move the joystick you should see the values change from 0 to 1023 depending on its position.

Joystick that can be interfaced with Arduino, Raspberry Pi, and other Microcontrollers and Embedded Boards

Joystick for custom designed/experimental RC planes, RC Cars.

Joystick for CNC Machines or other industrial controls

Electronic games pads similar to PlayStation.

Using the board to connect the joystick is reduced to the following pins:



Figure 93 (Joystick)

- VCC – supply voltage (5 V).
- GND– ground of the system.
- VRx – the analog output for the vertical axis (in the middle position, the value is equal to half of the supply voltage VCC/2).
- VRy– Analog output for the horizontal axis (in the middle position, the value is equal to half of the supply voltage VCC/2).
- SW– Digital signal button, in case of activation, it reaches the value of the GND, deactivated – is in thefloating status – so you need to add a pulling-up resistor to power.

4.6.4 Headphones and Microphone

Wearing Smart glasses and headphones at once can be heard on the ears. That's why our smart glasses will be invented with built-in headphones and microphone. They're essentially a form of wireless open headphones that allow you to hear the outside world.



4.6.5 Power Bank

The Raspberry Pi needs an input voltage of around 5 Volts, which is supplied through the USB type-C port. In fact, the input voltage should be a little bit higher than 5 Volts.

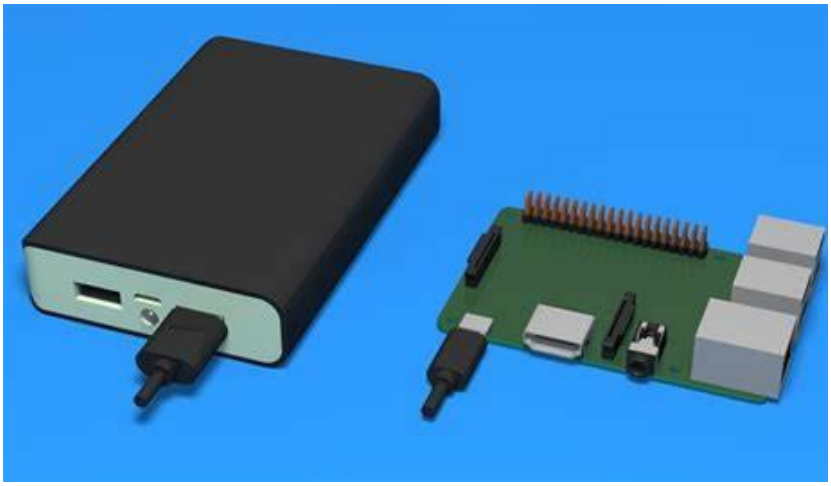


Figure 94 (Power Bank)

4.7.0 3D Printing:

3D printing falls under a section of digital fabrication that creates physical objects from digital 3D models by adding material layer by layer. It is also known as additive manufacturing because it builds objects by adding material rather than subtracting it like traditional manufacturing processes.

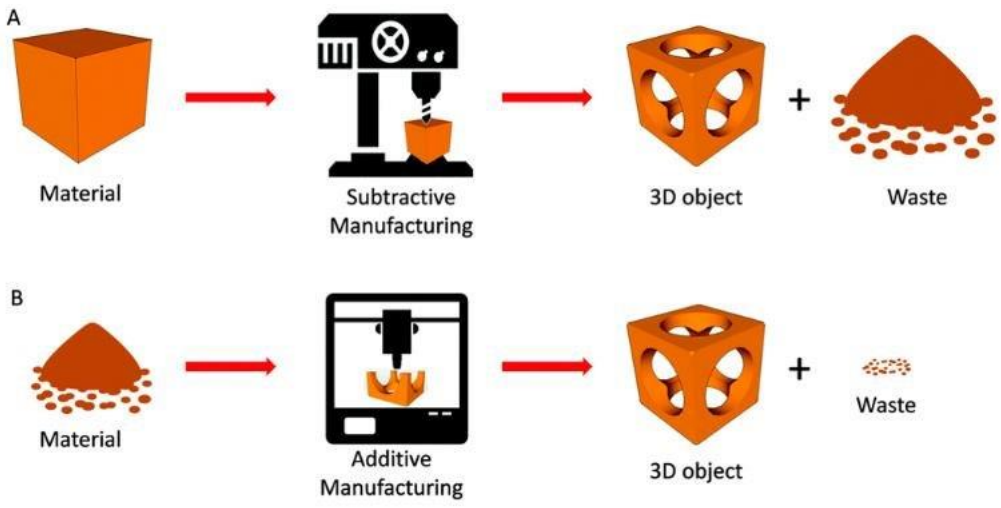


Figure 95 (Subtractive and Additive manufacturing)

4.7.1 3D Software:

There are many different software tools available like Tinkercad is free and works in a browser. Tinkercad offers a built-in feature to export your model as a printable file such as STL or OBJ for the next step which is called slicing.

4.7.2 CAD & CAN

3D CAD (computer-aided design) model is created or downloaded from a 3D model library. The 3D model is then sliced into thin horizontal layers using a software program called a slicer. The slicer generates a set of instructions, called CAM (computer-aided manufacturing) or G-code, that tell the 3D printer how to create each layer of the object.

The 3D printer then reads the G-code and starts to build the object by melting or extruding a material, such as plastic, metal, or resin, and depositing it layer by layer according to the instructions in the G-code. As each layer is completed, the print bed moves down or the print head moves up to create the next layer, until the entire object is complete.

4.7.3 3D Printing Techniques

There are many different types of 3D printing technologies, each with its own strengths and weaknesses. Some of the most common 3D printing technologies include FFF (Fused Filament Fusion) or FDM (Fused Deposition Modelling), DLP (Digital Light Processing), and SLA (Stereolithography).

4.7.4 Fused Filament Fusion (FFF)

Fused Filament Fabrication is an additive manufacturing technology, it's now the most popular 3D printing process, and in FFF a continuous filament of a thermoplastic material is used. Through a moving, heated printer extruder head, molten material is forced out of the print head's nozzle and is deposited on the workpiece Layer by layer to form the desired Product.

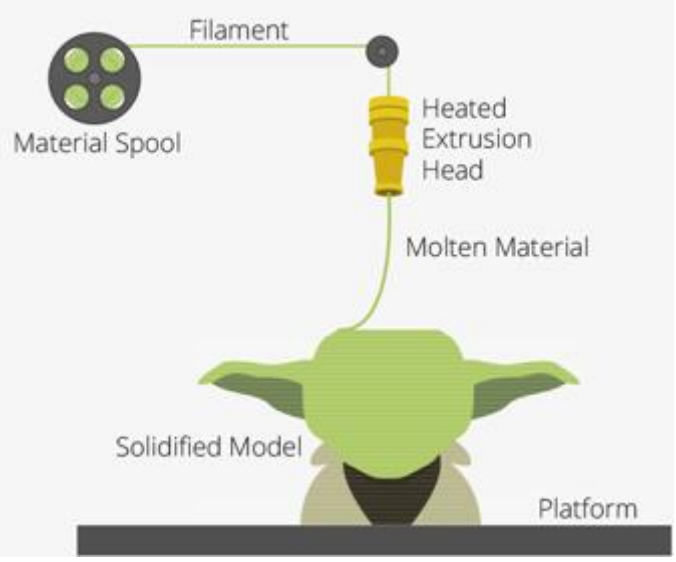


Figure 96 (FDM)

4.7.5 Digital Light Processing (DLP)

Digital Light Processing displays the image of the 3D model onto the liquid polymer. The exposed liquid polymer hardens and the build plate moves down and the liquid polymer is once more exposed to light. The process is repeated until the 3D model is complete and the vat is drained of liquid, revealing the solidified model.

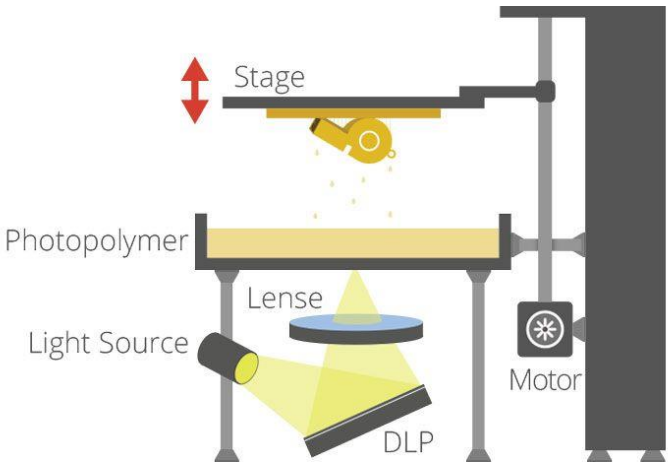


Figure 97 (DLP)

4.7.6 Selective Laser Sintering (SLS)

Selective Laser Sintering is an additive manufacturing technique that uses a high-power laser (for example, a carbon dioxide laser) to fuse small particles of plastic, metal (direct metal laser sintering), ceramic or glass powders into a mass that has a desired 3-dimensional shape.

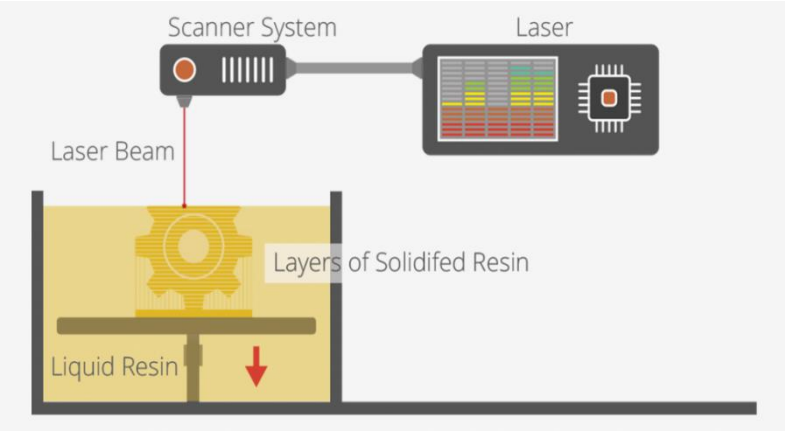


Figure 98 (SLS)

	Sinterit Lisa (SLS)	SLA	FDM
No need for support	✔	–	–
Freedom of form	●●●●●	●●●●○	●●○○○
Multiple parts printing at once	✔	–	–
Feature details	●●●●○	●●●●●	●●○○○
Printing materials	Powders	Liquids	Filaments
Temperature resistance	●●●●●	●●○○○	●●●●●
Durability	●●●●●	●●○○○	●●●○○
Printing of moving parts	✔	–	–

Figure 99 (SLS, SLA and FDM specification)

4.7.7 FDM 3D Printing material



Figure 100 (FDM materials specification)

4.7.8 3D Model for Raspberry Pi Camera

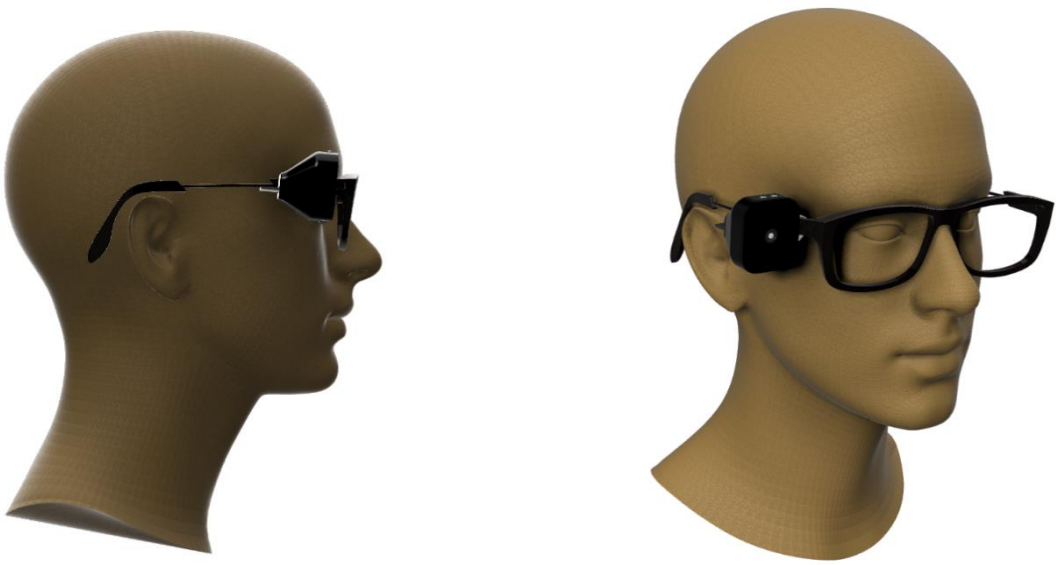


Figure 102 (Raspberry Pi 3D model)

4.7.9 3D Model for Joystick



Figure 103 *(Joystick 3D model)*

Chapter 5: Out Put

5.1.0 Hardware:

5.1.1 Paddle OCR:5.1.2 Yolov8s object detection:

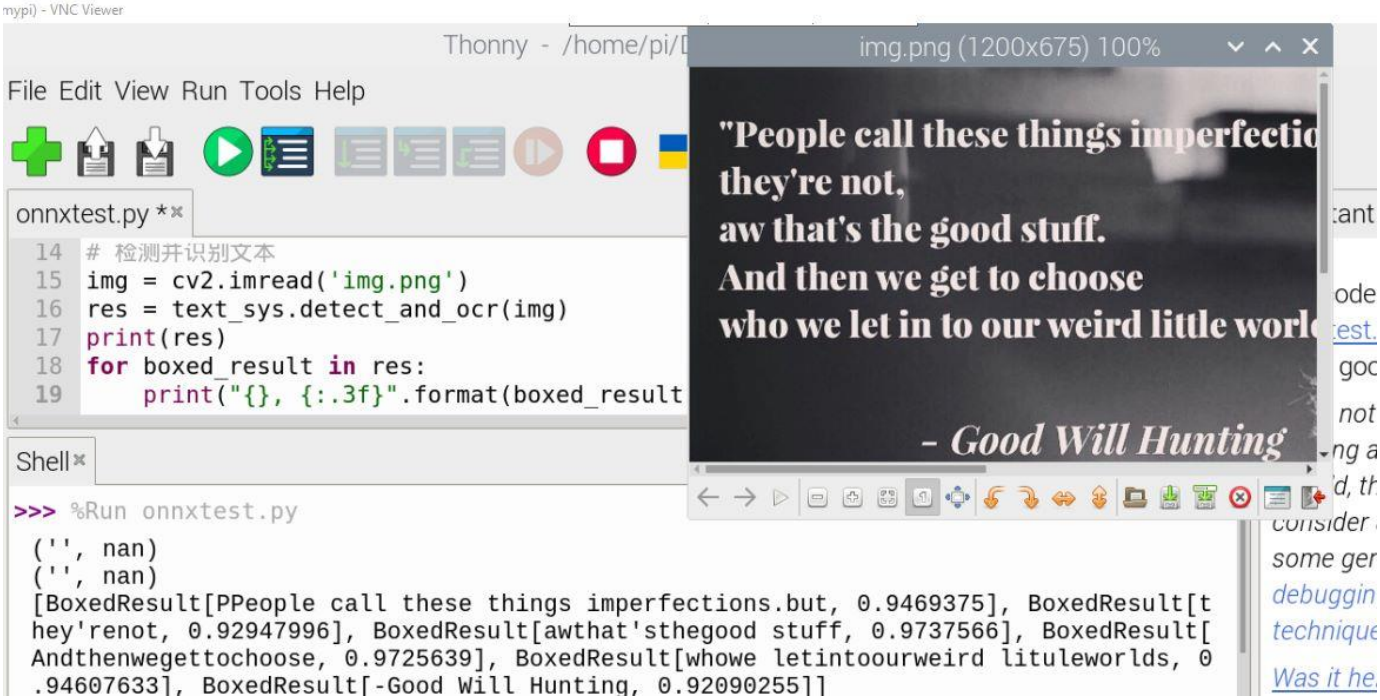


Figure 103(output ocr)

5.1.2 Face detection:

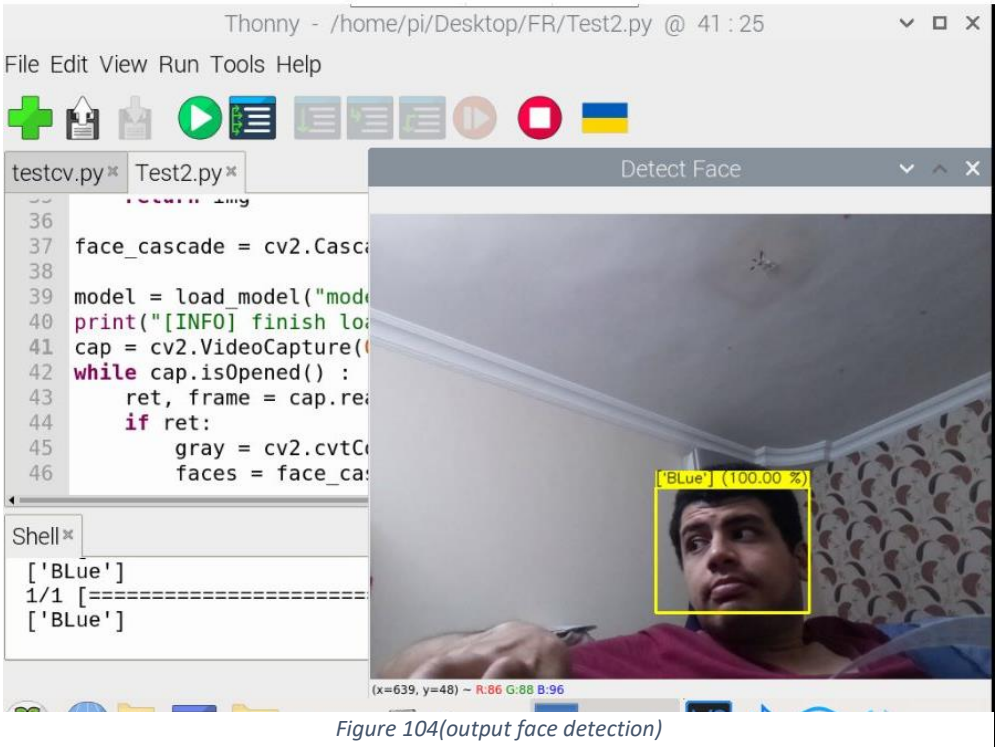


Figure 104(output face detection)

5.1.3 Object detection

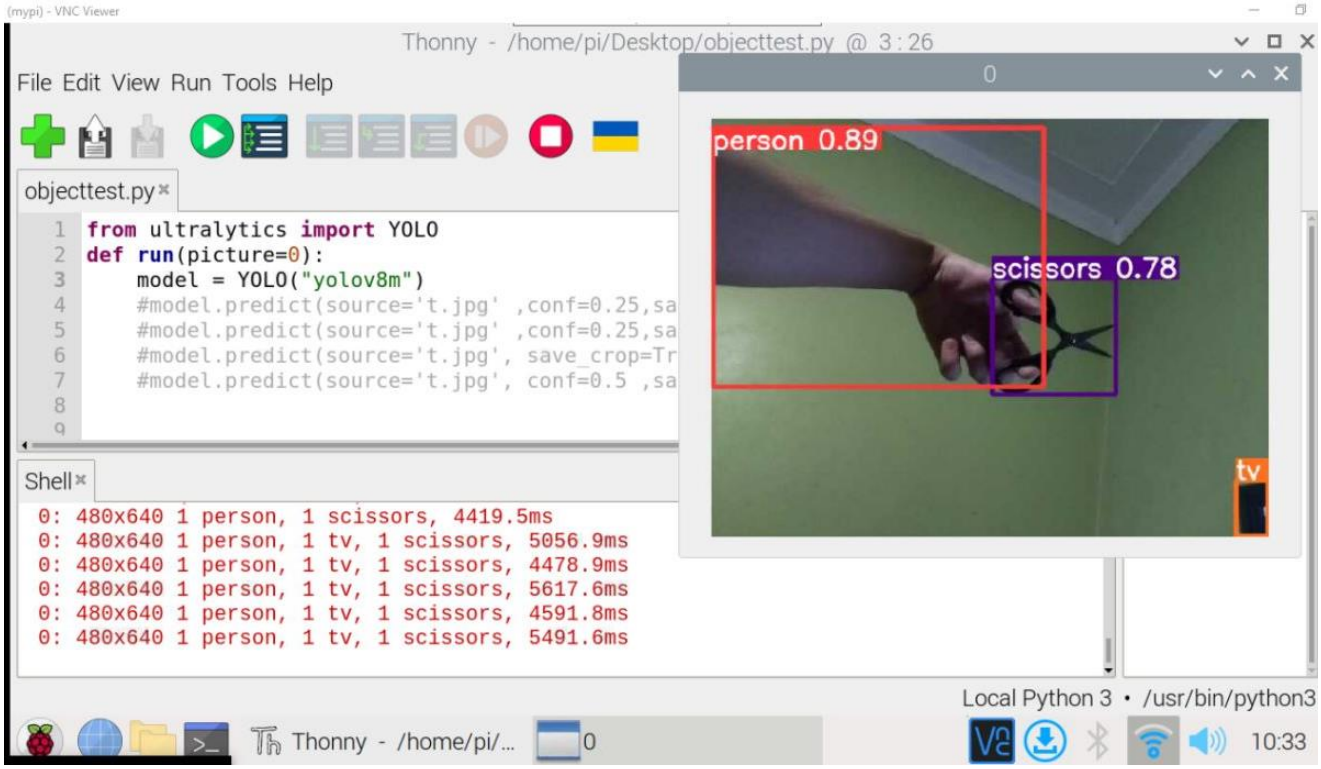


Figure 105(output object detection)

5.2.0 Software:

Mobile application code in GitHub:

http://github.com/sondos-sameer/Too-see

Web code in GitHub:

https://github.com/99zarka/tosee-build

5.2.1 Front end:

5.2.1.1 GitHub-pages

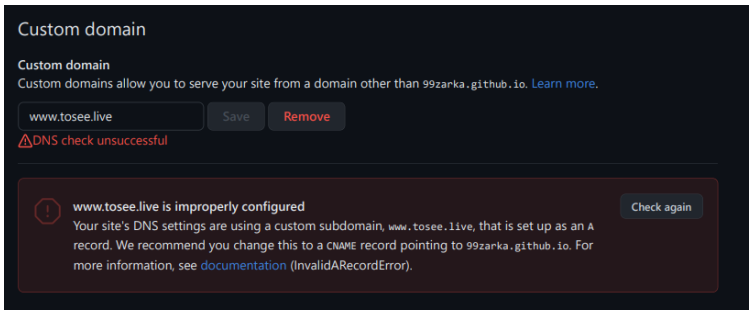


Figure 106 (adding custom domain)

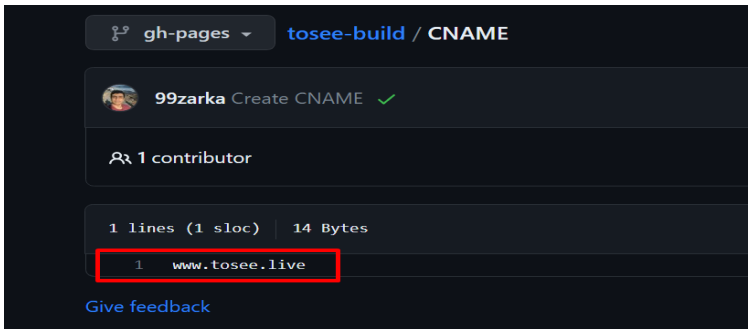


Figure 107 (create cname record)

5.2.1.2 Name.com

A	tosee.live	185.199.108.153 GitHub Pages	300	N/A	EDIT DELETE	CREATED: 2023-03-26
A	tosee.live	185.199.109.153 GitHub Pages	300	N/A	EDIT DELETE	CREATED: 2023-03-26
A	tosee.live	185.199.110.153 GitHub Pages	300	N/A	EDIT DELETE	CREATED: 2023-03-26
A	tosee.live	185.199.111.153 GitHub Pages	300	N/A	EDIT DELETE	CREATED: 2023-03-26
A	www.tosee.live	185.199.108.153 GitHub Pages	300	N/A	EDIT DELETE	CREATED: 2023-03-26
A	www.tosee.live	185.199.109.153 GitHub Pages	300	N/A	EDIT DELETE	CREATED: 2023-03-26
A	www.tosee.live	185.199.110.153 GitHub Pages	300	N/A	EDIT DELETE	CREATED: 2023-03-26
A	www.tosee.live	185.199.111.153 GitHub Pages	300	N/A	EDIT DELETE	

Figure 108(git-hub records)

5.2.2 Back end:

5.2.2.1 MS Azure

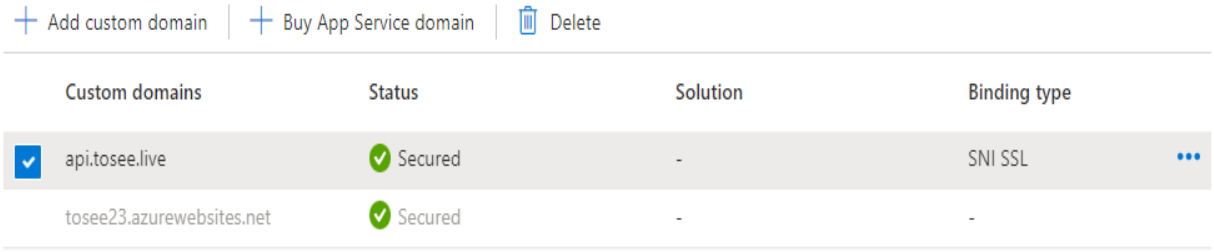


Figure 109(sub domain)

5.2.2.2 Name.com



Figure 110 (M.S Azure records)

5.2.2.3 URLS before applying our custom domain:

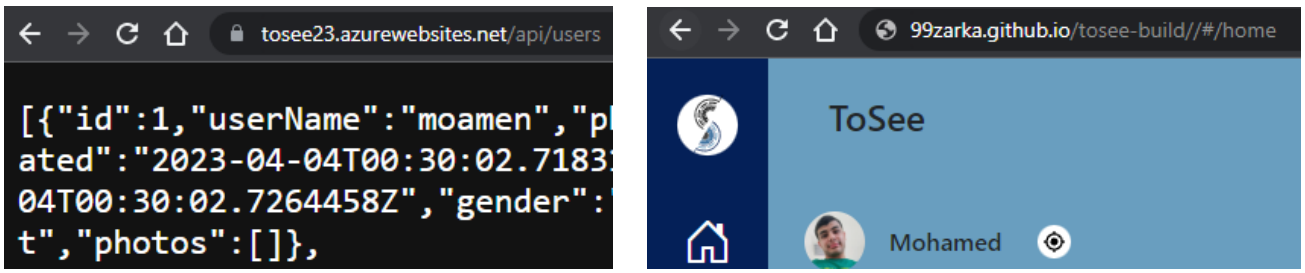


Figure 111(URL before domain)

5.2.2.4 URLS after applying our custom domain:

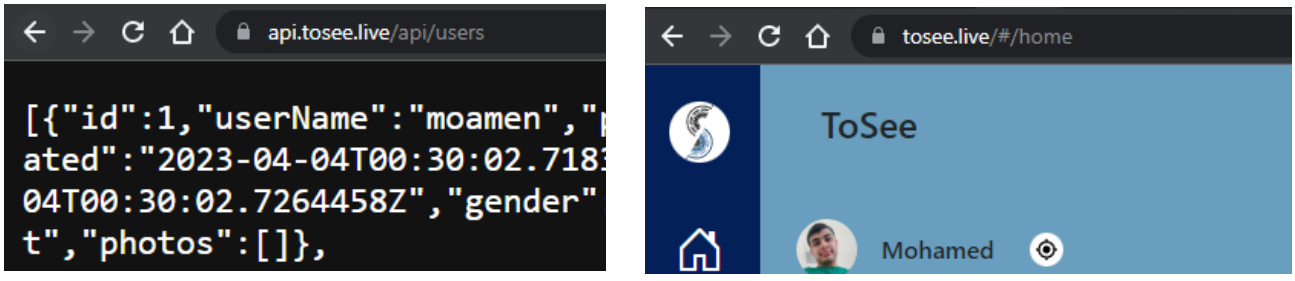


Figure 112 URL after domain

Conclusion

As the current prototype of the smart glasses for the blind requires an external processing unit, it may not be the most efficient solution for mass production. To implement the project efficiently, a semiconductor manufacturer could integrate the processing unit directly into the glasses, thus reducing the size and weight of the device. This would make it more practical and user-friendly for the visually impaired community.

With mass production, the cost of production would decrease, making the product more affordable and accessible to a larger audience. Moreover, the integration of additional sensors for healthcare monitoring could further expand the project's reach and attract a broader customer base, including those who are not visually impaired but require healthcare monitoring.

In summary, the integration of the processing unit into the glasses by a semiconductor manufacturer would increase the efficiency of the project and make it more practical for mass production. This, coupled with the inclusion of additional sensors for healthcare monitoring, could help the project reach a larger audience and attract a broader customer base.

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